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REVIEW ARTICLE



A systematic review of Lattice Boltzmann techniques for compressible flow modelling

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ABSTRACT

Accurately extending the Lattice Boltzmann Method (LBM) to compressible flows remains non-trivial, owing to the strong compressibility introduces large density and temperature variations, steep gradients and shock discontinuities, and stringent stability requirements, while thermodynamic consistency and a robust treatment of the energy equation (and related transport properties) are not straightforward within standard formulations. The paper aims to provide a systematic review of LBM-based approaches for compressible flow simulations, conducted with PRISMA guidelines. 109 research papers from SCOPUS database, published between 2012 and 2024, were filtered by inclusion and exclusion criteria. The selected literature is analysed from both bibliometric and technical perspectives, including publication trends, journal and geographic distributions, and – more importantly – methodological strategies adopted to handle compressibility based on the flow regime. The manuscript highlights how the energy equation is addressed and evaluates the associated choices of collision operator and turbulence modelling, together with the reported Mach and Reynolds number and introduces summary tables acting as a decision-oriented practical guide for method selection. The findings indicate a strong acceleration of efforts after 2020 and a consolidation of research focus on the Mach 0.5–5 range, alongside clearer specialization in method selection for higher-speed regimes. In particular, current trends favour multi-speed lattices, Double Distribution Function (DDF) frameworks, and hybrid LBM–CFD couplings, often combined with advanced stabilization mechanisms (e.g. entropic or multi-relaxation formulations and high-resolution shock-capturing updates for energy transport) to improve robustness in presence of shocks. In conclusion, the review synthesizes the state of the art, clarifies the main modelling bottlenecks, and highlights open gaps motivating future developments towards stable, thermodynamically consistent, and computationally efficient LBM tools for high-speed compressible flow applications. The purpose is to help researchers to recognize relevant references useful for their specific interests and to identify the most used strategies based on flow regime, configuration, and LBM approach.

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compressible flow; high
Mach number; CFD

1. Introduction

The Lattice Boltzmann Method (LBM), which evolved from the Lattice Gas Automata (LGA) method, has been recognized as an interesting tool for fluid dynamics simulations, due to its unique approach of utilizing the Boltzmann Transport Equation (BTE) rather than directly solving the macroscopic conservation laws of fluid mechanics. Unlike the conventional approaches, which directly address the conservation equations of macroscopic properties (i.e. mass, momentum, and energy), LBM employs the solution of the BTE to simulate fluids composed of fictive particles. The advantages of LBM simulation include ease of implementation, inherent strong potential for parallel processing in calculations, low numerical dissipation and dispersion error,

and proven high accuracy in in flow simulation (Perumal & Dass, 2015; Timm et al., 2016). Additionally, it has been shown that LBM can accurately reproduce the behaviour predicted by the compressible Navier-Stokes (NS) equations by determining macroscopic variables from the Boltzmann equation, achieving results comparable to a fourth-order central scheme with a four-stage Runge–Kutta method while computing 12.3 times faster than traditional NS solvers (Bayat et al., 2024; Succi, 2001). Moreover, recent progress in local-variable-based transition–turbulence models – for example, the Amplification-Factor Transport (AFT) model and other Reynolds-Averaged Navier-Stokes (RANS)-based schemes – has improved transition predictions in subsonic and transonic boundary layers (Jiakuan et al.,

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2025). This method defines the Probability Distribution Function (PDF) as the prospect of spotting particles at a particular position and with a specific velocity at a specified time. This function governs the evolution of fluid particles undergoing sequential streaming and collision processes across a discrete lattice during LBM simulations. The macroscopic fluid properties, including density and velocity, are acquired by taking moments of the Distribution Function (DF) (Perumal & Dass, 2015; Succi, 2001; Timm et al., 2016).

In general, for continuum systems, the LBM is expected to produce results consistent with hydrodynamic equations. The Lattice Boltzmann (LB) method, however, should result in outcomes that are comparable to those of other renowned methods, such as Molecular Dynamics (MD) or Monte Carlo (MC) simulations, in non-continuum systems, such as boundary layers where the Knudsen number is high. In the intermediate region between these two limiting cases – where hydrodynamic equations are no longer valid, and MD and MC methods become computationally prohibitive – LB modelling and simulation remain feasible. This viability is due to the inherent adaptability of the LB theoretical framework for describing complex systems characterized by spatially and temporally varying deviations from equilibrium (Xu et al., 2012). Nonetheless, the results produced by the LB method should be rigorously evaluated based on physical principles and analyses. The Chapman–Enskog analysis can be employed to expand the PDF around its equilibrium state as a Taylor series with respect to the Knudsen number. For a Knudsen number close to zero, indicating a system close to equilibrium, deviations from equilibrium become negligible, and the LB model converges with the Euler equations. By considering first-order terms in the Knudsen number while neglecting second-order terms – representing small deviations from equilibrium – the LB model aligns with the NS equations. Conversely, as the system moves further from equilibrium and second-order terms in the Knudsen number become significant, the LB model surpasses the descriptive capabilities of the NS equations (Bayat et al., 2024; Kruger, 2016; Xu et al., 2012).

Since its inception, the LBM has been particularly well-suited for simulations of weakly compressible and nearly incompressible complex flows (Succi, 2001), with successful applications spanning magnetohydrodynamics (Chen et al., 1991; Vahala et al., 2008), to flows of suspensions (Ladd, 1994), flows with phase separation (Cristea et al., 2009; Gunstensen et al., 1991; He et al., 1999; Sbragaglia et al., 2007; Shan & Chen, 1993; Sofonea et al., 2004; Swift et al., 1995; Xu et al., 2003; Xu et al., 2005; Xu et al., 2006), thermal flows containing heat transfer process (D’Orazio et al., 2004; D’Orazio

& Succi, 2004; Goodarzi et al., 2018; Karimipour et al., 2012; Karimipour et al., 2015), and flows through porous media (Succi et al., 1989; You-Sheng et al., 2008). Moreover, in the realm of particulate flows, continuum models of polydisperse granular segregation in hopper discharge demonstrate the challenges of capturing advection, diffusion and segregation; insights from such studies can inform the development of LBM frameworks for particulate and multiphase systems (Deng et al., 2020). The development of multiple-distribution-function LBM models has further enhanced its capability to simulate complex multi-component and multi-phase interactions, ensuring mass conservation and accurately capturing interfacial dynamics (Qiao et al., 2024; Xu et al., 2025). The versatility of LBM is evident in its application to high-Reynolds-number turbulent flows in complex rod-bundle geometries and to multiphase heat-transfer problems such as Leidenfrost-effect suppression on structured thermal pillars. These Graphics Processing Unit (GPU)-accelerated and central-moments LBM frameworks demonstrate how high-order and hybrid schemes can tackle diverse physical processes, providing insights for future compressible-flow models (Wang & Duan, 2025; Zou et al., 2025). However, as LBM techniques were originally designed for these nearly incompressible flows, they faced significant challenges when extended to fully compressible – those involving significant density and temperature variations or shock waves – flow regimes, where phenomena such as shock waves, sound propagation, and large density variations occur. In these regimes, traditional LBM models often struggled due to the limitations of standard lattice structures such as D2Q9 and D3Q19, which fail to accurately capture compressibility effects, particularly at high Mach numbers (Kruger, 2016; Renard, Feng, et al., 2021; Renard, Wisocq, et al., 2021). Emerging engineering problems such as high-speed water-entry of projectiles and the compression process in multi-stage wobble-plate compressors underscore the need for robust, compressible-flow solvers. Although these studies use continuum and fluid-structure-interaction approaches, they highlight scenarios where future LBM developments could make an impact (Liu et al., 2025; Tong et al., 2025).

Foundational research before 2012 shows that early attempts to extend the LB method to compressible flows were ingenious but ultimately limited. Alexander and co-workers (Alexander et al., 1992) introduced a compressible LB model by tuning the sound speed, which allowed higher Mach numbers but only for nearly isothermal flows. Guangwu et al. (1999) proposed a discrete-velocity model with three energy levels to recover the compressible Euler equations. Sun and collaborators (Sun, 1998; Sun & Hsu, 2003) then developed adaptive LB schemes

in which particle speeds vary with local Mach number and internal energy and extended these to two and three dimensions. Despite their innovations, all of these models retained the standard ‘propagation + collision’ framework and were either inconvenient to apply or numerically unstable, so very few practical results were reported. An alternative finite-difference LB approach offered more flexible lattice velocities, yet it still struggled with instability at Mach numbers greater than one; subsequent remedies such as entropy-based or flux-limiter schemes were largely confined to low-Mach regimes (Xu et al., 2012). By 2012, therefore, modelling compressible flows – especially those with shocks or discontinuities – remained a major obstacle for LBM, underscoring the need for innovative approaches like the additional-viscosity single-relaxation-time and multiple-relaxation-time frameworks that aim to improve stability while preserving the method’s efficiency.

This challenge has driven the need for innovative modifications and new methodologies that can extend LBM’s capabilities to simulate fully compressible flows with greater accuracy and stability.

One of the key approaches developed is the Multispeed (MS) method (Alexander et al., 1993), which enhances the accuracy of LBM by expanding the number of discrete velocities in the lattice, allowing for better recovery of the Navier–Stokes–Fourier (NSF) equations. This straightforward extension of LBM involves augmenting the number of discrete velocities of the lattice, but it often necessitates off-lattice methods (Watari & Tsutahara, 2003; Wilde et al., 2019) or other numerical schemes (Kataoka & Tsutahara, 2004), such as Finite Difference Methods (FDMs) or Finite Volume Methods (FVMs). This approach allows for the simulation of higher Mach number flows and improves accuracy in compressible regimes. However, These are computationally demanding compared to the standard collide-and-stream-based LBM. Moreover, severe instabilities arise when coupling the MS lattice with the standard collide-and-stream Bhatnagar-Gross-Krook (BGK) algorithm (Siebert et al., 2008).

Another significant advancement is the Double Distribution Function (DDF) method (He et al., 1998), which involves the utilization of an additional set of DF’s to represent energy. This method allows accurate modelling of thermal compressible flows at high Mach numbers but often eschews the standard collide-and-stream method in favour of robust numerical schemes (Feng et al., 2016; Guo et al., 2015; Xu & Huang, 2010). Some studies, however, have successfully applied the DDF model on a standard D2Q9 lattice with appropriate techniques (Frapolli et al., 2016b; Renard, Feng, et al., 2021; Saadat et al., 2019).

A third notable innovation is the Hybrid Lattice Boltzmann Method (HLBM) (35), which resembles the DDF method but calculates temperature using a Finite Difference (FD) or Finite Volume (FV) scheme instead of an additional population of DF’s. This approach is particularly effective in scenarios involving hydrodynamics under the Boussinesq approximation, where density variations appear only in a forcing term, with temperature treated as a passive scalar (Feng et al., 2018; Lallemand & Luo, 2003). To model fully compressible flows, an ideal gas coupling between the momentum and energy equations is imperative (Renard, Feng, et al., 2021). HLBM provides more accurate results in certain compressible flow regimes, but the computational demands of these additional solvers significantly increase the overall cost of the simulation (Feng et al., 2018; Lallemand & Luo, 2003).

Compared to traditional LBM models, these methods have a greater ability to model complex compressible flow phenomena, such as shock waves and sound propagation. Additionally, the hybrid methods offer flexibility in combining the strengths of both LBM and classical numerical solvers, enabling more accurate resolution of the energy equation. Moreover, the incorporation of techniques such as thermal Hermite expansions of the Equilibrium Distribution Function (EDF) has been crucial in accommodating compressibility in LBM models, though challenges such as flow stability and over-dissipation remain when handling local temperature variations (Dellar, 2014; Hosseini et al., 2020; Shan, 2019). Beyond compressible-flow applications, high-order analyses of LBM schemes, such as the recent truncation analysis for conservative Allen–Cahn models, highlight the importance of temporal- and spatial-difference corrections for improved accuracy and stability (Xu et al., 2023).

In this systematic review, we synthesize the procedural methodologies proposed to address the main bottlenecks in compressible LBM – namely thermodynamic consistency, robust energy-equation treatment, stability under steep gradients/shocks, and the reliable recovery of transport properties across wide Mach- and Reynolds-number regimes. This article is intended to provide researchers with a tool to identify suitable LBM techniques for their target compressible-flow simulations, by summarizing the advantages and disadvantages of the reviewed techniques together with the key ranges of relevant parameters. The reviewed literature is organized according to how the energy equation is handled in practice. This naturally leads to three overarching strategies, as follows:

- (i) single-distribution / multi-speed formulations (SDF/MS) that embed thermodynamics into an extended kinetic model,

- (ii) Double Distribution Function (DDF) frameworks that evolve a separate energy/temperature distribution,
- (iii) hybrid LBM–CFD couplings that delegate all or part of energy transport to an auxiliary solver.

Because the innovation of each paper often lies in specific implementation choices rather than a single universal category, we deliberately preserve paper-by-paper procedural details so that distinctive contributions remain traceable. In addition, the comprehensive comparison table set is intended to function as a practical guide: researchers can use it to identify suitable techniques for compressible-flow simulations based on the target Mach-number range and flow regime and then – depending on their preferred overarching strategy (SDF, DDF, or Hybrid) – quickly locate the most relevant reference study to follow as an implementation guide. Finally, we have included summary tables to further support synthesis and decision-making for readers. This material is intended to provide a clearer ‘implementation-level’ picture of what each approach does and why the resulting behaviour can differ in compressible-flow applications. This structure is used consistently to interpret the reported Mach/Re envelopes, stabilization mechanisms, collision-operator choices, and (when applicable) turbulence/transition modelling decisions.

The results of the systematic review make evident that the literature is highly heterogeneous, and doesn’t show a uniform picture of accuracy, stability, or computational cost evaluations. At this stage of the progress in this field, it is not possible to provide a uniform quantitative ranking or a consistent comparison across the papers.

The purpose of this paper is to provide a traceable evidence map of compressible LBM literature as a structured tool to help researchers to identify which classes of compressible LBM approaches have been used in different regimes and configurations.

2. Governing equations

As is described, extending the LBM to compressible-flow regimes requires that it consistently recover the compressible NSF equations. Despite the substantial differences among existing approaches – where the governing equations and mathematical structures can vary markedly depending on the chosen technique – this section provides a concise overview of the common mathematical foundations, mentioning both the relevant macroscopic governing equations and the principal LBM formulations developed for compressible flows.

2.1. Macroscopic governing equations

The compressible NSF equations presented here follow the standard derivation for a perfect gas with Fourier heat conduction and Newtonian viscosity.

For a Newtonian, calorically perfect gas obeying the ideal-gas law, the compressible NSF system can be written in conservative form as:

$$\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho u) = 0 \quad (1)$$

$$\frac{\partial (\rho u)}{\partial t} + \nabla \cdot (\rho u \otimes u) + \nabla P = \nabla \cdot \Pi \quad (2)$$

$$\frac{\partial (\rho E)}{\partial t} + \nabla \cdot [(\rho E + P)u] = \nabla \cdot (\lambda \nabla T) + \nabla \cdot (\Pi \cdot u) \quad (3)$$

where ρ is the density, u the flow velocity, $E = e + \frac{1}{2}u^2$ the specific total energy, P the pressure and T the temperature. The system is closed by the thermodynamic relations $P = \rho RT$ and $e = C_v T$, with R the gas constant and C_v the specific heat at constant volume.

The viscous stress tensor for a Newtonian fluid is:

$$\Pi = \mu \left[\nabla u + (\nabla u)^T - \frac{2}{3}(\nabla \cdot u)I \right] \quad (4)$$

and the heat flux obeys Fourier’s law $q = -\lambda \nabla T$. Here μ is the dynamic viscosity and λ is the thermal conductivity; for an ideal gas with constant Prandtl number $Pr = \mu C_p \lambda$ the two transport coefficients are related by $\lambda = \mu C_p / Pr$.

2.2. Lattice–Boltzmann equation

In LBM the state of the fluid is represented by a set of discrete DF’s $f_i(x,t)$ associated with a finite number of particle velocities (c_i). For each direction i , the evolution of f_i is governed by a discretised Boltzmann equation with a single-time-relaxation (BGK) collision operator:

$$\frac{\partial f_i}{\partial t} + c_i \cdot \nabla f_i = -\frac{1}{\tau_f} (f_i - f_i^{eq}) \quad (5)$$

where τ_f is a relaxation time linked to the fluid viscosity. The equilibrium distribution f_i^{eq} is chosen to ensure that the low-order moments of f_i recover the macroscopic density and momentum:

$$\rho = \sum_i f_i, \rho u = \sum_i c_i f_i \quad (6)$$

Using Chapman–Enskog expansion one can show that Equation (5) reduces to the NSF equations, when the equilibrium distribution reproduces sufficient velocity moments of the Maxwellian.

3. Materials and methods

This part outline the literature review approach and bibliometric analysis that were conducted. To ensure a comprehensive and reliable dataset, Scopus was chosen as the primary source. This decision was based on its extensive collection of research articles related to technological advancements, making it a more suitable choice compared to alternatives like Web of Science (Cabeza et al., 2020). On the other hand, resources such as Google Scholar and ResearchGate were not included due to concerns about the accuracy and reliability of their bibliometric data (Falagas et al., 2008). As previously mentioned, in light of the widespread adoption of the numerical approaches and the inherent advantages of LBM, significant enthusiasm has emerged for leveraging LBM in simulating diverse regimes, notably compressible flows, which are prevalent in gas-based industries. In this regard, the bibliometric analysis presented in the following sections explores various aspects – such as procedure, pros, cons, and more – of innovative approaches that employ the LBM for modelling compressible flows.

This systematic review of the application of LBM for simulating compressible flows was conducted in accordance with the Preferred Reporting Items for Systematic reviews and Meta-Analyses (PRISMA) guidelines (Gurevitch et al., 2018; Liberati et al., 2009; Page et al., 2021). As is well known, this method of reviewing the open literature is proposed to support a comprehensive and transparent search and reporting workflow, clearly delimited by the defined inclusion and exclusion criteria, and to provide an impartial and objective analysis for conclusive findings, avoiding bias. Based on this approach, the following structured procedure was followed for article selection. Initially, a preliminary identification phase was conducted based on the exclusion criteria, utilizing the titles of the publications. Subsequently, a screening step ensued, wherein the research abstracts underwent a meticulous evaluation to identify eligible papers according to the criteria of interest. Finally, studies deemed relevant following the analysis of the abstracts were selected for a comprehensive reading of the manuscript. Specific inclusion and exclusion criteria were demarcated and implemented throughout the identification, screening, and inclusion steps to select the studies of interest for the review, which were eventually fully-analysed.

Specifically, the electronic database SCOPUS was used for this study, and with the help of Boolean operators, a combination of keywords was used to find

relevant articles. More specifically, the following keywords were linked by the operator 'OR': 'LBM', 'Lattice Boltzmann', 'Lattice Boltzmann Method'; those keywords were linked by the operator 'AND' to the following keywords, again linked by the operator 'OR': 'Compressible Flow', 'Compressibility', 'High Compressibility', 'High-Compressibility', 'Compressibility Effect', 'Supersonic Flow', 'Transonic Flow', 'High Mach Number', 'High Ma Number', 'High Mach', 'High Ma'.

This study focuses on numerical solutions based on LBM for simulating compressible flows where the Mach number exceeds 0.3. To prioritize recent research, studies published from 2012 onwards (a span of 12 years), with a focus exclusively on English-language articles, have been considered as the target population. It should be pointed out that in their 2012 review article titled *Lattice Boltzmann Modeling and Simulation of Compressible Flows*, (Xu et al., 2012) Xu et al. concentrated on compressible flow simulations using the Boltzmann method conducted inside their research group.

Inclusion criteria:

- The research must have employed at least one method from the LBM family.
- The research must involve the simulation of flow with a Mach number greater than 0.3.

Exclusion criteria:

- Articles whose publication year is before 2012.
- Conference articles.
- The language is not English.
- Full text not available.

The division of tasks entailed an initial phase in which each of the three authors independently selected relevant material, followed by a subsequent phase in which all three authors collectively reviewed all studies. In instances of uncertainty, the senior author made the final decision.

4. Results

4.1. Formation of target database and related details

During the identification stage of this study, an initial total of 1104 articles were retrieved. Of these, 486 unique entries remained after the removal of duplicates, with 618 being removed. In regards to desires filters among the unique articles that remained, as shown in Figure 1:

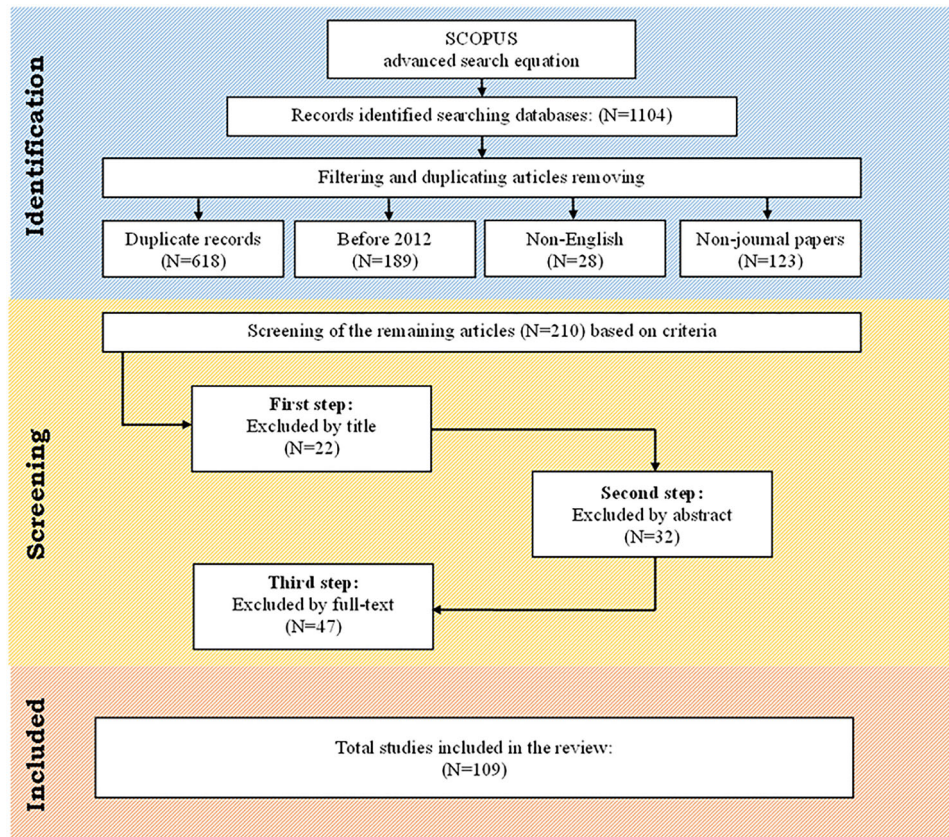


Figure 1. PRISMA flowchart for studies included in this review.

- Language: 28 were published in a language other than English.
- Publication Year: 189 were published before 2012.
- Type of Publication: 123 were published under a title other than scientific research or presented at conferences.

Since the aim of this review was to analyse journal research articles in English published after 2012, the initial dataset of 486 articles was refined using specific exclusion criteria. This filtering process led to the exclusion of 276 articles, leaving 210 that aligned with the predefined inclusion parameters. It should be noted that, some of these excluded articles, included more than one exclusion criterion. As detailed in Figure 1, 22 of these 210 articles were excluded based on their title, as they failed to meet the inclusion criteria. Additionally, another 32 were excluded after evaluating their abstracts. Consequently, 156 articles were shortlisted for further consideration. After an in-depth analysis of the full texts, 47 articles were excluded because they did not meet the inclusion criteria. The entire selection process followed the PRISMA guidelines and is illustrated in the diagram presented in Figure 1.

The final 109 articles reviewed are summarized in Table 1. This table provides the reference number, the name of the first author, the collision and turbulence model used, and the maximum Mach and Reynolds numbers simulated. The rows in this table are arranged alphabetically by the name of the first author. In order to maintain consistency in the article and facilitate comparison with the other parts, each article is given a specific code number.

As noted above, this research focused on the last 12 years. With the advent of new methodologies and advancements in computing power, interest in applying this method has radically increased. In fact, almost 55% of the studies were conducted after 2020, which means the last 38% of the investigated period, as depicted in the graph in Figure 2. It should be noted that the searching step was terminated in February 2024.

As shown in Figure 3, in terms of the affiliation country of the first author, $N = 37$ articles (Chen, Wang, et al., 2023; Chen, Yang, et al., 2023; Chen & Zhang, 2012; Feng et al., 2015; Feng et al., 2016; Feng & Tao, 2014; Guo et al., 2023; He et al., 2012; He et al., 2013; Huang, Jin, et al., 2023; Huang, Chen, et al., 2023; Li, Niu, et al., 2018; Li, Yuan, et al., 2018; Li & Shan, 2023; Li & Zhong, 2015a; Li & Zhong, 2015b; Li & Zhong, 2016; Lin et al., 2014;

Table 1. General specifications of the references and problem details studied.

Code	Ref.	1st Author	Collision Model	Turbulence Model	Max Re	Max Ma
#1	Atif et al. (2018)	Atif M	BGK	Without	512	*
#2	Bhadauria et al. (2021)	Bhadauria A	BGK	Without	1.5×10^4	0.85
#3	Bhadauria and Karlin (2024)	Bhadauria A	BGK	Without	100	5
				Very Large Eddy Scale model (VLES)		
#4	Casalino et al. (2022)	Casalino D	BGK	Without	2.2×10^7	1.35
#5	Chen, Wang, et al. (2023)	Chen J	BGK	Spalart-Allmaras	11.72×10^6	12.43
#6	Chen, Yang, et al. (2023)	Chen J	BGK	Spalart-Allmaras	6.5×10^6	10
#7	Chen and Zhang (2012)	Chen Q	BGK	Without	*	*
#8	Coratger et al. (2021)	Coratger T	HRR-p	LES	11.72×10^6	1.4
#9	Coreixas and Latt (2020)	Coreixas C	BGK	Without	*	2
#10	Coreixas et al. (2017)	Coreixas C	BGK & PR & RR	Without	10^6	0.57
#11	Dorschner et al. (2018)	Dorschner B	BGK (tunable adiabatic exponent) two-relaxation time	Without	400	100
#12	Dorschner et al. (2016)	Dorschner B	Entropic Multirelaxation Time (KBC)	Without	10,000	1.5
#13	Esfahanian and Ghadyani (2015)	Esfahanian V	BGK	Without	*	1.959
#14	Farag et al. (2021)	Farag G	HRR; Pressure-based BGK, Density-based regularized BGK	Without	800	4
#15	Farag et al. (2020)	Farag G	HRR-p	Without	800	1.5
#16	Feng, Boivin, et al. (2019)	Feng Y	HRR	Without	10^5	1
#17	Feng et al. (2020)	Feng Y	HRR	Without	10,000	1.2
#18	Feng, Guo, et al. (2019)	Feng Y	HRR	Without	10,000	1.69
#19	Feng et al. (2015)	Feng Y	BGK	Without	*	0.35
#20	Feng et al. (2016)	Feng Y	BGK	Without	*	0.35
#21	Feng and Tao (2014)	Feng Y	BGK	Without	*	10
#22	Frapolli et al. (2020)	Frapolli N	Quasi-equilibrium model combined with the entropic collision	Without	11.7×10^6	3
#23	Frapolli et al. (2016a)	Frapolli N	Entropic	Without	3×10^6	1.5
#24	Frapolli et al. (2015)	Frapolli N	Entropic	Without	3×10^6	1.5
#25	Frapolli et al. (2016b)	Frapolli N	Entropic	Without	400	1.8
#26	Ghadyani and Esfahanian (2014)	Ghadyani M	BGK	Without	*	2.9
#27	Ghadyani et al. (2015)	Ghadyani M	BGK	Without	*	10
#28	Ghaffarian and Hejranfar (2019)	Ghaffarian A	BGK	Without	5000	2
#29	Guo, Feng, Jacob, et al. (2020)	Guo S	HRR	Without	800	1.5
#30	Guo et al. (2021)	Guo S	BGK	Without	800	1.2
#31	Guo, Feng, and Sagaut (2020)	Guo S	HRR- ρ	LES	10,000	4.22
#32	Guo et al. (2023)	Guo Z	BGK	Without	1600	1.25
#33	Hanada and Kataoka (2021)	Hanada T	Collisionless	Without	*	5.1
#34	Hanada and Kataoka (2015)	Hanada T	Collisionless	Without	*	4.2
#35	He et al. (2012)	He B	BGK	Without	*	0.9
#36	He et al. (2013)	He Y	BGK	Without	*	10
#37	Hejranfar and Ghaffarian (2020)	Hejranfar K	BGK	Without	500	1.4
#38	Hejranfar and Ghaffarian (2016)	Hejranfar K	BGK	Without	*	1.4
#39	Hejranfar et al. (2020)	Hejranfar K	BGK	Without	*	0.8
#40	Hosseini et al. (2019)	Hosseini S.A	SRT & LKS & SRT-RR	Without	*	3.46
#41	Hosseini et al. (2020)	Hosseini S.A.	SRT & SRT-RR & MRT	Without	2000	4.1
#42	Huang, Jin, et al. (2023)	Huang H	Modified BGK based on the Li's model (Li & Zhong, 2015b)	Without	1000	3
#43	Huang, Chen, et al. (2023)	Huang X	BGK	Without	*	*
#44	Jalali Khouzani and Kamali Moghadam (2020)	Jalali Khouzani H	BGK	Without	*	*
#45	Huang et al. (2023)	Jalali Khouzani H	BGK	Without	5000	0.8

(continued).

Table 1. Continued.

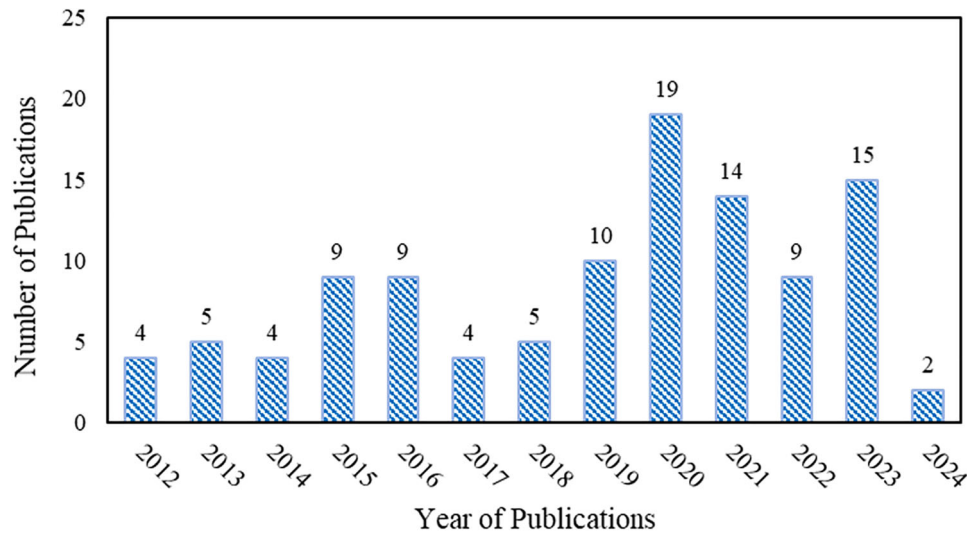
Code	Ref.	1st Author	Collision Model	Turbulence Model	Max Re	Max Ma
#46	Kallikounis et al. (2022)	Kallikounis N.G	BGK	Without	*	80
#47	Kallikounis et al. (2021)	Kallikounis N.G	Entropic	Without	1000	1.6
#48	Kallikounis and Karlin (2024)	Kallikounis N.G	Entropic	Without	1000	30
#49	Kamali Moghadam et al. (2021)	Kamali Moghadam R	BGK	Without	*	2
#50	Kataoka and Hanada (2019)	Kataoka T	Collisionless	Without	500	0.8
#51	Krivovichev and Bezrukova (2023)	Krivovichev G.V	BGK	Without	*	*
#52	Krivovichev and Marnopolskaya (2020)	Krivovichev G.V	BGK	Without	*	*
#53	Latt et al. (2020)	Latt J	Dynamic BGK	Without	10^4	1.6
#54	Li and Zhong (2015b)	Li K	BGK	Without	1000	2
#55	Li and Zhong (2016)	Li K	BGK	Without	10,000	0.8
#56	Li and Zhong (2015a)	Li K	MRT	Without	2×10^5	5
#57	Li et al. (2023)	Li W	RR	VLES	1.172×10^7	0.8
#58	Li, Niu, et al. (2018)	Li Y	BGK	Without	*	10
#59	Li, Yuan, et al. (2018)	Li Y	Collisionless	Without	*	10
#60	Li and Shan (2023)	Li Z	BGK	Without	1800	*
#61	Lin et al. (2017)	Lin C	BGK	Without	*	*
#62	Lin et al. (2014)	Lin C	BGK	Without	*	*
#63	Liu et al. (2019)	Liu Q	BGK	Without	*	11
#64	Liu et al. (2022)	Liu Z	BGK	Without	2×10^5	2
#65	Nasab et al. (2023)	Nasab H.H	BGK	Without	10,000	1.16
#66	Nejat and Abdollahi (2013)	Nejat A	BGK	Without	*	10
#67	Noah et al. (2021)	Noah K	BGK	LES	5.7×10^5	0.6
#68	Pae et al. (2023)	Pae K.N	BGK	Without	*	1
#69	Qin, Wu, Gu, et al. (2022)	Qin J	Collisionless	Without	*	2
#70	Qin, Wu, and Ma (2022)	Qin J	BGK	Without	5000	2
#71	Qiu et al. (2018)	Qiu R	BGK	Without	296,000	2.9
#72	Qiu et al. (2020)	Qiu R.F	BGK	Without	200	2.5
#73	Qiu et al. (2019)	Qiu R.F	BGK	Without	296,000	2.9
#74	Qiu et al. (2017)	Qiu R.F	BGK	Without	*	5.92
#75	Renard, Feng, et al. (2021)	Renard F	HRR	Without	800	1.5
#76	Renard, Wissocq, et al. (2021)	Renard F	BGK, PR, RR, HRR	Without	*	0.95
#77	Saadat et al. (2020)	Saadat M.H	DDF-BGK with quasi-equilibrium model (Karlin et al., 2013)	Without	10,000	1.5
#78	Saadat et al. (2019)	Saadat M.H	DDF-BGK with quasi-equilibrium mode (Karlin et al., 2013)	Without	800	1.3
#79	Saadat et al. (2021)	Saadat M.H	DDF-BGK with quasi-equilibrium mode (Karlin et al., 2013)	Without	800	1.2
#80	Saadat and Karlin (2020)	Saadat M.H	DDF-BGK with quasi-equilibrium mode (Karlin et al., 2013)	Without	12,000	0.85
#81	Safdari Shadloo (2019)	Safdari Shadloo M	BGK & MRT	Without	*	20
#82	Kumar and Maheshwari (2020)	Sashi Kumar G.N	Onsager BGK (Mahendra & Singh, 2013)	Without	*	5
#83	Şeker et al. (2023)	Şeker A.E	BGK	VLES	*	0.85
#84	Shim (2013)	Shim J.W	BGK	Without	*	*
#85	Shirsat et al. (2022)	Shirsat A.U	BGK	Without	*	30
#86	Chang et al. (2014)	Shu C	Collisionless	Without	9950	100
#87	Sun et al. (2018)	Sun Y	BGK	Without	5000	4
#88	Thyagarajan et al. (2023)	Thyagarajan K	MRT	Without	10^7	1.8
#89	Tran et al. (2022)	Tran S.B.Q	DDF-BGK with quasi-equilibrium mode (Karlin et al., 2013)	Without	10^7	1.8
#90	Tsutahara (2012)	Tsutahara M	BGK	Without	1000	0.7
#91	Liyue et al. (2022)	Wang L	BGK	Shear Stress Transport (SST)	*	0.75
#92	Wang and Zhang (2013)	Wang P	BGK	Without	*	*
#93	Wang et al. (2015)	Wang Y	BGK	Without	5000	0.8
#94	Watari (2016)	Watari M	BGK	Without	*	2

(continued).

Table 1. Continued.

Code	Ref.	1st Author	Collision Model	Turbulence Model	Max Re	Max Ma
#95	Wilde et al. (2021)	Wilde D	BGK	Without	1600	3
#96	Wilde et al. (2020)	Wilde D	BGK	Without	800	1.7
#97	Wissocq et al. (2023)	Wissocq G	RR	Without	*	1
#98	Xu et al. (2015)	Xu A	MRT	Without	*	2.30
#99	Yang et al. (2016a)	Yang L.M	Collisionless	Without	1.835×10^5	8.03
#100	Yang et al. (2016b)	Yang L.M	BGK	Spalart–Allmaras	5×10^6	0.76
#101	Yang et al. (2012)	Yang L.M	Collisionless	Without	*	10
#102	Yang et al. (2013)	Yang L.M	Collisionless	Without	*	10
#103	Yang et al. (2019)	Yang T	BGK	Spalart–Allmaras	*	*
#104	Yazdi et al. (2020)	Yazdi H	BGK with intermolecular force	Without	*	*
#105	Yoo et al. (2023)	Yoo H	HRR	LES	3.2×10^6	1.55
#106	Zhan et al. (2021)	Zhan N	Collisionless	Without	*	2
#107	Zhao et al. (2020)	Zhao S	HRR-p	Without	*	0.8
#108	Zhao et al. (2022)	Zhao X	BGK	k-Omega SST	11.72×10^6	0.85
#109	Zipunova et al. (2021)	Zipunova E	BGK	Without	*	100

Note: The * symbol denotes that the article provides no information on the specified topic.

**Figure 2.** The number of studies published in the last 12 years related to review.

Lin et al., 2017; Liu et al., 2019; Liu et al., 2022; Liyue et al., 2022; Qin, Wu, Gu, et al., 2022; Qin, Wu, & Ma, 2022; Qiu et al., 2017; Qiu et al., 2018; Qiu et al., 2019; Qiu et al., 2020; Sun et al., 2018; Wang & Zhang, 2013; Xu et al., 2015; Yang et al., 2012; Yang et al., 2013; Yang et al., 2016a; Yang et al., 2016b; Yang et al., 2019; Zhan et al., 2021) were from China, $N = 18$ articles (Bhadauria et al., 2021; Bhadauria & Karlin, 2024; Coreixas & Latt, 2020; Dorschner et al., 2016; Dorschner et al., 2018; Frapolli et al., 2015; Frapolli et al., 2016a; Frapolli et al., 2016b; Frapolli et al., 2020; Kallikounis et al., 2021; Kallikounis et al., 2022; Kallikounis & Karlin, 2024; Latt et al., 2020; Saadat et al., 2019; Saadat et al., 2020; Saadat et al., 2021; Saadat & Karlin, 2020; Thyagarajan et al., 2023) were from Switzerland, and $N = 16$ articles (Coratger et al., 2021; Coreixas et al., 2017; Farag et al., 2020; Farag et al., 2021; Feng, Boivin, et al., 2019; Feng, Guo, et al., 2019; Feng et al., 2020; Guo, Feng, Jacob, et al., 2020; Guo, Feng, & Sagaut, 2020; Guo et al., 2021; Renard, Feng, et al., 2021; Renard, Wissocq, et al., 2021; Safdari Shadloo,

2019; Wissocq et al., 2023; Yoo et al., 2023; Zhao et al., 2020) were from France. In the subsequent steps, the frequency of the country of affiliation of the first author included Iran (Esfahanian & Ghadyani, 2015; Ghadyani et al., 2015; Ghadyani & Esfahanian, 2014; Ghaffarian & Hejranfar, 2019; Hejranfar et al., 2020; Hejranfar & Ghaffarian, 2016; Hejranfar & Ghaffarian, 2020; Jalali Khouzani & Kamali Moghadam, 2020; Jalali Khouzani & Kamali-Moghadam, 2023; Kamali Moghadam et al., 2021; Nasab et al., 2023; Nejat & Abdollahi, 2013; Yazdi et al., 2020), Germany (Casalino et al., 2022; Hosseini et al., 2019; Hosseini et al., 2020; Wilde et al., 2020; Wilde et al., 2021), and Japan (Hanada & Kataoka, 2015; Hanada & Kataoka, 2021; Kataoka & Hanada, 2019; Tsutahara, 2012; Watari, 2016).

Another aspect to take into account is the frequency of articles published in journals (Figure 4). The survey revealed that journals such as 'Physical Review E', with 16% (Atif et al., 2018; Coreixas et al., 2017; Dorschner et al., 2016; Feng et al., 2020; Frapolli et al., 2016a; Guo

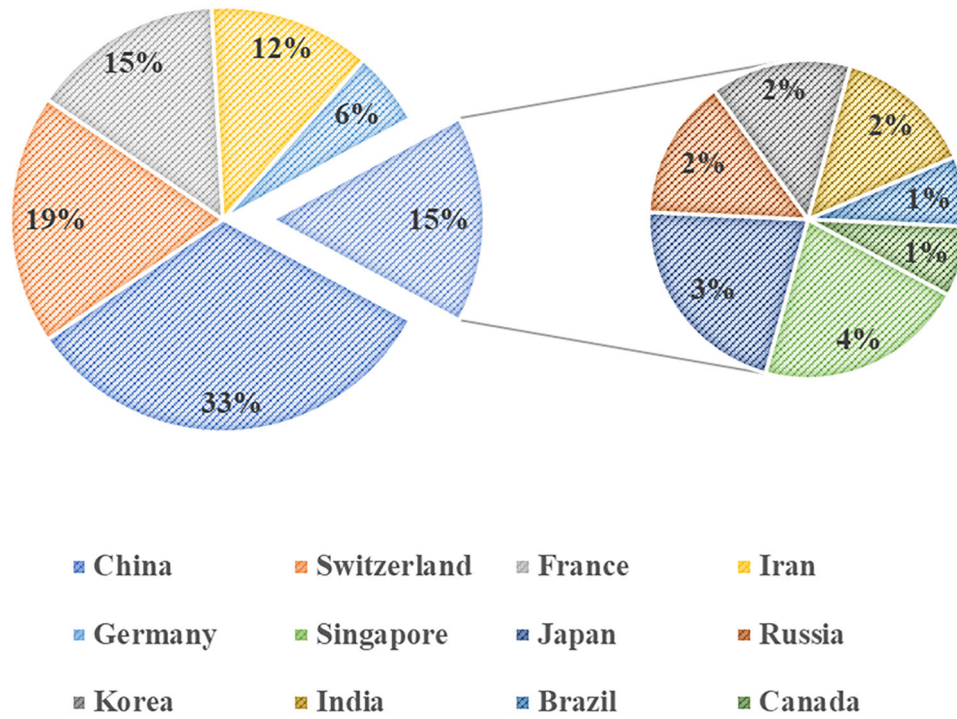


Figure 3. The percentage frequency of relevant articles is based on the country of affiliation of the first author.

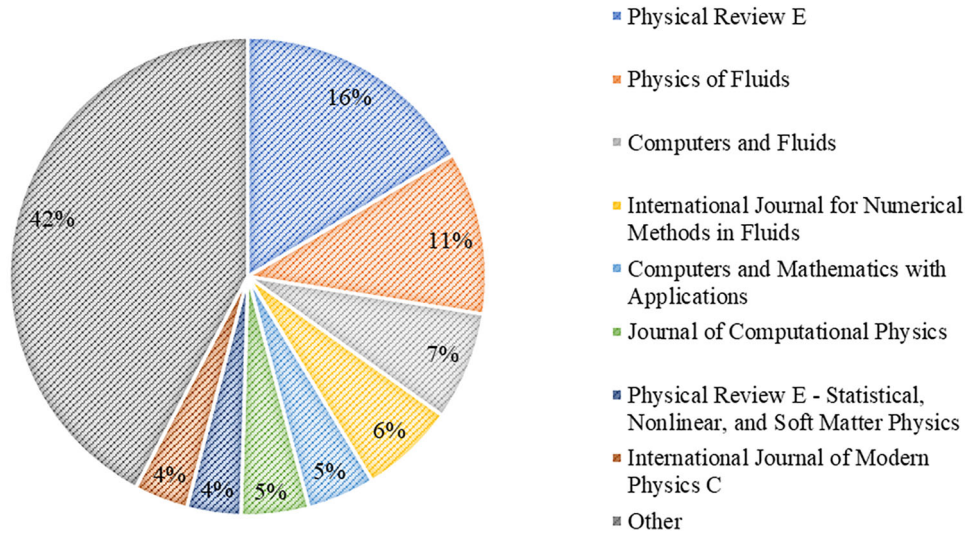


Figure 4. The percentage frequency of relevant articles is based on the journal in which they were published.

et al., 2023; Hejranfar et al., 2020; Hosseini et al., 2019; Kallikounis et al., 2021; Kallikounis et al., 2022; Kallikounis & Karlin, 2024; Saadat et al., 2019; Saadat et al., 2020; Shirsat et al., 2022; Wilde et al., 2020; Wilde et al., 2021; Yang et al., 2019; Yoo et al., 2023), 'Physics of Fluids', with 11% (Bhadauria et al., 2021; Coratger et al., 2021; Cor-eixas & Latt, 2020; Farag et al., 2020; Farag et al., 2021; Feng, Guo, et al., 2019; Guo, Feng, & Sagaut, 2020; Saa-dat et al., 2021; Saadat & Karlin, 2020; Şeker et al., 2023; Thyagarajan et al., 2023; Zhao et al., 2020), and 'Comput-ers and Fluids' (Bhadauria & Karlin, 2024; Chen, Wang, et al., 2023; Feng et al., 2016; Guo et al., 2021; Kumar &

Maheshwari, 2020; Renard, Feng, et al., 2021; Tran et al., 2022; Yang et al., 2013) were the most popular choices for publishing articles in this field.

4.2. Procedure, innovation of the techniques, object of the research

Table 1 reveals clear, method-dependent envelopes in both Mach and Reynolds numbers for compressible LBM. All reviewed approaches can reliably reach tran-sonic conditions ($Ma \approx 0.8 - 1.2$), but their extension

to higher Mach numbers differs substantially. In addition, Reynolds-number coverage is similarly application-driven: many studies focus on laminar or mildly turbulent regimes ($Re \sim 10^2 - 10^4$). Table 2 provides a concise description of the methodological processes utilized and the innovations implemented in each article.

To categorize the studied problems in the reviewed literature, eight general categories were defined, as illustrated in the Figure 5. The frequency distribution of the studied topics indicates that most researchers have shown a strong interest in problems related to shock wave structures and their interactions. Following this, the

Table 2. The procedure, innovation and pros and cons of the methods are presented in related references.

Code	Method	Procedure & Innovation	Advantages	Disadvantages
#1	Multispeed Thermal LBM	An energy-conserving LB model on a body-centred cubic lattice has been introduced for simulating thermohydrodynamics. The discrete EDF is selected to ensure consistency with the NS equations, conserving both mass and momentum.	Nonlinear stability across various parameters is demonstrated by the proposed model, making it ideal for simulating complex fluid dynamics, including turbulent Rayleigh-Bénard convection. The model's accuracy is enhanced by avoiding artificial closure on the third-order moment, unlike standard LBMs.	Although the model demonstrates high accuracy, its error increases with greater temperature deviations, showing 1.8% error at 50% deviation versus 0.4% at 5%.
#2	SLLBM	In this article, a Semi-Lagrangian Lattice Boltzmann Method (SLLBM) propagation method is employed as the chosen off-lattice scheme. A polynomial blended function approach is adopted to maintain mesh conformity, where a moving body influences mesh motion within a defined region.	Using an Arbitrary Lagrangian-Eulerian (ALE) formulation with polynomial blending functions enables the efficient recording of compressible flows and the treatment of multiple moving objects with minimal computational overhead.	This method, while inefficient for slender or closely spaced bodies, employs a weak solid-fluid coupling with staggered time-stepping, potentially causing an added-mass effect. Additionally, its off-lattice nature requires interpolation for population reconstruction, as characteristic lines' departure points no longer coincide with lattice nodes.
#3	PonD	The implementation of boundary conditions for simulating complex boundaries at high speeds using the Particle on Demand (PonD) approach, a kinetic theory-based method for high-speed compressible flows, is described. Challenges in computing missing populations within the semi-Lagrangian interpolation stencil were addressed using a non-equilibrium extrapolation scheme combined with inverse-distance weighting, enabling the construction of all populations at the identified boundary nodes.	This approach ensures stable and accurate boundary conditions, which are crucial for simulating complex flow phenomena like shock-vortex interactions and flow over two-dimensional cylinders in confined spaces.	The increasing computational complexity in LBM due to adding lattice velocities to preserve moments and improve isotropy is a common limitation. While PonD addresses this by adjusting discrete velocities based on local conditions, this adaptability can introduce computational overheads, especially in managing velocities and ensuring accurate simulations across diverse conditions.
#4	Transonic LB	In this study, the Rossiter-Heller formula for cavity resonance frequencies in rectangular cavities under subsonic and supersonic flow conditions is revised. The time-lag coefficient and acoustic backward propagation time are expressed as functions of the cavity aspect ratio and free-stream Mach number.	The new formula, based on corrective terms related to the Mach number and cavity aspect ratio, requires no additional parameters, enhancing resonance frequency prediction accuracy. The transonic LB solver overcomes D3Q39 lattice symmetry limits by computing fluid temperature via FD solutions of the total energy equation on the same Cartesian mesh.	The D3Q19 formulation can reach only slightly above 1 Mach numbers. For higher Mach numbers, you need to use a D3D39, which is also available in PowerFLOW (the one used for the cavity paper).

(continued).

Table 2. Continued.

Code	Method	Procedure & Innovation	Advantages	Disadvantages
#5	RLBFS	In this 3D Rotated Lattice Boltzmann Flux Solver (RLBFS), the NS equations are discretized using the FVM, with fluxes evaluated at each cell interface. Numerical stability is enhanced by obtaining convective fluxes through a hybrid approach, utilizing the D1Q4 LB model in two perpendicular directions, decomposed by the outer normal vector of each cell interface, while viscous fluxes are conventionally calculated using the second-order central scheme.	The proposed method effectively simulates supersonic flows around complex geometries with acceptable accuracy.	A notable limitation of this method is the current level of spatial accuracy. This method achieves second-order spatial accuracy, which may restrict the ability to capture details in complex flow fields, especially at high Mach and high Reynolds numbers.
#6	RLBFS	The innovation of this research lies in the development of a RLBFS that enhances stability in simulating compressible flows with intense shock waves. The RLBFS decomposes the cell interface's outer normal direction into two perpendicular velocity difference directions, with fluxes evaluated accordingly.	The RLBFS offers enhanced stability and reliability in simulating compressible flows with intense shock waves, avoiding unphysical oscillations and shock instability.	The method's primary limitations lie in its complexity and grid alignment dependency. While it stabilizes flux calculations by decomposing normal vectors perpendicularly, it remains sensitive to grid stretching, potentially compromising stability at high stretch ratios.
#7	FDLBM-G	A multi-step method for solving compressible flow problems using an improved FD-LBM, characterized by reduced dissipation, is presented. In this coupled DDF LB model, the evolution of the density and total energy DFs is solved using an implicit-explicit FDLBM, incorporating implicit-explicit Runge-Kutta schemes for time discretization and a Weighted Essentially Non-Oscillatory (WENO) scheme for space discretization.	The presented method introduces a new smoothness indicator that improves accuracy, reduces dissipation, enhances discontinuity capture, and refines simulations of compressible flows using the FD-LBM.	A limitation of the presented FD-LBM is that the standard LBM's collision transfer process is abandoned, which results in more complex programme development. Additionally, artificial viscosity is introduced for different time discretization schemes, leading to increased numerical errors.
#8	HRR-p LBM	A pressure-based compressible LB model incorporating mesh transitions, a Vreman subgrid turbulence model, and a shock-sensor is developed. Mass and momentum conservation equations are solved using a classical LBM solver coupled with a FV resolution of the entropy equation.	This method effectively manages complex geometries.	When encountering discontinuities at Mach numbers above 1.5, only the HRR-P LBM coupled with the total energy equation in a conservative format accurately produces the correct Rankine-Hugoniot jump relations. Although a model incorporating this feature has been proposed and validated for academic configurations, extending it to complex applications remains unresolved.
#9	ALBM	Adaptive LBMs that eliminate reliance on spatial interpolations are proposed. Velocity discretization is refined using domain decomposition, and collision handling is enhanced by Grad's approximation.	The method presented in this article demonstrates numerical stability and effectively handles nonlinear phenomena without requiring extensive interpolation.	This method is limited to simulating flows with a Prandtl number of 1. The root-finding solver's stability restricts temperature adjustments, limiting reference temperature changes for higher Mach numbers.

(continued).

Table 2. Continued.

Code	Method	Procedure & Innovation	Advantages	Disadvantages
#10	RR-LBM	This method extends the Recursive Regularization (RR) step to high-order LBMs within the Hermite basis framework, focusing on the regularization of pre-collision Velocity Distribution Function's (VDFs). The equilibrium state is expanded into a polynomial via Taylor expansion, non-hydrodynamic contributions induced by the streaming step are filtered, and non-equilibrium coefficients are computed through the Chapman-Enskog expansion by leveraging the recursive properties of the Hermite polynomial basis.	The RR procedure improves numerical stability compared to the Projection-based Regularization (PR) procedure in both standard and high-order LBMs.	This method's fixed specific heat ratio and Prandtl number ($Pr = 1$) significantly limit its application to fluids with a Prandtl number deviating from unity.
#11	PonD	A novel fluid dynamics formulation has been introduced, conceptualizing it as a kinetic theory with customizable particles. Semi-Lagrangian advection is employed to evaluate populations at a monitoring point by tracing back to a departure point using a reconstruction formula.	This method eliminates flow rate and temperature limitations, ensures error-free collision for numerical stability even with the simplest BGK model, and is adaptable to body-fitted unstructured meshes, not limited to space-filling lattices.	In this approach, the propagation step is computationally more intensive than the lattice propagation LBM.
#12	Entropic Lattice Boltzmann Method (ELBM)	A grid refinement technique has been developed, involving sequential steps. Equilibrium populations for LBM are computed by minimizing the entropy function with conservation constraints, using Lagrange multipliers.	This method keeps the number of populations in the multispeed lattice relatively small while reducing errors in the high Mach regime. Using locally embedded refined blocks within the computational domain significantly optimizes simulations by reducing computational effort.	The nonequilibrium parts depend on two independent relaxation parameters, presenting challenges in rescaling the f and g populations. Additionally, applying grid refinement techniques at the appropriate time and location can be difficult.
#13	Kataoka and Tsutahara (KT) Model (Perumal & Dass, 2015)	An innovative approach is presented in this research to enhance the stability and accuracy of the compressible LBM for simulating inviscid flows in high Mach number regimes. The limitations of the traditional discrete-velocity model by Kataoka and Tsutahara, which faced stability issues and numerical oscillations near shock wave discontinuities, are addressed by introducing a Modified Interpolation-Based (MIB) shock-detecting sensor that dynamically switches between second-order and higher-order filters, improving robustness and accuracy.	The presented method, utilizing a MIB sensor, enhances the numerical stability of compressible LB models, particularly in simulations with discontinuities like shocks, expansion waves, and contact discontinuities. Additionally, it prevents oscillations, a common challenge in simulating compressible flows.	This method is limited to simulating inviscid flows. Implementing the MIB sensor and numerical filters, particularly on mesoscopic variables, increases computational cost.

(continued).

Table 2. Continued.

Code	Method	Procedure & Innovation	Advantages	Disadvantages
#14	HRR-LBM	The proposed method integrates segregated, improved density-based, and pressure-based LBM into a unified framework. Forcing terms are introduced to partially balance numerical errors, resulting in a general formulation that encompasses both pressure-based and density-based methods.	The robustness and reliability of simulations increase by adjusting free fields in the equilibrium distribution, adopting the Monotonic Upstream-centred Scheme for Conservation Laws (MUSCL)-Hancock method for discretization, and applying supplementary regularization.	The methodology uses forcing terms to partially balance numerical errors, indicating a limitation in the model's inherent accuracy, as external adjustments are necessary to achieve the desired precision in simulations.
#15	HRR-p LBM	A novel hybrid model has been introduced employing a pressure-based predictor-corrector LB solver for the mass and momentum equations. The collision model has been modified to adopt a pressure-based approach, replacing the previous density-based HRR method.	The methodology effectively controls spurious pressure waves without requiring bulk viscosity and allows the use of an athermal collision kernel, eliminating sensitivity to case-dependent reference temperature definitions.	The approach alters the second-order moment of the population, necessitating a correction term to obtain a stress tensor comparable to traditional methods, highlighting a limitation in directly capturing the stress tensor without adjustments.
#16	HRR-LBM	A development of an HRR thermal LB model was presented, designed for simulating high subsonic and inviscid compressible flows. The governing equations, including energy conservation in entropy form, were considered.	The method offers low computational cost, acceptable stability, and accuracy. Its low dissipation and ability to simulate compressible flows make it an ideal option for aeroacoustic studies.	The proposed model simulates compressible flows at Mach 1, but its performance above Mach 1 remains untested and requires further investigation. Additionally, the model may encounter challenges in scenarios where viscosity plays a significant role.
#17	HRR-LBM	An innovative three-dimensional grid refinement technique within the HRR-LBM for compressible high-speed flows is presented. A multidomain approach, employing a communication algorithm for information exchange between fine and coarse domains at transition nodes, is utilized.	This paper presents an efficient grid refinement method for compressible aerodynamics simulations, effectively damping nonhydrodynamic modes while ensuring robustness and accuracy on both uniform and multi-resolution grids.	This method's limitation lies in the increased algorithm complexity and numerical cost in three-dimensional simulations due to the required rescaling procedure at multiple layer interface nodes.
#18	HRR-LBM	Efficient boundary conditions for compressible LBM methods, particularly the newly proposed HRR-LBM for compressible aerodynamics, are defined and evaluated. Simple extrapolation-based far-field boundary conditions, characteristic boundary conditions, and the absorbing sponge layer approach are extended for open boundaries in HRR-LBM.	The implementation of various interpolation, extrapolation, and absorbing sponge layer methods for solid walls, inlets, and outlets accurately simulates subsonic to supersonic nonreflecting boundary conditions and complex immersed solid walls, demonstrating proposed HRR-LBM's effectiveness in reducing meshing costs for complex geometries in steady and unsteady compressible aerodynamics.	The HRR collision model in the LBM incorporates multiple stages and specific configurations to manage open boundaries and complex boundary conditions. This research integrates various boundary conditions to simulate compressible flows and accommodate intricate wall geometries.
#19	Thermal compressible LB model	A three-dimensional model on standard lattices is introduced, utilizing a higher-order density DF to solve the flow field and an energy DF to compute the temperature field. The EOS for a thermal perfect gas is recovered through higher-order Hermite polynomial expansions in the NSF equations.	The method simulates thermal compressible flows using a higher-order density DF for the flow field and an energy DF for the temperature field, enhancing previous models. Through higher-order Hermite polynomial expansions, it recovers the thermal perfect gas equation of state in the NSF framework, accurately deriving EDF's for lattice models (D3Q15, D3Q19, D3Q27) with respect to discrete lattice symmetries.	The proposed three-dimensional LBM simulates thermal compressible fluid flows with large temperature differences and density variations in the low Mach number limit and also the third-order moment is not well restored in D3Q19 due to insufficient symmetry, with this defect becoming more pronounced when considering higher-order terms.

(continued).

Table 2. Continued.

Code	Method	Procedure & Innovation	Advantages	Disadvantages
#20	Multi-dimensional thermal LB model	A multi-dimensional DDF thermal LB model for fully compressible flows is developed, incorporating a FV scheme for temporal and spatial discretization. Third-order density and second-order total energy EDFs are employed on regular lattices.	The model effectively manages unsteady flows, fluid-solid/fluid interactions, and numerical dissipation, while its orthogonality feature ensures strong performance in multi-dimensional simulations.	The truncated approximation of EDFs must remain positive, but this is not ensured a priori. The deviation in third-order velocity moments, due to low-symmetry in lattice models, is corrected by adding an external force to the density distribution equation, restoring the missing cubic terms.
#21	Fully compressible multispeed LBM	A structured approach to simulating compressible thermal flows using a multispeed LB model is presented. Initially, a one-dimensional LB model, comprising an EDF, a D1Q5 lattice, and an evolution equation, was constructed.	Deriving EDFs from the moment system using a simple algebraic method simplifies computation, enhances model efficiency, and accurately captures complex flow features, including first and second Mach shocks, while also offering the potential for analysing compressible thermal flows.	*
#22	ELBM	In this article, the ELBM is employed to simulate thermal or fully compressible flows using higher-order networks. Shifted lattices are utilized to enhance the Mach number.	The methodology in this research efficiently treats shock waves interacting with turbulence, enables high-fidelity simulation of compressible turbulent flows, and seamlessly handles various compressible flow scenarios.	The methodology used in this research is limited by the finite set of discrete velocities, restricting the range of Mach numbers and temperatures in compressible LB simulations. Beyond this range, negative weights and equilibrium populations may arise, potentially causing numerical instability.
#23	ELBM	A systematic approach centred on the ELBM is outlined for simulating complex flows, including thermal and quasicompressible scenarios. Key steps involve population propagation to update discrete velocities, the implementation of accurate boundary conditions for solid wall interactions, and the conservation of mass, momentum, and energy using Lagrange multipliers while minimizing the discrete H function.	The presented method, characterized by simplicity and robustness, simulates a broad range of applications – from low Mach number and thermal flows to transonic and supersonic regimes – while accommodating arbitrarily complex obstacles using Cartesian meshes without grid refinement, turbulence models, tuning parameters, or shock front tracking.	In this method, minimizing the discrete entropy function, approximated as a polynomial in velocity, has two limitations: limited positivity (where the equilibrium distribution becomes negative at high Mach numbers) and inaccuracies at high flow velocities. Loss of positivity compromises the physical validity and introduces numerical instabilities.
#24	ELBM	An ELBM for compressible flows is introduced, utilizing a multispeed lattice to approximate the moments of the Maxwell-Boltzmann DF. Constructed through a hierarchy of tensor products of one-dimensional velocity sets, the lattice faithfully represents compressible flow dynamics.	This method ensures simulation fidelity, particularly in handling high Reynolds number flows and discontinuities like shock waves, without relying on ad hoc techniques such as tuning parameters or artificial dissipation.	A notable limitation of the ELBM model for compressible flow in the standard propagation-relaxation form is that it is restricted to a fixed adiabatic exponent (γ) of 5/3, corresponding to a monatomic gas. This limitation is inherent because the equilibrium distribution is based on the Maxwell-Boltzmann distribution for monatomic gases.

(continued).

Table 2. Continued.

Code	Method	Procedure & Innovation	Advantages	Disadvantages
#25	ELBM	The method presented demonstrates the Galilean invariance of LBM by enabling simulations in a comoving Galilean reference frame. This is accomplished using ELBM and introducing shifted lattices, which extend the operating range of LBM simulations, particularly for gas dynamics, without altering the standard propagation-collision algorithm.	The methodology in this paper offers Galilean invariance, enabling simulations across diverse flow regimes without altering the LB algorithm or increasing computational costs.	This method uses the development of the shifted lattice approach to improve performance in transonic and supersonic regimes as significant deviations occur when flow velocities or temperatures stray from the set reference conditions, and then using symmetric lattices leads to reduced accuracy. The shifted lattice approach can increase the complexity of the algorithms.
#26	Yan's LBM model	Yan is an LBM model with a multi-speed and multi-energy level and 17-particle velocities arranged in two standard D2Q9-type lattices. A complementary scheme is proposed for simulating inviscid compressible flows using LB models, incorporating numerical filters to enhance stability.	The proposed approach effectively removes spurious oscillations from shocks using numerical filters and a shock detection sensor, without impacting smooth or high-gradient regions.	This method does not account for viscosity, which presents challenges in scenarios where viscosity is not negligible.
#27	KT Model (Perumal & Dass, 2015)	An approach based on the discrete velocity KT model was proposed to simulate compressible inviscid flows using the FDLBM model. Numerical experiments revealed that the second-order FD scheme for spatial derivatives in the KT model induces Gibbs oscillations near discontinuities.	The sensor used in this article reduces numerical oscillations more effectively than the original compressible LB model and shows better performance with lower-order interpolations for larger values of artificial viscosity.	This model simulates inviscid compressible flows but faces challenges in scenarios where viscosity plays a significant role.
#28	SDLBM	In the article, the compressible form of the discrete Boltzmann-BGK equation with the Watari model is considered. A systematic approach to addressing viscous compressible flow problems is presented, primarily using the third-order Spectral Difference Lattice Boltzmann Method (SDLBM) coupled with an implicit dual-time stepping method that does not require matrix inversion.	Computations show that the SDLBM provides stable solutions for freestream Mach numbers up to about 2. While the tests are conducted on structured meshes, the SDLBM can extend to unstructured meshes.	For freestream Mach numbers exceeding 2, where strong shocks or discontinuities occur, suitable filters or limiters must be applied to stabilize the SDLBM. The method requires further development and assessment, as its computational cost and CPU time may exceed those of traditional SD methods or FD and FV LBMs.
#29	HRR-LBM	The method was designed for simulating three-dimensional compressible flows using a D3Q19 lattice, integrating the combined HRR collision operator with an optimized collision kernel, Hermitian basis, and correction conditions. Furthermore, energy conservation was innovatively implemented by solving the entropy conservation equation in a non-conservative form.	The D3Q19 grid with 20 degrees of freedom is employed as the minimum required for compressible aerodynamics within LBM framework. Utilizing the non-conservative form of the entropy equation for energy conservation minimizes calculation costs and reduces higher-order errors in mass conservation.	In the context of entropy conservation, the evolution of density is quickly corrupted by spurious entropy production, rendering this method unsuitable for supersonic flows characterized by strong non-equilibrium effects.

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Table 2. Continued.

Code	Method	Procedure & Innovation	Advantages	Disadvantages
#30	HLBM	Modified consistent energy equations in entropy and internal energy forms were introduced to reduce error terms and enhance consistency between conservative and non-conservative formulations of the energy conservation equation. LBM was applied for mass and momentum conservation, while the energy conservation equation was solved using either FVM or FDM.	The corrected and consistent energy equations proposed in this method demonstrate robustness and accuracy within the HLBM framework, enabling efficient simulation of compressible flows.	The proposed method is sensitive to thermodynamic variables, making it challenging to align conservative and non-conservative energy conservation equations. Additionally, the hybrid approach combining LBM with FVM or FDM for energy conservation increases computational complexity, potentially raising computational costs compared to using a single method.
#31	iHRR- ρ LBM	An improved standard thermal LBM model for compressible flows, known as the iHRR- ρ model, is proposed. The method employs a modified HRR collision model enhanced with correction terms, effectively filtering out unphysical modes during simulations without adjusting the σ parameter.	The proposed method exhibits low dissipation while maintaining simplicity, making it particularly beneficial for simulating compressible turbulent flows.	The simplifications in this article may not fully capture the complex dynamics of turbulent flows, as these processes differ fundamentally and may require separate optimization for improved accuracy.
#32	DUGKS	A reformulation of the Discrete Unified Gas Kinetic Scheme (DUGKS) has been introduced, incorporating WENO Flux Reconstruction (FR) and a modified source term. This specialized FV-LBM for the compressible NS equations in DDF formulation delineates mass and momentum transport via the kinetic equation for a density DF, while energy transport is governed by an energy DF.	The incorporation of the Van Leer limiter and WENO interpolation within the DUGKS framework mitigates additional numerical dissipation, preserving accuracy at local smooth extremes and ensuring well-resolved flow fields even in discontinuities also this method significantly reduces computational costs.	This method primarily addresses rarefied gas flows across various regimes, especially non-continuum flows. However, when applied to continuous compressible flows, managing numerous discrete velocities notably impacts computational efficiency.
#33	Sone's model (Timm et al., 2016) coupled by devising a discrete molecular-velocity model	Based on Sone's approach, this method was developed using the collisionless kinetic equation in the streaming process, with the DF modified to its Chapman-Enskog type at each time step. Discrete molecular velocities were utilized, eliminating the need to store velocity DF's.	This method reduces numerical dissipation by using small Δt to simulate flows with low relaxation time, and it calculates macroscopic variables without storing the velocity DF.	This method exhibits weak damping, particularly in cases of rough spatial mesh distribution.
#34	Sone's model; Free molecular type kinetic equation (Timm et al., 2016)	In this research, based on Sone's original idea, the development and application of a new free-molecular-type kinetic scheme for simulating two-dimensional supersonic flows were investigated. The proposed scheme solves the kinetic equation using a finite number of molecular velocities, ensuring that the macroscopic variables – such as density, flow velocity, and pressure – derived from the solution satisfy the desired fluid-dynamic-type equations.	This scheme, applicable to various cases, captures shock waves sharply without requiring spatial fitting methods.	The design's performance in 3D scenarios with complex geometries requires confirmation, particularly since it simulates supersonic flows below Mach 5.0 without spatial schemes like TVD or ENO, indicating a limitation in its applicability to higher Mach numbers, potentially restricting its use in hypersonic flow simulations.

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Table 2. Continued.

Code	Method	Procedure & Innovation	Advantages	Disadvantages
#35	Shan Xiaowen's model (Succi, 2001)	An improved LBM model for simulating compressible fluid flow is detailed, where boundary conditions are managed using a non-equilibrium extrapolation method. This method divides the DF into equilibrium and non-equilibrium parts, ensuring second-order accuracy, providing smoother transitions, and reducing errors.	This method effectively simulates one- to three-dimensional flows within the subsonic to transonic range.	The code's accuracy in supersonic flows remains unconfirmed, which may lead to challenges in these scenarios.
#36	IMEX FD-LBM	A 3D finite-difference LB model was developed to simulate compressible flows with shock waves. The model, based on the DDF approach, utilizes a density DF for the flow field and a total energy DF for the temperature field.	The present model features discrete equilibrium density and total energy DF's derived from Hermite expansions of continuous equilibrium distributions, enabling the simulation of high-Mach-number compressible flows with a flexible specific-heat ratio. An IMplicit-EXplicit (IMEX) finite-difference technique solves the discrete kinetic equations, enhancing computational convergence rate and numerical stability.	The IMEX FD scheme applied here helps with stability, but the computational complexity remains a concern.
#37	SDLBM	The development and application of the unstructured SDLBM for simulating compressible flows on unstructured meshes are presented. The compressible form of the discrete Boltzmann equation with the Watari model, a second-order upwind finite-difference LBM, is considered.	This method provides stability and accuracy while handling inviscid flows by adjusting parameters like relaxation time and wall boundary conditions, enabling precise simulations for both inviscid and viscous flow problems.	Stabilizing the solution algorithm to accurately predict stronger shocks, discontinuities, and curved-edge cells for representing curved wall boundaries in the flowfield requires further development and assessment.
#38	SDLBM	A method was presented employing a SD scheme to solve the compressible form of the LBE for inviscid compressible flows on structured grids. The compressible LBE, incorporating a Single Relaxation Time (SRT) and utilizing the BGK approximation for collision modelling, based on Watari's 2D model, was defined.	A sensitivity study examines how the accuracy of the SDLBM solution depends on numerical parameters. It finds that selecting a very small relaxation time and time step improves flow field predictions but can cause instability and oscillations near shocks due to reduced numerical dissipation and also this method may not be suitable for flow regimes in which viscosity is important.	A sensitivity study examines how the accuracy of the SDLBM solution depends on numerical parameters. It finds that selecting a very small relaxation time and time step improves flow field predictions but can cause instability and oscillations near shocks due to reduced numerical dissipation.
#39	ALE-FVLBM	In this method, the two-dimensional FV-LBM is formulated within the ALE framework on an unstructured, body-fitted mesh to accurately model the body shape and incorporate mesh movement due to body motion. The multiple particle speeds model proposed by Watari has been used.	This method remains stable against flow field discontinuities with varying specific heat ratios, accurately models body shape, and effectively accommodates mesh movement resulting from body motion.	The method applies poorly to flows with strong turbulence and viscosity effects because its solution accuracy depends on interpolation for exchanging information between moving and base networks and it neglects viscosity.

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Table 2. Continued.

Code	Method	Procedure & Innovation	Advantages	Disadvantages
#40	*	This study was dedicated to the concept of shifted stencils within the context of the LBM. In this regard, three exact EDFs – namely, second- and fourth-order polynomial Hermite expansions and entropic EDF – as well as three collision operators: the SRT, Lattice Kinetic Scheme (LKS), and fourth-order RR are examined.	This approach enables LBM to handle arbitrarily high nondimensional velocities, extending its validity range. Additionally, using standard first-neighbor stencils with shifted lattices minimizes memory consumption and processing costs.	The solver's applicability is constrained by shifted stencils, limiting it to flows with velocities centred on the stencil shift. This can cause instability near stagnant walls, especially when using first-neighbor standard stencils, requiring careful consideration of the flow domain and velocity profiles.
#41	Thermal LBM	The effect of local temperature on the behaviour and numerical properties of the LB solver for polynomial EDFs was systematically studied. Using von Neumann formalism, the linear stability domain for various EDF orders and advanced collision models, including regularized and Multi Relaxation Time (MRT) operators, was established.	Moment-matching EDFs demonstrate superior computational efficiency over Hermite expansion-based EDFs. Among various collision operators, the MRT model in temperature-scaled Hermite polynomial space exhibits the broadest stability domain.	The linear stability domain depends on the EDF order and collision models used, imposing constraints on stability that must be managed.
#42	Shifted lattice model	The potential energy DF, based on D3Q25 density DF, has been introduced to create a new DDF model. The MUSCL FV scheme was adopted in this method.	This model registers discontinuities, ensures stability, and allows for an adjustable Prontel number by employing a shifted lattice approach for coordinate conversion, thereby enhancing the model's feasibility.	One notable limitation of this model is that it achieves a maximum Mach number of up to 3, restricting its applicability for simulating flows with higher Mach numbers common in supersonic and hypersonic regimes.
#43	AMR-RLBFS	The methodology integrates Adaptive Mesh Refinement (AMR) to dynamically adjust the computational grid, enhancing efficiency and accuracy, with RLBFS for improved flux calculation at cell interfaces. The FVM is utilized to discretize and solve the governing equations, updating flow variables at cell centres, while the D1Q4 model is employed to evaluate inviscid fluxes at each cell interface.	The presented method efficiently achieves high numerical accuracy on feature-specified adaptive meshes while using relatively low computational resources.	The AMR method introduces complexities in managing mesh at interfaces between different refinement levels, leading to challenges in ensuring conservation across these interfaces and potential errors if not properly managed.
#44	AFV-LBM	The innovation of this research is the development of the AFV-LBM by combining LBM with the adjoint method for optimizing unsteady incompressible and compressible inviscid flows in one- and two-dimensional problems. This methodology leverages the Circular Function (CF) for estimating gradient vectors, facilitating optimization tasks effectively.	The proposed method efficiently handles both compressible and incompressible inviscid flows, accurately managing discontinuous compressible flow fields with strong shock waves. It achieves fast convergence without numerical instability, even with large design step sizes.	This method may not directly apply to viscous flows without modifications to account for viscosity, and the assumptions used may limit its accuracy in scenarios where these assumptions are invalid.
#45	AFV-ALBE	The first step involves solving LBE to determine flow field variables, including density, velocity components, and energy around the airfoil. Subsequently, the adjoint equation, derived from the LBE, is solved to calculate the sensitivity of a predefined cost function with respect to flow variables and airfoil shape.	The developed code in this article uses an optimization algorithm combining the adjoint method and LBM for airfoil shape optimization, significantly improving the efficiency of obtaining the objective function gradient, independent of the number of design variables.	Introducing a new lattice based on the CF concept presents challenges, as this approach, though robust for specific applications, inherently limits optimization to aerodynamic shapes.

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Table 2. Continued.

Code	Method	Procedure & Innovation	Advantages	Disadvantages
#46	Modified PonD	In this modified PonD method, Grad's PR of particle populations is combined with reference frame transformations to enhance stability and accuracy. Two discretization schemes, the FVM based on DUGKS and the Semi-Lagrangian Scheme, have been implemented in this study.	The revised reference frame transformation enhances the flexibility of this method in numerical implementation, improving the stability and accuracy of simulations and enabling the PonD model to effectively handle extreme cases of compressible flows with strong discontinuities.	The proposed method employs Grad's projection for regularization to enhance stability and accuracy, but its reliance on a specific moment system may limit applicability if the system is not well-suited to the simulated flow conditions.
#47	Multiscale SLLBM	The development of a multi-scale LBM is presented, which adaptively refines the velocity space of particles to efficiently and accurately simulate fluid flows, particularly in regions with high Mach or Knudsen numbers. Adaptive refinement allows higher-order models to be utilized only when necessary.	This method adaptively refines particle velocity space, utilizing higher-order models only where necessary, and efficiently couples discrete velocity sets, significantly reducing computational requirements.	The existence of overlap regions, where both lower and higher-order models provide an accurate description of the flow field, can present significant challenges.
#48	PonD	The focus of this work is on the properties of the reference frame, where kinetic equations are formulated in an adaptive reference frame defined by local fluid velocity and temperature. A consistent representation of populations across different reference frames is achieved through reference frame transformation.	The method in this paper ensures model accuracy throughout the system, eliminating the need to explicitly calculate forcing terms.	The direct formulation of a kinetic equation with adaptive velocities introduces additional forcing terms that include derivatives of particle speeds and imposes constraints on computational demands for frame transformations, which may be prohibitively demanding for elliptic-type equations.
#49	Hybrid- LBM	A axisymmetric compressible LBM has been developed for the numerical simulation of supersonic inviscid flow using the FVM. The method incorporates CF's with specific source terms to recover the Euler equations in cylindrical coordinates.	The 3rd-order algorithm effectively captures shock discontinuities in compressible axisymmetric supersonic flow around complex geometries.	This approach is unsuitable for phenomena lacking axial symmetry or involving complex 3D interactions that cannot be simplified into an axisymmetric framework.
#50	LBM based on Sone's model (Timm et al., 2016)	An LB model for the compressible N-S equations is introduced, based on Sone's kinetic system, and adapted as an effective numerical solver. Discrete molecular velocities and a Chapman-Enskog-type velocity DF are employed.	The method allows for the simulation of compressible flow and sound propagation around complex geometries, enabling the free selection of flow parameters, including three transport coefficients. It ensures that specific moments of the velocity DF satisfy predefined relations, leading to accurate solutions of the compressible N-S equations.	The proposed model accurately solves the compressible NS equations for any parameter set, provided that the time interval is sufficiently small.
#51	Multi-speed	A numerical procedure for stability analysis, based on constructing stability domains in parametric space, is proposed. Discrete velocity kinetic models used in LBM simulation of compressible flows, with varying specific heat ratios in the equation-of-state, are analysed.	This study simplifies the analysis of linear differential equations by focusing on zero solutions to precisely determine stability domains. The approach applies to various fields, including multiphase and magnetohydrodynamic simulations, and reveals how input parameters affect stability, aiding model optimization.	The method identifies that certain models, like the one with 65 equations, while stable, are not optimal for practical computations due to their complexity, which increases computational cost and implementation difficulty.

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Table 2. Continued.

Code	Method	Procedure & Innovation	Advantages	Disadvantages
#52	Optimized FDLBM	In this paper, an optimization approach for FD schemes used in solving the Linearized Acoustic Equations (LAE) within a computational splitting procedure is proposed. The schemes, dependent on a scalar dimensionless parameter, approximate terms with spatial derivatives.	The methodology proposed in this paper optimizes finite-difference schemes for the advective stage of LBM by minimizing dispersive and dissipative characteristics, leading to reduced numerical dispersion and improved convergence.	The proposed method's validity remains unconfirmed in both 2D and 3D cases, as well as across different physical scenarios.
#53	Modified quadrature-free LBM	The model presented in this article is based on a numerical equilibrium extended to reproduce an arbitrary number of moments of the Maxwell-Boltzmann distribution. A 13-moment system and BGK collision operator, incorporating shifted lattices and a kinetic sensor, are employed to locally increase viscosity in response to deviations from equilibrium.	This method effectively handles discontinuities in the zero-viscosity limit and performs well in low-viscosity and supersonic regimes. Its adaptable kinetic sensor enables local increases in viscosity based on deviations from equilibrium, making it suitable for various physical scenarios.	In this method, error levels strongly depend on the chosen moment and lattice. Although restricting the model to high Reynolds number flows can balance diffusive errors, convective errors persist and significantly affect the physics.
#54	Modified multi-speed model	In this study, the third-order MUSCL FV scheme was employed, introducing potential energy as an independent DF into the MS model, leading to a new DDF compressible flow model of LBM. After initializing DF's and macroscopic quantities, numerical flux on an interface was calculated, and collision operators were computed.	This method applies to flows with high-pressure ratios, allows adjustability in the Prandtl number and specific heat ratio, and uses the third-order MUSCLs FV scheme, enhancing conservation, flexibility, and accuracy in LBM, making it suitable for non-uniform grids.	This method becomes unstable at Mach 3, indicating its limitations at high Mach numbers.
#55	Potential energy DDF LBM	A DDF LBM is selected to accurately model compressible flow dynamics. Body-fitted grids, including a C-type grid for airfoil simulations, are meticulously prepared to match object geometries, ensuring precise resolution of surface and wake flows.	This method, characterized by acceptable accuracy, stability, and low numerical dissipation, is suitable for studying sound generation and propagation in fluid flows.	The third-order MUSCL FV scheme and second-order IMEX Runge-Kutta scheme for time advancement provide robust numerical frameworks but introduce computational complexity and require careful calibration of numerical parameters to ensure stability and accuracy.
#56	MRT-LBM	An approach for the development and application of a multiple-relaxation-time LBM for high-speed compressible flows is presented. The governing equations are discretized using a third-order MUSCL FV scheme, with nine relaxation coefficients in the MRT model adjusted based on physical principles to enhance accuracy.	The MRT scheme is preferred over the SRT scheme in this research due to its advantages in physical principles, parameter selection, and numerical stability, making it suitable for simulating compressible flows with strong shocks and for use on non-uniform grids.	This method increases computational costs and involves using nine adjustable relaxation coefficients in the MRT model. Although adjusting these coefficients can significantly improve simulation results, determining the optimal values requires a deep understanding of the physical background, which complicates parameter selection.
#57	LBM utilizing hybrid lattices	A double-sided surfel-based HLBM with arbitrary interface coupling for different lattice regions is proposed. This method utilizes a transparent interface of double-sided surfels to separate and connect computational regions between different reference frames, ensuring conservation laws and accurate flow transitions.	This hybrid lattice method is accurate, stable, and robust, allowing users to define TS regions with arbitrary shapes to encompass high-speed flows while maintaining the low numerical dissipation characteristics of the HS solver, thereby enhancing aeroacoustic simulation accuracy.	The interface between high-subsonic and transonic flow regions uses double-sided surfels to ensure smooth transitions and uphold conservation laws, requiring careful management to prevent numerical disturbances, which are challenging to implement.

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Table 2. Continued.

Code	Method	Procedure & Innovation	Advantages	Disadvantages
#58	WENO-LBFS	The innovation of this research lies in the development and application of a hybrid WENO scheme integrated with a Lattice Boltzmann Flux Solver (LBFS) for simulating complex fluid dynamics. The FD WENO scheme has been incorporated into four types of LBFS – namely, inviscid LBFS, viscous LBFS, and two HLBFS models – to simulate compressible flows effectively.	The hybrid WENO- LBFS scheme integrates a switch function to automatically transition between inviscid and viscous LBFS, combining the advantages of WENO and HLBFS. This approach offers high accuracy, resolution, and efficiency, accurately solving smooth regions and effectively capturing shock waves without oscillations.	The viscous WENO-LBFS scheme is less accurate than the inviscid version for smooth problems but offers better resolution and captures complex wave structures more effectively in shock wave problems.
#59	WENO-LBFS	This article presents a WENO Scheme- LBFS for simulating compressible flows, specifically addressing challenges such as shock waves. The method integrates several essential components: it utilizes conventional Euler equations to model inviscid compressible flow, focusing on the conservation of mass, momentum, and energy.	The fifth-order accuracy of the WENO-LBFS scheme makes it an acceptable choice for simulating compressible flows with shock waves and discontinuities, considering factors such as numerical errors and resolution near these features.	This method, while computationally costly, may encounter challenges in scenarios where viscosity effects are significant.
#60	LBM forcing scheme	A systematic discretization scheme for the body force term within LBM framework, accurate to second-order in both space and time, is introduced. This scheme applies to moments of arbitrary order.	The proposed method accurately captures strong temperature variations in flows and can be easily adapted to complex collision models like the spectral multiple relaxation time model, enhancing its applicability.	The order of expansion is crucial for accuracy in the recovered macroscopic equations. Insufficient expansion, particularly of the force term, can lead to errors such as inaccurate heat flux.
#61	DBM	A reactive multi-component Discrete Boltzmann Model (DBM) combined with a one-step overall chemical reaction is presented, considering both chemical reactions and external forces. A two-dimensional model is proposed, featuring adjustable parameters that control discrete velocities and internal energies through additional degrees of freedom.	This model dynamically calculates detailed nonequilibrium quantities, providing accurate information for designing devices that operate in transient or extreme conditions away from equilibrium.	The model assumes simplified one-step reactions in chemical kinetics, while its accuracy has not been confirmed with these simplifications for the actual complex kinetics.
#62	Polar coordinate LB kinetic model	A polar coordinate LB kinetic model for compressible flows is presented, with a method indicated to recover the continuum DF from the discrete DF. A hybrid scheme, similar yet distinct from operator splitting, is proposed, and temporal evolution is calculated analytically.	The LB kinetic model presented here effectively investigates nonequilibrium effects, considering deviations from thermodynamic equilibrium and employing accurate microscopic radial boundary conditions.	The radial boundaries must be carefully treated according to the specific situation, as this adds challenges in various scenarios.
#63	Hermite-Expansion Based LBM	The method presented in this article is used to simulate high-Mach-number compressible flows with shock waves through a Hermite-expansion-based LB method within the double-distribution-function approach. Discrete EDFs are obtained from Hermite expansions of continuous EDFs, with discrete velocity sets derived from Gauss-Hermite quadrature abscissas.	This approach allows for efficient and stable time-stepping, particularly crucial in simulations involving disparate flow speeds, such as those with both fast and slow dynamics.	Implementing this method is mathematically complex, especially for high-Mach-number flows, due to the need for high-order Hermite polynomials and high-degree Gauss-Hermite quadratures, which demand significant computational resources.

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Table 2. Continued.

Code	Method	Procedure & Innovation	Advantages	Disadvantages
#64	Cell-centred FV LBM	A method for simulating high-speed viscid compressible flows on unstructured grids is presented. The particle DF is based on the CF, with the corresponding total energy DF adopted.	The method's parallel performance on the Tianhe-2 supercomputer demonstrates significant scalability across thousands of processor cores, suggesting its potential for fast, high-fidelity simulations of large-scale, high-speed viscid compressible flows in complex computational domains.	The computational complexity of the FV LBM and its high computational costs, particularly when handling large-scale problems on unstructured grids, is addressed through a parallel algorithm to mitigate these demands. Still, the computational cost remains a significant consideration for practical applications.
#65	ALE-FVLBM	In this study, a central difference unstructured FVLBM within an ALE framework is applied to simulate compressible viscous flows over moving bodies. The two-dimensional compressible LB equation, based on the Watari model, is discretized using a second-order cell-centred FVM with a central averaging scheme for numerical fluxes on unstructured quadrilateral meshes.	This method, while capable of simulating complex geometries and moving bodies, provides a reasonable resolution of the viscous region near the wall boundary and effectively handles both steady and unsteady viscous compressible flow problems.	The proposed method's primary limitation is its application solely to laminar flows. While it has been suggested that the method could be extended to turbulent flows and applied to both stationary and moving bodies in more practical, real-world flow scenarios, such an extension has not been undertaken in this study.
#66	KT Model (Perumal & Dass, 2015)	To simulate compressible inviscid flows, the M-TVD method, which combines the advantages of the Total Variation Diminishing (TVD) and modified Lax-Wendroff (M-LW) methods, is presented. The TVD scheme is first employed for space discretization of the convective term in the LBE.	The M-TVD model used in this article employs a differentiable switch, maintaining robustness, low computational cost, and simplifying the handling of various Mach numbers, including strong shock waves.	The artificial viscosity in this method primarily serves to eliminate oscillatory behaviour in the solution, which may otherwise cause instability.
#67	Hybrid KT Model (Bayat et al., 2024; Jiakuan et al., 2025)	A method combining LBM with Large Eddy Simulation (LES) is proposed to study subsonic turbulent jet flows and investigate jet noise physics in near-field and far-field regions. The approaches of Shan and He, utilizing a higher-order expansion of the Maxwell-Boltzmann EDF, and Kataoka and Tsutahara, using an alternative functional form, are incorporated into an LES framework.	This method effectively predicts the acoustic field of subsonic turbulent jets in both near and far fields and accurately simulates fluid flow at high Reynolds numbers.	This method's capability to simulate compressible turbulent flows at high Mach numbers, particularly those involving shock waves, is questionable.
#68	LBM with Tait EOS	The acoustical field of a ballistic shock wave therapy device was simulated using LBM and validated experimentally. To address weak compressible fluid flow, the EDF was expressed differently from the standard Hermite series.	Incorporating the Tait EOS with standard Boltzmann allows the proposed methodology to simulate compressible flow, avoiding the mathematical complexity of other approaches.	This approach applies only when the shock wave is relatively weak. While it predicts shock wave propagation, it does not account for complex reflections, diffractions from medium disparities, or the thermal field generated by the ballistic source.

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Table 2. Continued.

Code	Method	Procedure & Innovation	Advantages	Disadvantages
#69	FR-LBFS	A high-order method based on the FR approach combined with the LBFS is presented for solving inviscid compressible flow problems. The FR approach is employed to solve the macroscopic Euler equations, while convective fluxes at the cell interface are evaluated using a one-dimensional compressible LBFS model without free parameters.	Compared to macroscopic equation-based flux solvers, the current FR-LBFS simplifies flux calculations due to inherent convective terms in the LBE. The method effectively captures shocks and demonstrates high accuracy for flows on both structured and unstructured grids, consistently aligning with analytical and benchmark data.	The correctness of the performance of this method in compressible flows where viscosity is important has not been confirmed.
#70	FR-LBFS	A method based on the LBFS has been presented for simulating viscous compressible flows. A high-order LBFS scheme was used to compute the inviscid flux, and a FR approach was employed for spatial discretization.	This method achieves higher-order accuracy, with the FR approach unifying nodal Discontinuous Galerkin (DG) and SD frameworks, demonstrating its flexibility and adaptability. Additionally, it achieves the same accuracy with fewer meshes compared to LBM.	*
#71	Coupled DDF-LBM (two-relaxation-time kinetic model)	A LB model on a Hermite basis for compressible viscous flows is presented and developed within the double-distribution-function approach with an adjustable specific heat ratio and Prandtl number. The EDF is determined using Hermite expansion.	The results demonstrate that the presented model effectively simulates shocks and compressible flows in high-order speed discretization.	This model requires high-order speed discretizations to adequately capture shocks, which increases computational costs.
#72	Two-relaxation-time kinetic model	In this study, the DDF-LBM was employed to simulate unsteady shock wave/boundary layer interaction in a shock tube. To enhance the numerical performance of compressible LBM, the third-order Weighted NND (WNND) scheme, based on the stencils of the second-order NND scheme, was introduced.	This approach operates within a DDF framework, allowing for adjustable specific heat ratios and ensuring accurate boundary treatment.	Numerical stability remains a challenge when simulating high-speed viscous flow using this method.
#73	Two-relaxation-time kinetic model	The implementation of compressible LB simulations on GeForce GPUs was presented in this research. The DDF-LBM was performed on both the GTX 1080 GPU and the E5-2620v4 CPU for comparison.	This article demonstrates that the GPU-executed method effectively handles certain compressible viscous flows, showing that SP operation on GPUs without ECC memory provides sufficient precision.	When simulating compressible flows with shock waves and the Riemann problem using single precision operations on GeForce GPUs, numerical fluctuations are observed, leading to deviations in results, especially in the presence of large gradients or flow discontinuities common in compressible flow problems.
#74	Two-relaxation-time kinetic model	Two LB models based on a coupled double-distribution-function approach for compressible flows have been developed, allowing for adjustable specific-heat ratio and Prandtl number. The primary difference between the models lies in their discrete equilibrium density and total energy DF's.	This method, with its numerical stability and adjustable Prandtl number, effectively simulates compressible flows involving heat transfer and varying thermal properties.	The use of various EDFs for density and total energy impacts the model's consistency, thereby affecting its numerical stability so that one of the presented models failed to handle compressible flows with shock waves, highlighting its limitation in certain physical scenarios.

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Code	Method	Procedure & Innovation	Advantages	Disadvantages
#75	HLBM	A HLBM has been proposed, combining LBM algorithm for solving mass and momentum equations with a FVM for the entropy-based energy equation. A novel discretization of the lattice closure correction term was introduced, enabling supersonic flow simulations.	The proposed HLBM extends LBM's application to supersonic flows, demonstrating good stability and effectively simulating viscous supersonic flows.	The accuracy and stability of the proposed HLBM depend heavily on the numerical parameter σ , whose optimal value is not yet clearly established.
#76	HLBM	An original von Neumann analysis of the HLBM with an ideal gas thermodynamic closure has been conducted. Stability on a standard lattice with various one- and two-dimensional HLBM has been explored, and multiple instability mechanisms at relatively high Mach numbers have been identified.	This paper highlights original modal transfers from the entropy mode to the shear mode in both the primitive and conservative formulations, enabling access to the macroscopic composition of modes. Additionally, the stability of the method improves by using appropriate corrective terms and addressing the acoustic CFL constraint.	The study's limitations include the lack of quantitative correlation between the qualitatively validated spectral analyses and experimental observations, necessitating further investigation. Secondly, only a subset of HLBM classes was examined in this study, and additional methods, such as multispeed approaches or DDFs, were not explored in this paper to determine whether the mode transfer is related to the conservative formulation or is specific to the hybrid method.
#77	SLLBM	In this article, the semi-Lagrangian scheme is applied to simulate compressible flows. As propagation occurs on an arbitrary non-uniform grid, interpolation is required to reconstruct the populations at the departure points.	The presented method allows arbitrary time step selection and efficiently captures small-scale flow structures near wall boundaries using body-conforming meshes; moreover, compared to standard LB models requiring larger velocity sets, this model is simpler and more efficient.	The present model is valid as long as correction terms remain small, which needs choosing the shift velocity frame appropriately.
#78	Thermal DDF LBM with shifted lattice	A two-population LB model is introduced, utilizing exact space discretization of the propagation step, a key feature of LB methods. The model is extended to supersonic flows with shock waves using shifted lattices.	The method presented in this article, compared to multispeed lattice models, offers simpler boundary conditions, lower computational cost, and accurately predicts the speed of sound, confirming its temperature independence. Additionally, by introducing correction terms in the kinetic equations, the methodology effectively eliminates spurious terms in the momentum and energy equations.	The model can extend to supersonic flows using shifted lattices, but at higher speeds, the deviations and required correction terms become large, indicating a limitation in its applicability to high-speed flows without significant adjustments.
#79	SRT-BGK with correction terms	The model is based on a two-population approach and utilizes SRT BGK kinetic equations. These equations, derived from LBM, incorporate first- and second-order approximations, the product-form extended equilibrium for the momentum lattice, and an equilibrium pressure tensor with an appropriate correction term.	This proposed method eliminates the need for interpolation, preserving the simplicity and efficiency of standard lattices while removing spurious terms in the stress tensor inherent in the standard LBM during high-speed flow analysis.	The nonunique correction terms used to eliminate spurious components in the stress tensor may affect the model's robustness and accuracy under varying flow conditions. Additionally, this method has not been validated for Mach numbers exceeding 1.5.

(continued).

Table 2. Continued.

Code	Method	Procedure & Innovation	Advantages	Disadvantages
#80	ALE-LBM	In this article, the ALE technique is employed, mapping the kinetic equations from a moving physical domain to a fixed computational domain. Correction terms are incorporated, and grid shifts are applied.	The model uses an unstructured mesh fitted to the body to resolve small-scale flow structures near the walls, demonstrating that the Geometric Conservation Law (GCL) problem, which requires special treatment in ALE formulations of NS solvers, does not arise, thereby satisfying the GCL exactly.	This method, which is limited to two-dimensional problems, requires a blending function to construct the mapping function for problems involving deformation or the relative motion of multiple objects. Additionally, because the transformed equations must be solved in the ALE method, exact propagation in the space-filling lattice is no longer possible.
#81	KT Model (Perumal & Dass, 2015)	The application of LB models for simulating compressible flows, with a focus on stability and accuracy enhancements through discretization techniques, is discussed. A two-dimensional numerical framework, based on incorporating M-LW and M-TVD schemes, has been utilized.	The combination of the MRT model and novel discretization techniques enhances the robustness of the original LBM, effectively preventing oscillations in compressible flow problems.	Between the two proposed schemes, the M-LW scheme potentially exhibits instability, particularly in areas with shocks.
#82	LAMB-CNS	In this research, the single-component Onsager-BGK kinetic model proposed by Mahendra and Singh (Xu et al., 2012) is extended to a multi-species system. The resulting solver, named the Lattice-Boltzmann Multi-specie-flux Based Compressible NS solver (LAMB-CNS), derives viscous compressible flow equations for multi-species diffusion problems and is implemented within a cell-centred FV framework based on the LBM.	This multi-species model accurately adds entropy and accommodates polyatomic gases with varying molecular weights, Prandtl numbers, and specific heat ratios.	This model employs a single temperature approach and does not account for turbulence.
#83	Mesh refinement	In this study, modifications to the front and back walls and their combination on the sound pressure level response of the M219 cavity at supersonic regime were investigated using SIMULIA PowerFLOW software, which employs LBM. The turbulence model, a combination of the RANS and LES techniques, was utilized for accuracy, with a Cartesian mesh strategy and VR enhancing grid resolution in local areas of interest.	This study investigates turbulence structures using the VLES method, which efficiently combines RANS and LES strengths, offering competitive accuracy and reduced computational demands.	*.
#84	Higher-order LBM	This method, which presents a set of uniform polynomial equations generating multidimensional on-lattice higher-order models of the LB theory, has been formulated to create LB models for one-, two-, and three-dimensional spaces while maintaining a compact number of discrete velocities. These derived models are subsequently applied to simulate thermal compressible flows.	This method, which presents an analytical approach that retains an on-lattice nature, results in a 95-velocity model with equal accuracy but fewer discrete velocities, showcasing the method's ability to enhance efficiency while maintaining high levels of accuracy. In addition, along with acceptable accuracy, it also provides the possibility of simulating thermal compressible flows.	The method's dependence on specific viscosity values to align with analytical solutions presents a limitation. For example, shock tube simulation results exhibit discrepancies with the analytical solution, attributed to viscosity differences.

(continued).

Table 2. Continued.

Code	Method	Procedure & Innovation	Advantages	Disadvantages
#85	KT Model (Perumal & Dass, 2015) WENO-based FD-LBM	The method used for simulating compressible flows involves discretizing the Discrete-Velocity Boltzmann Equation (DVBE) with the BGK approximation in both time and space. A third-order Runge-Kutta method is applied alongside a fifth-order WENO FD scheme.	By employing the third-order Runge-Kutta time-stepping method, numerical instabilities that can occur when higher-order discretizations are combined with lower-order time integration methods are avoided.	In this study, the dissipation term is determined by considering both dissipation error and stability criteria. This approach requires careful handling of dissipation terms to maintain stability and accuracy, particularly when simulating high-Mach-number inviscid flows.
#86	LBFS	In this paper, the LBFS is presented for simulating compressible and incompressible flows. The solver is based on the FVM discretization of governing differential equations, specifically the NS or Euler equations.	The results demonstrate the method's capability to simulate flows with high Mach numbers and complex geometries effectively.	Viscosity effects are not considered in this article, so the method's ability to address scenarios where viscosity is important remains unconfirmed.
#87	DG-LBFS	The high-order DG method, combined with a HLBFS, is employed to solve the compressible Euler and NS equations. Instead of using conventional flux methods like local Lax-Friedrichs or Roe, a HLBFS is used to evaluate the inviscid flux across cell interfaces.	The HLBFS effectively captures strong shocks and thin boundary layers by introducing a function that controls artificial viscosity. Numerical results from benchmark problems demonstrate that this numerical solver achieves accurate solutions even on a very coarse grid.	A key limitation of this approach is the use of a switch function to control numerical dissipation, which can introduce flaws. Additionally, only the inviscid flux at the cell interface is considered, while the viscous flux remains approximated by a conventional smooth function.
#88	Generalized Titarev's model (Kruger, 2016)	A DDF-LBM strategy was employed, proposing a family of LBMs for compressible flow simulations in the continuum regime. An exponential distribution-based collision step generalized Grad's post-collision DF's, with post-collision moments numerically imposed via a fast-converging Newton–Raphson procedure.	This method offers several advantages, including Numerical Collision (NC) in moment space, a tunable Prandtl number, the incorporation of external forces, polyatomic contributions via a second set of populations, and a shock-capturing technique utilizing local artificial viscosity through a second exponential DF. Additionally, a shock sensor improves dispersion and stability in simulations of discontinuous flows under inviscid conditions.	One significant limitation of the NC approach is that it may encounter stability issues at high Reynolds numbers. Although the model performs well in many cases, stability can degrade under high Reynolds conditions, especially for the highest mesh resolutions.
#89	Lagrange multipliers method	The presented method incorporates a high-accuracy estimation of local equilibria and employs a stabilization technique using local relaxation time and a shifted lattice. A two-population thermal LBM is used.	Despite using a relatively modest grid resolution, the current LBM model achieves acceptable fidelity in simulating supersonic flows at high Reynolds and Mach numbers.	The current two-dimensional simulation methodology does not model turbulence behaviour at high Reynolds numbers so its results in turbulent flow regimes are not expected to yield reliable solutions. Additionally, handling high Mach numbers by shifted lattices introduces computational costs and complexity.
#90	ALE-FDLBM	In this article, the FDLBM method, incorporating stable FD schemes, is utilized. To simulate moving bodies, the ALE formulation is integrated into the method, with the NS equations recovered via the Chapman-Enskog expansion.	This method accurately simulates various physics, including aeroacoustics, aerodynamics, and two-phase flows.	This method's validity above Mach 1 remains unconfirmed, and the boundary conditions fail to completely eliminate slip in the Knudsen layer, where molecular behaviour deviates from equilibrium.

(continued).

Table 2. Continued.

Code	Method	Procedure & Innovation	Advantages	Disadvantages
#91	Hybrid	In this research, the boundary surrogate model is proposed to simulate multi-scale global flow field of the airfoil covered with micro/nano grooved surface structure. This model replicates the effects of microscopic near-wall regions.	This innovative simulation method conserves computing resources while investigating turbulence control by micro/nano groove structures on airfoil surfaces.	The primary limitation of the proposed boundary surrogate model is its applicability only when riblets or grooves are within the viscous sublayer, which simplifies computation but restricts the model's use in more complex three-dimensional applications or cases involving substantial flow interactions beyond the viscous sublayer.
#92	Local equilibrium function	In this research, a method for constructing the local equilibrium function for LBM based on the assignment matrix for macroscopic conditions has been developed. This approach directly derives the discrete form of the equilibrium function without considering its continuous form, thereby eliminating free parameters.	This method in addition to satisfies energy statistic conditions and flexible specific-heat ratio, introduces an assignment matrix to directly assign macroscopic conditions to obtain the local equilibrium function, without considering the continuous form of its. This direct derivation from macroscopic conditions ensures that the model does not contain free parameters.	The method is specifically designed for one-dimensional LBM models, which benefits targeted applications but limits direct applicability to more complex two-dimensional or three-dimensional flow simulations without further adaptation or extension.
#93	LBFS	The refinement and examination of three LBFS versions, developed for isothermal, thermal, and compressible flows, have been investigated. The FVM was applied to solve the macroscopic governing equations, yielding solutions for macroscopic flow variables at cell centres.	The presented method in this article demonstrates reasonable computational efficiency, controlled numerical dissipation, and does not generate unphysical pressure oscillations for lid-driven cavity flows. It is also effective for simulating compressible flows on unstructured grids.	A key disadvantage of the method in this research is its computational efficiency; while LBFS achieves stability and accuracy on non-uniform grids, it requires significant computational resources due to FR steps involving cell-by-cell interpolation, though fewer nodes are required in LBFS compared to standard LBM on identical grids.
#94	Higher-order LBM	An advanced higher-order LBM for simulating rarefied gas flows, from slip to free molecule regimes, is introduced. Higher-order LBM models with EDFs incorporating higher-order terms have been constructed, ensuring improved accuracy in capturing nonequilibrium physics.	This method's acceptable stability and low numerical dissipation make it suitable for sound propagation studies as well as supersonic flow analysis.	Despite acceptable stability, models with an order higher than four significantly increase computational cost.
#95	SLLBM	Three-dimensional compressible flows were simulated using the SLLBM. Interpolation was employed to overcome the limitations of traditional explicit time integrators, enabling larger time-step sizes without requiring additional filtering or stabilizing techniques.	The velocity set of D3Q45 reduces the computational load in the first place, but it can also reduce errors since fewer interpolations are needed. A suited filtering and stabilization technique beyond the interpolation itself would potentially further develop the method.	The use of large time steps and interpolation in this method has the potential to create numerical instabilities in the presence of complex flow phenomena. Also, there is a higher effort required to implement the method (compared to the standard LBM).
#96	SLLBM	The method presented employs a semi-Lagrangian streaming step, restricted to static, non-moving reference frames at a fixed reference temperature. The DF values are calculated using a double population off-lattice approach by tracing fluid parcels' trajectories backward in time to their departure points.	The advantages of the presented method are its flexibility with time step size, lack of a dedicated time integrator as integration occurs along characteristics, and suitability for stretched or unstructured meshes. Additionally, it employs high-order polynomials, and achieves fourth-order spatial and second-order temporal convergence.	The proposed interpolation-based LBM does not inherently conserve mass, momentum, and energy during the streaming step, though high-order interpolation polynomials can mitigate, but not entirely eliminate, this issue.

(continued).

Table 2. Continued.

Code	Method	Procedure & Innovation	Advantages	Disadvantages
#97	HLBM	A conservative HLBM for multi-species reactive flows has been developed. This is achieved by integrating LBM with FV schemes, retaining the low dissipation properties of LBM for acoustic and vortical modes using the collide and stream algorithm, while explicitly controlling entropy and mass fraction transport through a carefully chosen FV scheme.	This method simulates thermal, compressible, and radioactive flows while allowing explicit specification of the gas heat capacity ratio and EOS, enhancing its applicability to polyatomic gases. Integrated with FV schemes, it ensures conservativity, controls linear stability, and preserves LBM's low dissipation for isentropic phenomena.	solving equations explicitly in this method increases computational costs by ignoring additional memory requirements.
#98	MRT-LKBM	In this study, a method for simulating combustion phenomena is presented, dynamically coupling the chemical energy released during combustion to the system by incorporating a chemical term into the LB kinetic equation. A Multi-Relaxation Time LB Kinetic Model (MRT-LBKM) is introduced, which satisfies 24 independent kinetic moment relations and employs a discrete velocity model with 24 velocities, allowing for the natural adjustment of thermodynamic nonequilibrium effects at each simulation step.	This model effectively simulates strong shock and detonation waves while allowing for the adjustment of specific heat ratio and Prandtl number.	The increased number of equations raises computational costs. Moreover, coupling chemical energy release with the LB kinetic equation may require modifications to the collision term to ensure the MRT model accurately represents macroscopic behaviour, adding complexity and potentially limiting its straightforward application without detailed analysis and adjustment.
#99	HLBFS	A HLBFS is proposed for the simulation of viscous flows, following Chapman-Enskog analysis. The NS equations are discretized using the FVM, with the fluid domain represented by discrete control volumes.	The method effectively simulates viscous flows by evaluating viscous flux and controlling numerical dissipation, enabling accurate capture of both strong shock waves and thin boundary layers.	The results show that Scheme I can't effectively simulate viscous boundary layers with adequate numerical dissipation for hypersonic flows. Although Scheme II provides stable solutions for strong shock waves, but it has higher numerical dissipation.
#100	LBFS	The method presented extends the LBFS for simulating 3D viscous compressible flows. Derivatives of conservative variables are first calculated and reconstructed at cell interfaces, where EDFs are then computed.	This method introduces a switch function to calculate the dimensionless relaxation time, controlling numerical dissipation and accurately capturing shock waves without spurious oscillations. Additionally, the implicit LU-SGS method enables the use of large CFL numbers and accelerates the convergence rate.	In this article, all simulations are performed with Mach numbers below 1, within the subsonic range. Thus, the method's competence at Mach numbers exceeding 1 requires further confirmation.
#101	FV-LBM	Three non-free parameter LBMs based on FV have been developed and compared. In the proposed FV-LBM, the normal convective flux at the cell interface is evaluated using a one-dimensional compressible LB model, while the tangential flux is calculated as in conventional Euler solvers.	The methodology applies a local time step and an implicit residual smoothing scheme to enhance computational efficiency. While the FV-LBM primarily uses a one-dimensional LB model for normal flux vectors, it is easily extended to multi-dimensional cases and introduces three non-free parameter models (D1Q3, D1Q4, and D1Q5).	This model is designed to simulate non-viscous flows. Therefore, this method can have limitations in simulating the flow of viscous flows.
#102	LBFS	A platform for deriving a non-free parameter LB model is presented, based on conservation forms of moments treated as an inverse problem. EDFs associated with lattice velocities are solved directly from these conservation forms.	This method physically determines lattice velocities, allowing the simulation of high Mach flows without artificial specification.	This work considers only the simulation of inviscid compressible flow.

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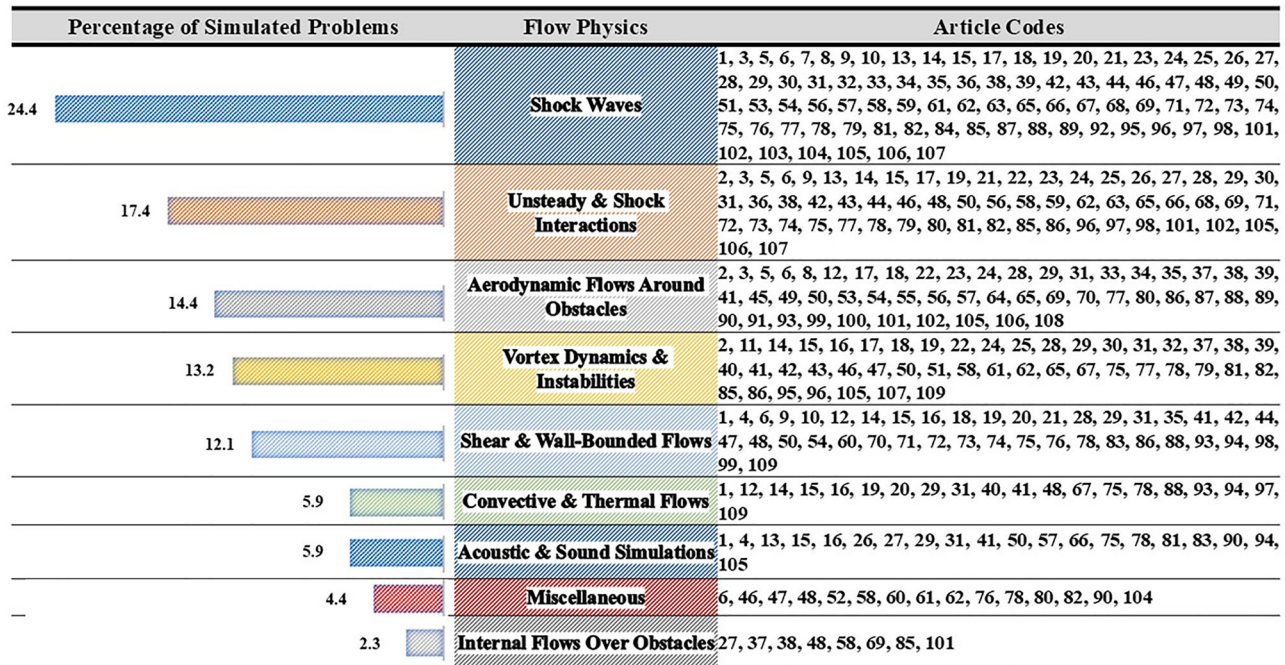
Table 2. Continued.

Code	Method	Procedure & Innovation	Advantages	Disadvantages
#103	LBFS	In this paper, a multicomponent LBFS is developed to simulate two-dimensional compressible viscous reacting flows. Building on an existing LBFS for single-component compressible flows, the solver discretizes multicomponent NS equations using the FVM and evaluates numerical flux at the cell interface via the local solution of the LBE.	The method used in this article accurately simulates multicomponent compressible viscous reacting flows with low numerical dissipation. Its effectiveness in mitigating the grid aspect ratio's impact on the switch function enhances its applicability to complex grids, reducing grid quality requirements in LBFS and allowing controlled numerical dissipation near unsmooth regions like shock waves and combustion fronts.	The Chapman-Enskog analysis shows that the equilibrium part corresponds to the inviscid flux, while the nonequilibrium part represents numerical dissipation. Accordingly, the equilibrium flux should dominate in smooth regions, and the nonequilibrium dissipation should primarily be added near shock waves and combustion fronts to ensure numerical stability.
#104	LBM with Phase-field approach	The innovation of this research lies in developing and validating a LBM tailored for simulating compressible two-phase flows, specifically high-density ratio liquid-gas flows. The effective intermolecular force is computed, and stability and accuracy are enhanced by solving the BE for a non-ideal fluid using the modified Lax-Wendroff method with artificial viscosity.	The methodology presented in this paper addresses critical issues in cavitation modelling, including pressure-driven phase change, gaseous phase compressibility, and the establishment of a consistent flow field to accommodate pressure variations across the computational domain, ensuring accurate representation of the relevant physical phenomena.	This methodology enables phase separation within LBM but requires assumptions about intermolecular forces, limiting its applicability to specific two-phase flow scenarios; moreover, it is unsuitable for simulating shock phenomena.
#105	HRR-LMB	An effective method for coupling LBM with the MUSCL-Hancock scheme in a rotating grid framework is proposed. The compressible LBM is integrated with rotating overset grids, known as the Chimaera method, sliding mesh, or moving reference frame.	This approach preserves the second-order accuracy of LBM and effectively simulates moving geometries in compressible flows at high Reynolds numbers.	Numerical errors introduced by the discrete forcing term and the coupling of LBM with the MUSCL-Hancock scheme remain limitations of the method despite mitigation efforts.
#106	Meshfree LBFS	A Least Square-based FD (LSFD) scheme is employed in this method to discretize the governing equations for simulating compressible inviscid flows. Point clouds, rather than traditional grids, are used.	This method accurately and efficiently calculates fluxes at virtual control points and eliminates the need for mesh generation, a common bottleneck in CFD development due to the difficulty of creating satisfactory meshes for complex geometries.	Ensuring solution consistency and convergence is challenging due to the lack of a structured grid, and implementing boundary conditions is difficult. Moreover, the computational cost of meshfree methods is often higher than traditional methods, particularly for large-scale problems, due to the need for searching and organizing points at each computational step.
#107	HLBM	A new numerical scheme has been developed to solve scalar equations coupled with a LBM in hybrid approaches. Its primary application involves solving energy equations alongside an LBM solver handling mass and momentum conservation.	The presented method eliminates the inconsistency between mass conservation in LBM solver and the scalar equation, ensuring accurate jump conditions for compressible flows, as validated by Riemann problems. Additionally, the scheme easily adapts to external forces, such as gravity, by modifying the computed flux.	Although its simple implementation allows dynamic inclusion of upwinding, the need for enhanced stability introduces additional numerical diffusion.

(continued).

Table 2. Continued.

Code	Method	Procedure & Innovation	Advantages	Disadvantages
#108	Implicit LBFS	The proposed method, operating within the FVM framework, is designed to obtain time-accurate results for unsteady flows. It is combined with the dual time-stepping technique to efficiently simulate these flows, with sub-iterations applied at the pseudo-time level.	The ability to simulate high Mach number flows and complex geometries on both structured and unstructured meshes with greater computational efficiency than explicit LBFS is demonstrated.	*
#109	RegPonD	In this paper, the RegPonD method is introduced to reduce the computational cost of the original PonD method, which stems largely from matrix inversions during the rescaling of discrete DF between gauges. To eliminate matrix inversions, an alternative method of DF rescaling is proposed, where the discrete DF is restored from moments, and the reference frame conversion is performed in moment space.	This method reduces the computational cost of gauge conversion in the original PonD for 3D simulations without compromising the excellent Galilean invariance of the original PonD.	Unlike the traditional LBM with its explicit, mass-conserving streaming step, the PonD method uses an implicit, non-mass-conserving streaming step. This off-lattice streaming requires interpolation, increasing computational complexity and potentially affecting stability, which depends on interpolation choices and the predictor-corrector scheme, warranting further research.

**Figure 5.** The investigated configurations in the references.

aerodynamics of flow around external bodies and vortex instabilities have received significant attention. It is noteworthy that a single paper may examine multiple problems; therefore, the reported frequency percentages are evaluated based on the total number of studied problems rather than the number of articles.

Since the choice of the velocity discretization model directly affects the order of recovered moments (and

therefore the achievable accuracy, isotropy, and stability) as well as the associated computational cost, it represents a central design decision in Boltzmann-based simulations. Figure 6 summarizes the frequency of the discrete-velocity sets adopted in the reviewed studies and, at the same time, reports how the published cases are distributed across problem dimensionality. Overall, approximately 8% of the reviewed studies

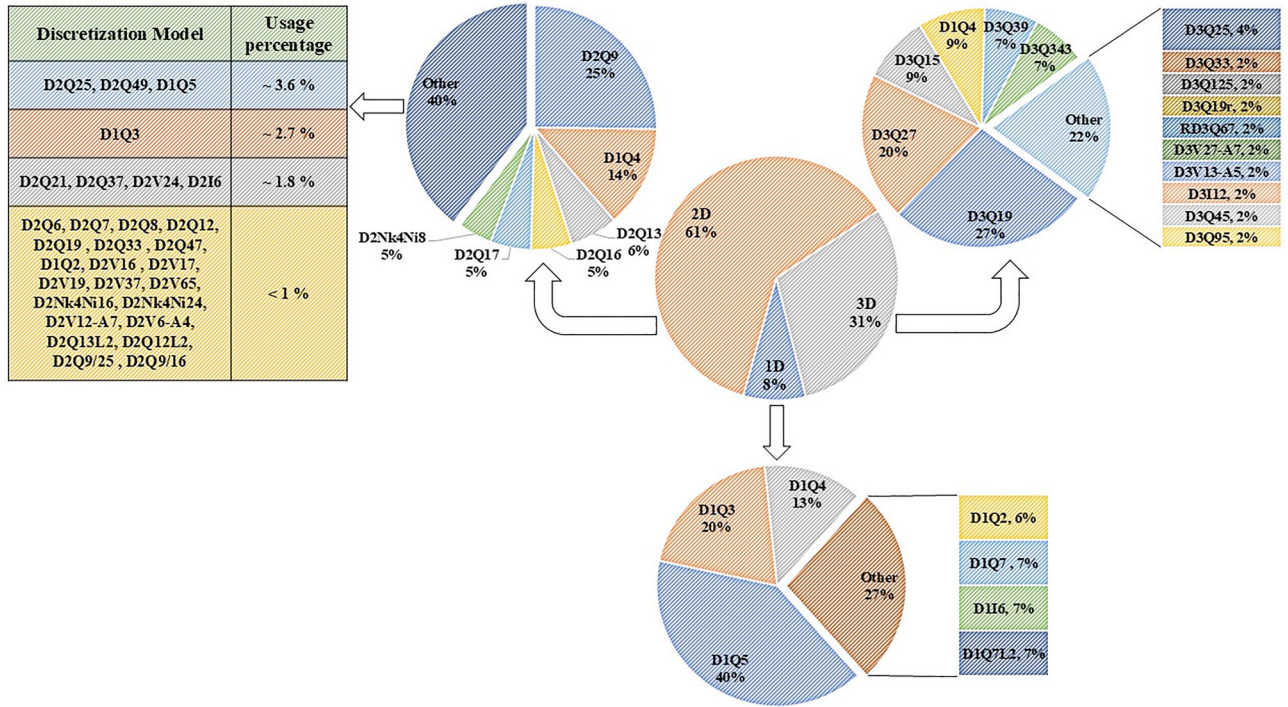


Figure 6. Discretization models used in the reviewed articles (frequency by article count). The reported trends reflect practical trade-offs between accuracy/isotropy and computational cost: compact lattices (e.g. D2Q9, D3Q19) dominate for efficiency, while extended multispeed stencils are mainly adopted for higher-Mach/shock-dominated regimes.

include one-dimensional configurations, whereas two-dimensional problems account for about 61% of the literature. Within the 2D subset, D2Q9 is the most commonly used lattice (about 25%), reflecting its favourable balance between efficiency and accuracy and the maturity of its implementations, particularly on structured cartesian grids where streaming is straightforward and boundary treatments are widely established. The next most frequently reported model is D1Q4, which in many studies appears in connection with flux evaluation within LBFS: in these hybrid formulations, the kinetic step is often reduced to a compact, effectively one-dimensional stencil at interfaces to compute numerical fluxes with lower overhead while retaining robustness. Three-dimensional problems constitute roughly one-third of the dataset, which is consistent with the substantially higher cost of fully 3D computations; among these studies, D3Q19 and D3Q27 are the predominant choices. This prevalence can be interpreted as a cost–accuracy compromise: D3Q19 is frequently selected to limit memory footprint and computational expense, whereas D3Q27 is employed when improved isotropy and additional robustness are desired, albeit at higher cost.

Beyond these mainstream models, the database shows that extended multispeed stencils are used selectively, typically in regimes where stronger non-equilibrium effects, shock dynamics, or higher-Mach compressible

features must be resolved. Several high-Mach two-dimensional demonstrations therefore adopt larger lattices (e.g. D2Q21, D2Q37, and related extended sets) to better support higher-order moment recovery and improve stability under sharp gradients. At the same time, the limited overall uptake of such high-order/large-stencil models in the broader literature highlights a practical constraint: although many advanced discretizations exist in principle, their increased population count amplifies memory traffic, algorithmic complexity, and wall-clock cost, which often makes them less attractive for routine studies compared with more compact lattices.

It is also apparent that many papers use canonical benchmark configurations – frequently in two dimensions – not only to study a particular physical scenario, but also to demonstrate the reliability and generality of a proposed scheme under controlled conditions (e.g. shock-tube families and shock–vortex/shock–bubble interactions as primary validation vehicles, often complemented by at least one external-flow case such as flow over an airfoil, cylinder, wedge, or projectile). Finally, it should be noted that a single article may report multiple test cases with different physical dimensions and/or velocity discretizations; accordingly, the percentages in Figure 6 are computed at the article level, based on the frequency of published references in which each model appears.

4.3. Thermal models

From the procedure and innovations of the methods reviewed, it can be inferred that addressing the energy equation in LBM is crucial to creating accurate modelling of thermal effects in compressible flows. The complexity of thermal transport in LBM appears due to the inherent coupling between temperature, density, and velocity fields, which can significantly impact the stability and accuracy of simulations. As mentioned, the primary approaches to solving the energy equation in LBM which are Single Distribution Function (SDF), DDF, and Hybrid methods. Each of these approaches exhibits distinct traits in terms of computational efficiency, numerical stability, and thermal accuracy.

Additionally, some researchers have employed innovative and unique techniques in this domain. Figure 7 summarizes the approaches employed in the reviewed articles based on how they handle the energy equation. Panel (a) reports the overall distribution of the identified strategies: the DDF approach accounts for 37% (Bhadauria et al., 2021; Bhadauria & Karlin, 2024; Chen & Zhang, 2012; Coreixas & Latt, 2020; Dorschner et al., 2016; Feng et al., 2015; Feng et al., 2016; Feng & Tao, 2014; Frapolli et al., 2015; Frapolli et al., 2016a; Frapolli et al., 2020; Guo et al., 2023; He et al., 2013; Huang, Jin, et al., 2023; Huang, Chen, et al., 2023; Kallikounis et al., 2021; Kallikounis et al., 2022; Kallikounis & Karlin, 2024; Krivovichev & Bezrukova, 2023; Latt et al., 2020; Li et al., 2023; Li & Zhong, 2015a; Li & Zhong, 2015b; Li & Zhong, 2016; Liu et al., 2019; Liu et al., 2022; Qiu et al., 2017; Qiu et al., 2018; Qiu et al., 2019; Qiu et al., 2020; Saadat et al., 2019; Saadat et al., 2020; Saadat et al., 2021; Saadat & Karlin, 2020; Thyagarajan

et al., 2023; Tran et al., 2022; Wang et al., 2015; Wilde et al., 2020; Wilde et al., 2021; Yazdi et al., 2020), followed by the SDF method at 30% (Atif et al., 2018; Chen, Wang, et al., 2023; Coreixas et al., 2017; Dorschner et al., 2018; Esfahanian & Ghadyani, 2015; Frapolli et al., 2016b; Ghadyani et al., 2015; Ghadyani & Esfahanian, 2014; Ghaffarian & Hejranfar, 2019; Hanada & Kataoka, 2015; He et al., 2012; Hejranfar et al., 2020; Hejranfar & Ghaffarian, 2016; Hejranfar & Ghaffarian, 2020; Jalali Khouzani & Kamali Moghadam, 2020; Jalali Khouzani & Kamali-Moghadam, 2023; Kataoka & Hanada, 2019; Krivovichev & Marnopolskaya, 2020; Li & Shan, 2023; Lin et al., 2014; Lin et al., 2017; Nasab et al., 2023; Nejat & Abdollahi, 2013; Noah et al., 2021; Safdari Shadloo, 2019; Shim, 2013; Shirsat et al., 2022; Tsutahara, 2012; Wang & Zhang, 2013; Watari, 2016; Xu et al., 2015; Yang et al., 2013; Yang et al., 2019), and the Hybrid method at 28% (Casalino et al., 2022; Chang et al., 2014; Chen, Yang, et al., 2023; Coratger et al., 2021; Farag et al., 2020; Farag et al., 2021; Feng, Boivin, et al., 2019; Feng, Guo, et al., 2019; Feng et al., 2020; Guo, Feng, Jacob, et al., 2020; Guo, Feng, & Sagaut, 2020; Guo et al., 2021; Hosseini et al., 2020; Kamali Moghadam et al., 2021; Kumar & Maheshwari, 2020; Li, Niu, et al., 2018; Li, Yuan, et al., 2018; Qin, Wu, Gu, et al., 2022; Qin, Wu, & Ma, 2022; Renard, Feng, et al., 2021; Renard, Wissocq, et al., 2021; Sun et al., 2018; Şeker et al., 2023; Wissocq et al., 2023; Yang et al., 2012; Yang et al., 2016a; Yang et al., 2016b; Yoo et al., 2023; Zhan et al., 2021; Zhao et al., 2020; Zhao et al., 2022); the remaining 5% are grouped as Other. Importantly, panel (b) shows that the preferred strategy varies with the flow regime: Hybrid is most frequent in transonic studies (40%), DDF dominates the supersonic regime (45.8%), whereas SDF is most prevalent in subsonic (42.9%) and

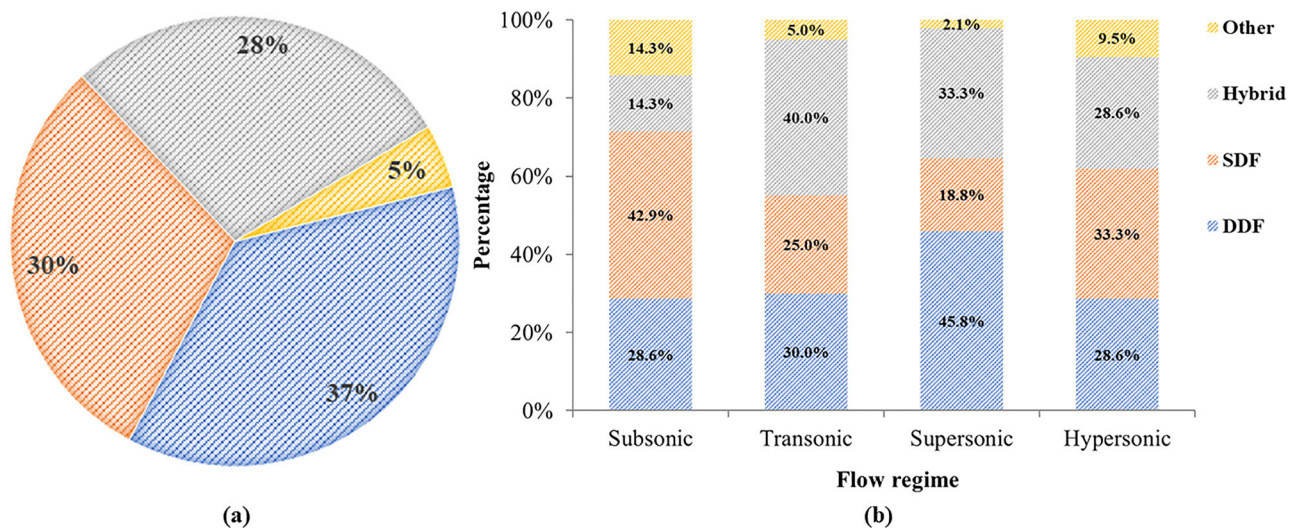


Figure 7. Energy-equation treatment strategies in the reviewed studies: (a) overall share across all studies, and (b) regime-wise share of DDF, SDF, Hybrid and Other (based on maximum Mach number).

hypersonic (33.3%) studies. (Studies with unreported maximum Mach number are excluded from the regime-wise breakdown). In this classification, $Ma = 0.3\text{--}0.8$ is defined as compressible subsonic, $0.8\text{--}1.2$ as transonic, $1.2 < Ma \leq 5$ as supersonic, and $Ma > 5$ as hypersonic (which also includes hypervelocity). It should be noted that these frequencies reflect prevalence in the literature and should not be interpreted as a direct cross-paper comparison of performance, since accuracy, stability, and cost metrics are strongly configuration-dependent and are not uniformly reported.

4.3.1. Single-distribution (multispeed) models

In the reviewed literature, MS (often categorized as SDF) compressible LB schemes extend the classical LBGK formulation by enlarging the discrete-velocity set and, in some cases, modifying the equilibrium distribution; in these models a single population per direction is used to represent both momentum and (implicitly) energy transport, which requires higher-order velocity terms in the equilibrium, commonly introduced via a third-order truncated Hermite form such as:

$$f_i^{eq} = w_i \rho \left[1 + \frac{c_i \cdot u}{c_s^2} + \frac{(c_i \cdot u)^2}{2c_s^4} - \frac{u^2}{2c_s^2} + \dots \right] \quad (7)$$

where w_i are lattice weights and c_s is the lattice sound speed. Achieving accurate recovery of the compressible NSF equations at elevated Mach numbers generally demands large velocity stencils (e.g. D2Q37 or D3Q67) together with careful truncation/regularization of the Hermite expansion, but the absence of an explicit energy equation implies that temperature variations are captured only approximately; correspondingly, in single-relaxation-time realizations the kinematic viscosity follows $\nu = c_s^2 (\tau_f - \frac{1}{2}) \delta t$; and many MS/SDF formulations exhibit an effectively fixed Prandtl number close to unity because the same relaxation mechanism governs both momentum and energy-like modes. From a robustness standpoint, high-Mach computations are frequently limited by lattice anisotropy and insufficient dissipation, motivating stabilization strategies such as adding bulk viscosity through an extra second-order diffusion term in the evolution equation, or adopting MRT and entropic collision operators to relax different moments at distinct rates while enforcing an H-theorem-type constraint. Overall, most MS/SDF studies in the dataset report stable and reasonably accurate performance up to approximately ($Ma \lesssim 3\text{--}4$), whereas only a small number of more aggressive variants reach $Ma \sim 10$ at the cost of very fine lattices and increased computational demand; reported Reynolds numbers span from laminar regimes ($\sim 10^2\text{--}10^5$) to moderate turbulent conditions depending on the application. In general, SDF schemes struggle

with large temperature gradients, and their stability at high Mach numbers is poor.

4.3.2. Double-distribution-function (DDF) models

In the DDF framework, compressible LBM is formulated by solving two kinetic equations: the conventional populations $f_i(x,t)$ for mass and momentum (relaxation with time τ_f as in Equation (5)), and an additional set $g_i(x,t)$ that transports internal energy/temperature via a BGK form:

$$\frac{\partial g_i}{\partial t} + c_i \cdot \nabla g_i = -\frac{1}{\tau_g} (g_i - g_i^{eq}) \quad (8)$$

A common choice for the energy equilibrium links g_i to the local kinetic energy associated with f_i :

$$g_i^{eq} = \frac{1}{2} (c_i - u)^2 f_i^{eq} \quad (9)$$

so that the zeroth moment recovers the internal-energy density $\rho e = \sum_i g_i$, and the coupled moments reproduce the full NSF system. Because τ_g are independent, the Prandtl number can be tuned through $Pr = \nu \backslash \alpha = \tau_f \backslash \tau_g$, although a recent compressible-LBM thesis notes that practical DDF implementations often rely on extended lattices or multi-speed stencils and that the standard BGK collision model can impose a fixed-Pr limitation, motivating more advanced collision models for robust high-Mach simulations (Rathnayake 2024). Across the reviewed studies, DDF configurations simulated benchmark problems in supersonic and hypersonic regimes, exhibiting improved stability and thermal accuracy in strong-shock regimes; however, these gains come at the cost of roughly doubling memory and computational expense, and many works further increase algorithmic complexity by updating the energy distribution with high-order FD/FV advection (frequently WENO) to enhance stability in shock regions. Generally, In practice, DDF achieves higher stability and better thermal accuracy, though at the cost of increased complexity and memory. Also, reported Reynolds numbers span laminar to transitional conditions, with some extensions to turbulent channel-flow cases.

4.3.3. Hybrid LBM-CFD coupling

In the reviewed studies, hybrid LBM-CFD approaches are characterized by an explicit splitting of the NSF system: the LBM (often via an isothermal LBE) advances the DF's f_i using Equation (5) to update ρ and u , while the total energy equation (Equation (3)) is integrated separately using a conventional FD/FV solver; at each coupling cycle, the macroscopic energy solver uses the current flow fields to obtain (T) and (P) (through

the equation of state), and these thermodynamic variables are then fed back into f_i^{eq} to initiate the next LBM time step. This partitioned workflow enables the use of high-resolution shock-capturing schemes in the energy (and, in some implementations, also momentum) update, yielding superior robustness for transonic and low-supersonic regimes with strong shocks (typically $Ma \approx 0.8$ – 2) compared with pure LBM, and, in specific formulations, effective resolution of strong shock waves and thin boundary layers – for example, in HLBM flux solvers where inviscid fluxes are computed by a one-dimensional LB model while viscous fluxes are approximated by standard discretizations. Although a few test cases report operation at higher Mach numbers, the dominant application domain remains realistic aerodynamic configurations, where stability and shock capturing are prioritized; the corresponding trade-off is that hybrid schemes are generally the most algorithmically complex and resource-intensive, due to added code infrastructure, inter-solver communication, and increased computational cost. Finally, beyond hybridization, the dataset also indicates that entropic LBM variants provide a promising complementary route to extend the accessible Mach range by modifying the lattice formulation without increasing the number of discrete velocities. In general, Hybrid methods offer the best shock resolution but sacrifice LBM's simplicity and parallel efficiency.

4.3.4. Summary of modelling strategies

While Figure 7 focuses on how energy-equation treatments are distributed across the different regimes classified by maximum Mach number, Figure 8 illustrates how it occurs over the years since 2012. As such, Figure 8 is intended to illustrate publication trends rather than to rank the approaches by intrinsic superiority. The trend suggests that, compared with the earlier years of the period, recent publications show a stronger preference for hybrid and DDF formulations relative to SDF schemes. Moreover, whereas early studies more often emphasized 'stress-testing' methodologies to identify stability and applicability limits – frequently through hypersonic benchmark problems – more recent work has shifted towards supersonic and transonic regimes, reflecting a greater focus on practically relevant conditions rather than extreme test cases. Note that, as reported in Table 1, the maximum Mach number simulated is not reported in some articles.

As we can see from Table 1 and Figure 8, each category affects the achievable Mach/Reynolds range and has characteristic advantages and disadvantages. All reviewed approaches can reliably reach transonic conditions ($Ma \approx 0.8$ – 1.2), but their extension to higher Mach numbers differs substantially: In comparison with the

SDF and DDF, more transonic regimes are simulated by HLBM. Meanwhile, the DDF seems to be more used for the supersonic regime. In addition, it seems that before 2018, most researchers were stressing the methods and looking for extreme Mach numbers, but since then, they have become more stable and focused on other aspects. Another interesting thing is that in recent years, the SDF method has not been very popular, while the hybrid has attracted more attention. More advanced developments – most notably entropic and PonD variants – push the envelope further, reporting $Ma \approx 30$ and even $Ma \approx 100$ in specialized cases, albeit typically as stress tests enabled by targeted stability mechanisms (e.g. dynamic velocity sets and entropy constraints).

At the lattice level, the dataset also underscores intrinsic limitations: standard D3Q19 implementations are effectively confined to $Ma \approx 1$ before accuracy and stability degrade, and even lattice extensions generally yield only modest gains unless paired with specially formulated models; accordingly, correctly capturing shock jumps beyond $Ma \approx 1.5$ is reported only for dedicated conservative/stabilized formulations. Also, regarding Reynolds-number coverage for hybrid/flux-solver families (e.g. RLBFS, FDLBM, WENO-LBFS), it generally operate in the $\leq 10^5$ range, but when combined with turbulence modelling or implicit dissipation, they can reach high-Re engineering flows around 10^6 to 10^7 . In addition, DDF approaches are mostly limited to laminar and transitional regimes. Although a single transonic BGK/VLES SDF formulation studied a case with $Re = 1.2 \times 10^7$, most of them were used to simulate flow with low Re numbers (mostly $\leq 10^3$).

These trends align with a consistent pattern linking energy-equation treatment to accuracy and robustness: SDF approaches, which handle energy implicitly within a single distribution, tend to under-predict thermal effects and struggle under large temperature gradients, with BGK-based variants becoming unstable as Ma increases; DDF formulations reduce errors in shock position and sound speed and are inherently more stable at high Ma due to added degrees of freedom and tailored energy collisions (often strengthened via MRT/HRR), and can therefore be more efficient overall than SDF at high Mach by avoiding prohibitively small time steps; hybrid (LBM + FD/FV) approaches explicitly solve the energy equation and thus deliver the highest accuracy in strong gradients (comparable to classical CFD), albeit with additional coupling overhead and moderate computational cost. Finally, the reviewed papers indicate that method selection correlates with discretization and benchmark choices: Cartesian structured grids remain dominant (with unstructured/adaptive meshes used in a small minority of cases, often in hybrid contexts),

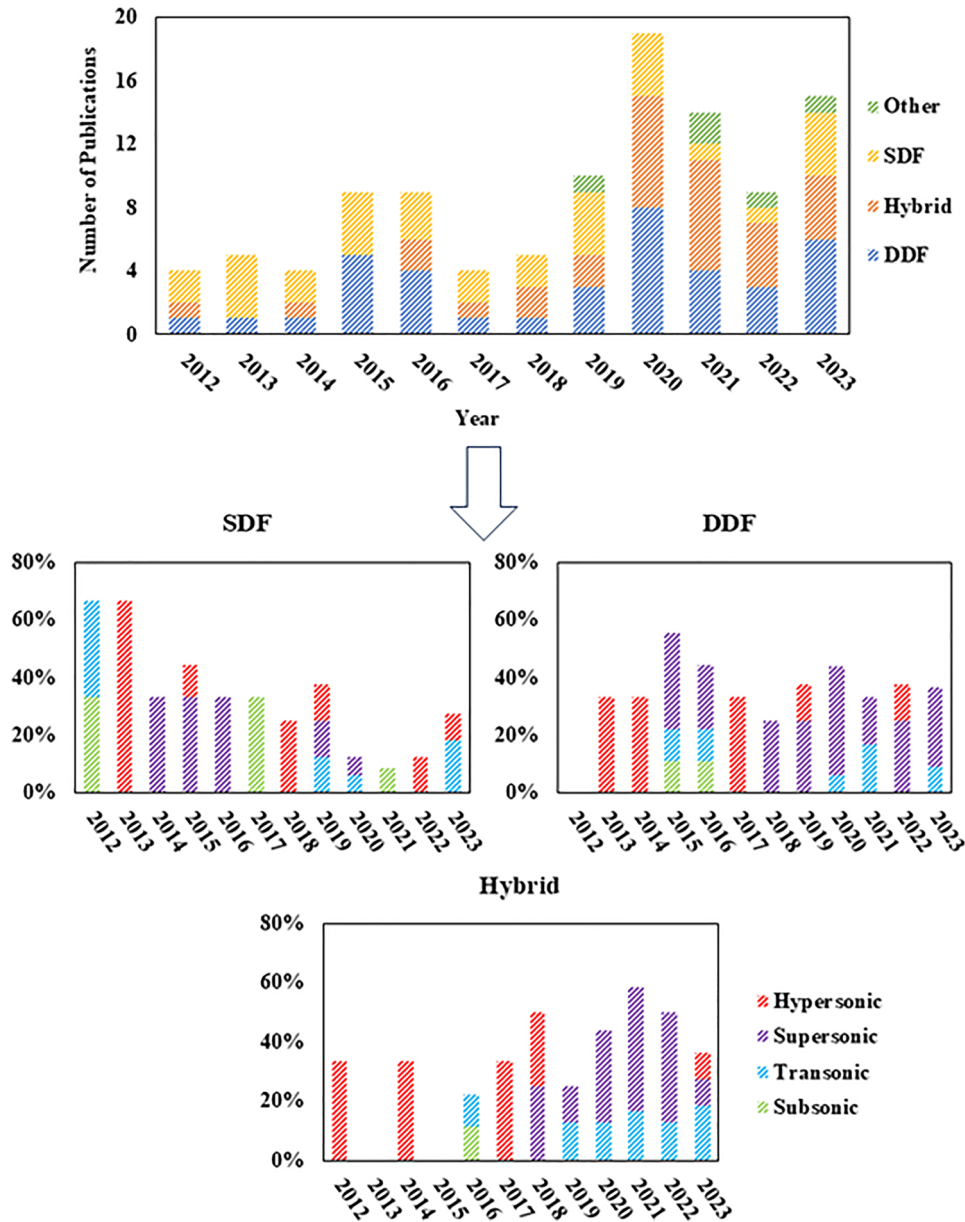


Figure 8. Distribution of the reviewed studies across flow regimes based on maximum Mach number, shown over time and grouped by energy-equation treatment approach.

mostly $Ma \geq 10$ demonstrations are reported on structured lattices, and validation is heavily anchored in shock-dominated benchmarks (shock tubes and shock interactions), followed by external aerodynamics; over 2012–2024, adoption is broadly balanced (DDF $\approx 37\%$, SDF $\approx 30\%$, Hybrid $\approx 28\%$), while the overall capability has advanced from near-transonic limits in early studies to multiple independent shock-stable approaches by 2024 – yet most post-2020 work remains concentrated in $Ma \approx 0.5 - 5$ as a practical compromise between fidelity, stability, and cost.

Table 3 summarizes the key features of the three main classes of compressible LBM described above.

A comprehensive comparative assessment of these methodologies is presented in Tables 4–6, highlighting their procedures, pros and cons, computational cost, implementation complexity, stability in high-Mach regimes, accuracy in resolving temperature gradients, and implementation complexity.

With regard to the choice of the collision model, it could be noted that it has a significant role on the stability and accuracy of LBM-based thermal simulations. In these reviewed studies, various models have been developed to address the limitations of traditional BGK approaches, particularly in the management of energy transport. It can be observed in Table 1 that the most

Table 3. Comparison of compressible LBM formulations.

Method	Energy handling	Key equations / relationships
SDF/MS	Implicit; energy encoded in the truncated equilibrium f_i^{eq}	Equation (5) with a single distribution. Equilibrium expanded in Hermite series [Equation (7)]. Viscosity $\vartheta = c_s^2 \left(\tau_f - \frac{1}{2} \right) \delta t$
DDF	Explicit; second distribution g_i carries temperature/energy	Two LBEs: Equation (5) for f_i and Equation (8) for g_i . Equilibrium g_i^{eq} given by Equation (9). Thermal diffusivity $\alpha = c_s^2 \left(\tau_g - \frac{1}{2} \right) \delta t$
HybridLBM-CFD	External; LBM for momentum, CFD solver for energy	LBM solves Equation (5) (often isothermal), while energy solver integrates Equation (3).

Note: Each method handles the energy equation differently, leading to distinct trade-offs between computational cost, stability and accuracy.

Table 4. Highlights of the various energy equation handling approaches.

Method	Procedure Highlights	Notable Innovations
SDF/MS	Uses extended velocity sets (beyond D2Q9/D3Q19). Stream-collide scheme same as standard LBM, but with more discrete speeds (often off-lattice directions). EOS often employed for pressure closure (e.g. ideal gas or Tait EOS for liquids) instead of solving energy equation.	High-order lattices (e.g. D3Q27, D3Q67) to recover higher-order moments (NSF) and handle high Mach. ELBM to improve nonlinear stability at high Mach. Shock detectors adding artificial bulk viscosity to smooth shock fronts.
DDF	Solves two Boltzmann equations: one for density-momentum (usual f 's), one for energy (typically g 's). Often uses separate lattices for f and g ; e.g. f on D2Q9, g on D2Q9 or D2Q5 (for temperature). May implement streaming for both, or use FD for the energy equation (some use advection-diffusion solver for g). Requires coupling via EOS: pressure and energy fields exchange via $p = \rho RT$ or energy conservation. Many apply MRT or flux limiters on the energy equation to capture shocks without Gibbs oscillations.	PonD: adaptive addition of discrete velocities during runtime to capture strong nonequilibrium features. TRT/MRT for energy: adjusting relaxation of energy mode separately to improve thermal stability. Coupled DDF-FV schemes: used a FV-WENO solver for the energy distribution (g -population) to capture shock gradients sharply. Adaptive chemistry coupling in some studies (allowing heat release in g -equation for combustions flows). Conservative DDF: ensuring g -populations conserve total energy (preventing drift in shock heating).
HLBM (Momentum-Energy Split)	Couples LBM with macroscopic solver: LBM handles momentum & continuity; a FD or FV solver handles energy (and sometimes momentum too, in flux solver approach). Requires exchange at each time-step: e.g. use LBM to compute fluxes or pressure for the FD solver, and vice-versa ensure LBM feels the thermal pressure force from energy solver. Often employs conservative formulation: e.g. solve total energy equation with FD, while LBM evolves deviations (to enforce correct shock thermodynamics). Implementation can be on uniform or adaptive grids; many use <i>block-structured mesh refinement</i> around shocks. Boundary conditions can be handled in the FD part (allowing easy use of far-field, no-slip, etc., similar to CFD).	LBFS: Shu and coworkers developed a method using LBM to obtain numerical fluxes for a FV solver (essentially replacing Riemann solver with LBM). This innovation allows complex geometries (unstructured grid FV) with LBM's low dissipation fluxes. Hybrid Recursive Regularization (HRR) + FD: Sagaut's group introduced an HLBM that regularizes LBM for momentum and uses FD for energy, ensuring stable coupling and linear stability up to Mach ~ 2 . Implicit and multi-scale coupling: Some hybrids run LBM with a larger time-step by coupling implicitly with the FD solver (improving efficiency for low Mach flows). Parallel coupling frameworks: Several innovations in how to efficiently couple two solvers in parallel were made (to handle the increased computation).

commonly used collision models for thermal LBM simulations are:

- **BGK:** A simple and computationally efficient model, but prone to numerical instability in high-Mach and high-temperature gradient flows.
- **MRT:** Introduces multiple relaxation times to improve numerical stability, making it reasonable for thermal simulations with complex boundary conditions.
- **HRR:** Decreases numerical instabilities in high-speed flows and improves energy conservation.

- **Entropic Models:** Ensure thermodynamic consistency by constraining entropy evolution, making them useful for fully compressible thermal flows.

The using of an efficient collision model depends on the specific requirements of the simulation, including stability, accuracy, and computational efficiency. Deployment of advanced collision models can significantly improve the capability of LBM for high-speed, thermally coupled flows.

Table 5. Pros and cons of various energy equation handling approaches.

Method	Advantages	Disadvantages
SDF	Simple stream–collide algorithm; low memory; fast per step; easy implementation; suitable for low or moderate Mach numbers	Poor stability at high Mach; lacks energy equation; oscillations near shocks; requires large velocity sets and often artificial viscosity or entropy fixes for stability; limited to 2D or simple 3D lattices
DDF	Handles energy equation explicitly; improved stability; better temperature prediction; adjustable Prandtl number; can reach higher Mach; extensible to strong shocks and complex thermodynamics	Doubles memory and cost; complex coupling between f and g distributions; often uses FD schemes; limited by lattice size; parameter tuning required; often coupled with higher–order advection schemes
Hybrid	Strong shock resolution; robust for transonic/hypersonic flows; flexible boundary conditions; allows unstructured grids and adaptive meshes; accommodates complex equations of state	High implementation complexity; coupled solvers require data exchange; moderate to high computational cost; loses pure LBM simplicity and explicitness

With regard to the Influence of temperature variations on LBM models, it is well known that they are an important part of determining fluid properties like density and pressure in compressible flows. The effectiveness of an LBM model in thermal applications depends on several factors:

- *Mach Number Dependence:* Standard LBM schemes face challenges in accurately capturing energy transfer at high Mach numbers. DDF and Hybrid approaches help mitigate this issue by providing more accurate temperature field representation.
- *Thermal Coupling with Hydrodynamics:* efficient energy transport modelling is vital for describing phenomena such as shock heating and heat diffusion, which dominate in supersonic and hypersonic flows.
- *Knudsen Number Sensitivity:* In non-continuum regimes, modifications to traditional LBM models may be essential to account for rarefied gas effects and deviations from equilibrium assumptions.

In general, the selection of the energy equation treatment in LBM can have a significant influence on the thermal simulation accuracy and efficiency. The appropriate method is chosen from the trade-off among thermal accuracy, numerical stability, and computational costs. Future research in this field needs to be concentrated on:

- Developing improved collision models optimized for energy transport.
- Exploring hybrid methodologies that integrate macroscopic solvers for heat transfer.
- Investigating adaptive mesh techniques to efficiently resolve strong temperature gradients in compressible flows.

By referring to these challenges, advancements in thermal LBM techniques can further improve the modelling of high-speed compressible flows in various engineering and scientific applications.

5. Current status, challenges, and future directions

5.1. Current status, challenges

Building upon what has been discussed, LBM is today a highly promising alternative to the traditional CFD methods for simulating compressible flows. Its mesoscopic nature, inherent parallelizability, and capability to model complex flow phenomena have made it more appealing in the domains of high-speed aerodynamics, shockwave interactions, and thermo-fluidic simulations. In line with the scope of this systematic review, the discussion below emphasizes cross-cutting challenges and trends rather than providing a theoretical critique of each individual formulation.

By 2024, multiple independent approaches exist that stably simulate shocks and even hypersonic flows. Although researchers simulated extreme Mach numbers, it should be stressed that these cases are often academic stress-tests. In practical terms, most studies after 2020 still focus on the Mach 0.5–5 range, but with much improved stability and accuracy compared to a decade ago. Recent advances to enabling the extension of the method to fully compressible regimes, are the development of multi-speed lattices, entropic and MRT collision

Table 6. Comparative analysis of energy equation approaches.

Method	Approx. computational cost	Stability at high Mach	Accuracy in Temperature Gradients	Implementation Complexity
SDF	Low	Moderate	Low; energy equation not solved	Low
DDF	High ($\approx 2 \times$ SDF)	High	High	High
Hybrid	Moderate–high	Very high (robust across shocks)	Moderate–high (uses CFD solvers)	Very high

models, hybrid approaches that combine LBM with NS solvers, and AMR techniques for enhancing computational efficiency. However, despite these achievements, there are still several challenges that need to be addressed before one can take full advantage of LBM for complex compressible flow problems.

One of the main restrictions of LBM for compressible flow simulations is numerical stability at high Mach numbers. The conventional velocity discretization in lattice methods puts a limit on the range of Mach numbers, leading to compressibility errors in the form of spurious pressure oscillations and lack of energy conservation. The issue is even worse in flows with intense shock waves, where the conventional collision models are not able to provide stability without introducing excessive numerical dissipation.

Another key aspect is how the energy equation is treated. Although SDF, DDE, and Hybrid methods all represent various compromises between computational efficiency, accuracy, and stability, it is challenging to capture strong temperature gradients, shock heating, and phase transition phenomena. Thermal model selection is application-dependent, and the development of special-purpose formulations is typically required.

Implementation of boundary conditions is another area needing improvement. Well-prescription of inflow conditions for supersonic flows, shock-wall interaction models, and non-reflecting boundary conditions are necessary for getting credible solutions. While recent attempts have suggested the use of hybrid methods combining LBM with NS solvers to enhance boundary treatments, more developments are needed in terms of better compatibility with complicated geometries and multi-scale simulations.

From the computational cost and scalability point of view, LBM has benefits in parallel computing but is still confronted with simulating compressible flows at high resolution. The prohibitively high computational cost of capturing fine-scale features and high-Mach interactions demands the advancement of more efficient algorithms. Although GPU-accelerated LBM solvers have demonstrated the potential to save computing time, it is yet to be determined if they are more efficient than conventional CFD approaches for simulations at large scales.

From the point of view of physical modelling, LBM turbulence modelling, especially for high-Reynolds-number flows, is still an open challenge. Early compressible LBM works (2012–2015) were mostly laminar. After ~2018, a few studies integrated LES-like approaches or ran DNS at higher Re. For instance, Dorschner et al. (2016) used an entropic LBM for a turbulent channel flow ($Re \sim 10^4$), and some hybrid approaches in 2020+ simulated high-Re external flows (using implicit

SGS models). The review data show about 10% of studies employed a turbulence model implying the majority were still laminar/DNS. But a small upward trend in Re capability is seen with the adoption of MRT/entropic collisions and hybrid methods, enabling exploration of turbulent compressible cases. The scale transition from mesoscopic to macroscopic with the preservation of physical fidelity is a fundamental issue that needs further investigation. One potential avenue for enhancing turbulence modelling in compressible flows is the hybridization of LBM with LES or RANS models.

In addition, by focusing on distribution of dimensionality, reviewed data confirms that 3D cases have gradually increased (most 3D papers appeared after 2016) as computational power grew. It is worth noting that, D2Q9 remains the most used velocity model (25% of cases) while advanced lattices and 3D models are less frequent – reflecting that many researchers first tried to extend the standard lattice before resorting to bigger models.

5.2. Future directions for development

To overcome these challenges, several directions of research need to be addressed:

- Improving Stability for High-Mach-Number Flows;

The development of higher-order lattice structures, entropy-based stabilization techniques, and RR methods can help LBM become more robust in supersonic and hypersonic flows.

- Enhancing Thermal and Multiphysics Capabilities

The coupling of LBM with radiation models, multi-phase solvers, and chemical reaction kinetics needs to be established in future research to extend its applicability beyond aerodynamics, particularly in energy and combustion sciences.

- Hybrid and Multi-Scale Approaches

Extending the application of hybrid solvers, in which LBM solves momentum equations and the energy equation is treated with traditional CFD methods, can lead to enhanced accuracy and stability. Multi-resolution models in which various parts of a simulation employ different LBM formulations could also offer a compromise between accuracy and computational efficiency.

- Computational Efficiency and Machine Learning Integration

The use of artificial intelligence and machine learning for parameter optimization, adaptive meshing, and automatic model selection has great potential to minimize computational expense without sacrificing high accuracy. Utilizing exascale computing capabilities and cloud-based LBM simulations can help in promoting large-scale aerodynamics and aerospace applications.

- Advancing Boundary Condition Treatments

The formulation of more general non-reflecting and absorbing boundary conditions, and more accurate immersed boundary techniques, can increase the capability of LBM for simulating realistic high-speed flow environments.

- Exploring Emerging Applications

Outside classical fluid dynamics, LBM can make an impact on biological fluid mechanics (e.g. blood flow simulation), electrohydrodynamics, plasma physics, and atmospheric science. Its application to these areas may instigate new theoretical and computational advancements.

6. Conclusion

This systematic review synthesizes recent procedural advances in the LBM's for compressible-flow simulation and contextualizes them through a database-driven assessment of performance envelopes and application patterns. Across the reviewed studies, all compressible LBM families demonstrate reliable operation in transonic regimes (approximately $Ma \sim 0.8$ – 1.2), while their ability to progress towards higher Mach numbers differs markedly; moreover, method adoption over 2012–2024 remains broadly balanced, with a slight prevalence of DDF formulations ($\approx 37\%$) over SDF/MS approaches ($\approx 30\%$) and hybrid strategies ($\approx 28\%$), indicating sustained parallel development rather than convergence to a single dominant paradigm.

The results further show that method choice is tightly coupled to shock robustness, thermal fidelity, and attainable flow regimes. Hybrid approaches have predominantly targeted flows near Mach 1 and are frequently selected for realistic aerodynamic configurations where conservative shock-capturing is essential, whereas MS/SDF and DDF formulations span a substantially wider Mach range, with many validations concentrated in moderate supersonic conditions ($Ma \approx 2$ – 5) and a smaller subset of specialized schemes demonstrating extreme capabilities (often $Ma \gtrsim 10$, and in a

few idealized stress tests even higher). Importantly, baseline lattices exhibit intrinsic limits (e.g. standard D3Q19 behaviour degrading beyond $Ma \approx 1$), and only specially formulated models have been shown to reproduce correct shock jumps beyond ($Ma \approx 1.5$), reinforcing that high-Mach success is typically contingent on explicit stabilization mechanisms (e.g. entropy constraints, dynamic/adaptive velocity concepts, regularization, or conservative coupling). At the same time, the treatment of energy emerges as a primary driver of accuracy: SDF approaches, which represent energy implicitly, tend to underpredict thermal effects and struggle with large temperature gradients, while DDF formulations – by allocating additional degrees of freedom to the thermal field – consistently report improved shock/temperature accuracy and enhanced high-Mach stability, albeit with increased computational and implementation costs; hybrid energy solvers achieve the highest accuracy in strong gradients (comparable to classical CFD), but at the price of extra coupled-PDE overhead and greater code complexity.

Reynolds-number coverage in the database is similarly broad and largely application-driven: many compressible LBM studies remain in laminar or mildly turbulent regimes ($Re \sim 10^2$ – 10^4), yet hybrid/flux-solver families extend to high Reynolds engineering conditions ($Re \sim 10^6$ – 10^7), when paired with implicit dissipation or turbulence modelling. SDF and DDF studies commonly remain laminar/transitional, with some demonstrations reaching turbulent regime using high-order shock-resolving advection (e.g. WENO) for the thermal update. Consistent with these trends, achieving very high Reynolds and robust shock stability typically requires advanced collision/turbulence treatments (e.g. MRT/entropic/LES), while only a minority of works ($\approx 10\%$) explicitly employ turbulence models, indicating that compressible LBM turbulence at high Reynolds remains an open frontier. Finally, a clear methodological–geometric trade-off is visible: structured Cartesian lattices dominate, unstructured/adaptive meshes appear in a small subset – often in hybrid contexts – and the most extreme Mach-number demonstrations are reported exclusively on structured grids, suggesting that researchers tend to avoid geometric complexity when pushing compressibility limits; correspondingly, shock-centric benchmarks (shock tubes and shock interactions) remain the most prevalent validation pathway, followed by external aerodynamics around canonical bodies (airfoils, cylinders, wedges).

Taken together, these findings indicate that compressible LBM has progressed from predominantly transonic demonstrations to a mature landscape where multiple independent routes can stably resolve shocks and, in

Table 7. Decision matrix.

Approach	Computational cost	Stability at High Mach	Accuracy in Strong Gradients	Implementation Complexity
SDF	Low: Minimal memory (1 distribution) and simple operations. Typically fastest per time-step.	Moderate: Stable for moderate Ma. Instabilities arise beyond transonic speeds without special treatment.	Low: Struggles with steep temperature or shock gradients (error-prone without high-order schemes).	Simple: Easiest to implement – akin to standard LBM with minor extensions.
DDF	High: $\sim 2\times$ the variables; often uses smaller time-steps or advanced schemes for energy solver.	High: Good stability into supersonic regime. Dedicated energy solver avoids thermal instabilities	High: Can accurately resolve shocks/gradients when coupled with proper schemes (approaches full NS fidelity).	Complex: Two distribution sets plus coupling. More tuning (relaxation rates, etc.) and coding effort.
Hybrid LBM	Moderate: Extra solver adds overhead, but often only one distribution in LBM (so $\sim \geq 1\times$ and $\leq 2\times$ cost). Parallelizable but heavier than pure LBM.	High: Most robust at high Mach. Can handle strong shocks and expansions (stability like traditional CFD).	Moderate: Captures primary gradients well (shock fronts via FD/FV). Small-scale accuracy depends on coupling (can be slightly dissipative).	Complex: Requires integrating two methods; careful coupling needed. Code and algorithm complexity are highest.

specialized settings, even hypersonic regimes; however, practical usage after 2020 still concentrates largely in the $Ma \sim 0.5\text{--}5$ range, emphasizing robustness and accuracy over extreme stress tests. Future research should therefore prioritize: (i) stability-preserving formulations that extend shock reliability without prohibitive lattice refinement, (ii) energy treatments that preserve thermal accuracy under strong gradients while controlling cost, (iii) scalable turbulence-capable frameworks for high-Reynolds compressible flows, and (iv) coupling and adaptivity strategies (including mesh/velocity adaptivity and modern hardware-aware parallelization) that enable complex geometries without forfeiting stability. Within this trajectory, data-driven and machine-learning tools are well positioned to support adaptive modelling choices (e.g. localized stabilization, surrogate closures, or parameter selection), but their impact will depend on careful integration with physically consistent collision and energy-transport design.

6.1. Decision matrix: cost vs. stability vs. accuracy vs. complexity

Finally, we compile a unified decision matrix comparing the three main approach types (SDF, DDF, Hybrid) across key criteria: computational cost, high-Mach stability, accuracy in resolving strong gradients, and implementation complexity. This reflects general trends identified in the Table 7.

This matrix guides decision-making for method selection. If one needs a quick, easy solution for mild compressibility, an SDF model is attractive due to low cost and simplicity – but one should not expect it to handle $Mach \gg 1$ or strong shocks with high fidelity. For simulations requiring higher Mach and better thermal accuracy, a DDF approach is often worth the cost: it offers significantly improved stability in high-speed flows and can resolve temperature fields and shock dynamics

more accurately, at the expense of more computation and implementation effort. Finally, for the most challenging scenarios (very high Mach numbers, complex geometry, high Re/turbulence), hybrid methods are recommended. They deliver the best stability and robust shock-capturing (comparable to established CFD solvers), and while they are complex and moderately costly, they may be the only viable option to simulate, for example, a Mach 5 flow around a real 3D vehicle or a high-Re transonic wing. These trade-offs were explicitly noted in the review: SDF is ‘low cost but moderate stability/accuracy’, DDF is ‘high cost but high stability/accuracy’, and Hybrid is ‘complex, moderate cost, high stability’. The appropriate choice thus depends on the requirements of stability vs. resources – a balance highlighted as a crucial consideration for future compressible LBM work.

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Author contributions

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References

- Alexander, F. J., Chen, H., Chen, S., & Doolen, G. (1992). Lattice Boltzmann model for compressible fluids. *Physical Review A*, 46(4), 1967. <https://doi.org/10.1103/PhysRevA.46.1967>
- Alexander, F. J., Chen, S., & Sterling, J. (1993). Lattice Boltzmann thermohydrodynamics. *Physical Review E*, 47(4), R2249. <https://doi.org/10.1103/PhysRevE.47.R2249>
- Atif, M., Namburi, M., & Ansumali, S. (2018). Higher-order Lattice Boltzmann model for thermohydrodynamics. *Physical Review E*, 98(5), 053311. <https://doi.org/10.1103/PhysRevE.98.053311>
- Bayat, A. S., Reza, M., & D'Orazio, A. (2024). Assessing accuracy and limitations in calculating drag coefficients using the Lattice Boltzmann method. In *The 9th International Conference on Knowledge and Technology of Mechanical, Electrical Engineering and Computer of Iran*.
- Bhadauria, A., Dorschner, B., & Karlin, I. (2021). Lattice Boltzmann method for fluid–structure interaction in compressible flow. *Physics of Fluids*, 33(10), 106111. <https://doi.org/10.1063/5.0062117>
- Bhadauria, A., & Karlin, I. (2024). High speed flows with particles on demand: Boundary conditions. *Computers & Fluids*, 271, Article 106152. <https://doi.org/10.1016/j.compfluid.2023.106152>
- Cabeza, L. F., Chàfer, M., & Mata, É. (2020). Comparative analysis of Web of Science and Scopus on the energy efficiency and climate impact of buildings. *Energies*, 13(2), 409. <https://doi.org/10.3390/en13020409>
- Casalino, D., Gonzalez-Martino, I., & Mancini, S. (2022). On the Rossiter-Heller frequency of resonant cavities. *Aerospace Science and Technology*, 131, Article 108013. <https://doi.org/10.1016/j.ast.2022.108013>
- Chang, S., Wang, Y., Yang, L., & Wu, J. (2014). Lattice Boltzmann flux solver: An efficient approach for numerical simulation of fluid flows. *Transactions of Nanjing University of Aeronautics & Astronautics*, 31, 1–15.
- Chen, J., Wang, Y., Yang, D., Chen, Q., & Sun, J. (2023). Development of three-dimensional rotated Lattice Boltzmann flux solver for the simulation of high-speed compressible flows. *Computers & Fluids*, 265, Article 105992. <https://doi.org/10.1016/j.compfluid.2023.105992>
- Chen, J., Yang, D., Chen, Q., Sun, J., & Wang, Y. (2023). A rotated Lattice Boltzmann flux solver with improved stability for the simulation of compressible flows with intense shock waves at high Mach number. *Computers & Mathematics with Applications*, 132, 18–31. <https://doi.org/10.1016/j.camwa.2022.12.003>
- Chen, Q., & Zhang, X. (2012). A novel less dissipation finite-difference Lattice Boltzmann scheme for compressible flows. *International Journal of Modern Physics C*, 23(11), 1250074. <https://doi.org/10.1142/S012918311250074X>
- Chen, S., Chen, H., Martnez, D., & Matthaeus, W. (1991). Lattice Boltzmann model for simulation of magnetohydrodynamics. *Physical Review Letters*, 67(27), 3776. <https://doi.org/10.1103/PhysRevLett.67.3776>
- Coratger, T., Farag, G., Zhao, S., Boivin, P., & Sagaut, P. (2021). Large-eddy lattice-Boltzmann modeling of transonic flows. *Physics of Fluids*, 33(11), 115112. <https://doi.org/10.1063/5.0064944>
- Coreixas, C., & Latt, J. (2020). Compressible Lattice Boltzmann methods with adaptive velocity stencils: An interpolation-free formulation. *Physics of Fluids*, 32(11), 116102. <https://doi.org/10.1063/5.0027986>
- Coreixas, C., Wissocq, G., Puigt, G., Boussuge, J.-F., & Sagaut, P. (2017). Recursive regularization step for high-order Lattice Boltzmann methods. *Physical Review E*, 96(3), 033306. <https://doi.org/10.1103/PhysRevE.96.033306>
- Cristea, A., Gonnella, G., Lamura, A., & Sofonea, V. (2009). A Lattice Boltzmann study of phase separation in liquid-vapor systems with gravity. arXiv preprint arXiv:0907.2778.
- Dellar, P. J. (2014). Lattice Boltzmann algorithms without cubic defects in Galilean invariance on standard lattices. *Journal of Computational Physics*, 259, 270–283. <https://doi.org/10.1016/j.jcp.2013.11.021>
- Deng, Z., Fan, Y., Theuerkauf, J., Jacob, K. V., Umbanhowar, P. B., & Lueptow, R. M. (2020). Modeling segregation of polydisperse granular materials in hopper discharge. *Powder Technology*, 374, 389–398. <https://doi.org/10.1016/j.powtec.2020.06.065>
- D'Orazio, A., Corcione, M., & Celata, G. P. (2004). Application to natural convection enclosed flows of a Lattice Boltzmann BGK model coupled with a general purpose thermal boundary condition. *International Journal of Thermal Sciences*, 43(6), 575–586. <https://doi.org/10.1016/j.ijthermalsci.2003.11.002>
- D'Orazio, A., & Succi, S. (2004). Simulating two-dimensional thermal channel flows by means of a Lattice Boltzmann method with new boundary conditions. *Future Generation Computer Systems*, 20(6), 935–944. <https://doi.org/10.1016/j.future.2003.12.005>
- Dorschner, B., Bösch, F., & Karlin, I. V. (2018). Particles on demand for kinetic theory. *Physical Review Letters*, 121(13), Article 130602. <https://doi.org/10.1103/PhysRevLett.121.130602>
- Dorschner, B., Frapolli, N., Chikatamarla, S. S., & Karlin, I. V. (2016). Grid refinement for entropic Lattice Boltzmann models. *Physical Review E*, 94(5), Article 053311. <https://doi.org/10.1103/PhysRevE.94.053311>
- Esfahanian, V., & Ghadyani, M. (2015). Improvement of the instability of compressible Lattice Boltzmann model by shock-detecting sensor. *Journal of Mechanical Science and Technology*, 29(5), 1981–1991. <https://doi.org/10.1007/s12206-015-0419-0>
- Falagas, M. E., Pitsouni, E. I., Malietzis, G. A., & Pappas, G. (2008). Comparison of PubMed, Scopus, Web of Science, and Google Scholar: Strengths and weaknesses. *The FASEB Journal*, 22(2), 338–342. <https://doi.org/10.1096/fj.07-9492LSF>
- Farag, G., Coratger, T., Wissocq, G., Zhao, S., Boivin, P., & Sagaut, P. (2021). A unified hybrid lattice-Boltzmann method for compressible flows: Bridging between pressure-based and density-based methods. *Physics of Fluids*, 33(8), 086101. <https://doi.org/10.1063/5.0057407>

- Farag, G., Zhao, S., Coratger, T., Boivin, P., Chiavassa, G., & Sagaut, P. (2020). A pressure-based regularized lattice-Boltzmann method for the simulation of compressible flows. *Physics of Fluids*, 32(6), 066106. <https://doi.org/10.1063/5.0011839>
- Feng, Y., Boivin, P., Jacob, J., & Sagaut, P. (2019). Hybrid recursive regularized thermal Lattice Boltzmann model for high subsonic compressible flows. *Journal of Computational Physics*, 394, 82–99. <https://doi.org/10.1016/j.jcp.2019.05.031>
- Feng, Y., Guo, S., Jacob, J., & Sagaut, P. (2019). Solid wall and open boundary conditions in hybrid recursive regularized Lattice Boltzmann method for compressible flows. *Physics of Fluids*, 31(12), 126103. <https://doi.org/10.1063/1.5129138>
- Feng, Y., Guo, S., Jacob, J., & Sagaut, P. (2020). Grid refinement in the three-dimensional hybrid recursive regularized Lattice Boltzmann method for compressible aerodynamics. *Physical Review E*, 101(6), 063302. <https://doi.org/10.1103/PhysRevE.101.063302>
- Feng, Y., Sagaut, P., & Tao, W. (2015). A three dimensional lattice model for thermal compressible flow on standard lattices. *Journal of Computational Physics*, 303, 514–529. <https://doi.org/10.1016/j.jcp.2015.09.011>
- Feng, Y., Sagaut, P., & Tao, W.-Q. (2016). A compressible Lattice Boltzmann finite volume model for high subsonic and transonic flows on regular lattices. *Computers & Fluids*, 131, 45–55. <https://doi.org/10.1016/j.compfluid.2016.03.009>
- Feng, Y., & Tao, W.-Q. (2014). A compressible thermal Lattice Boltzmann model with factorization symmetry. *Numerical Heat Transfer, Part B: Fundamentals*, 66(6), 544–562. <https://doi.org/10.1080/10407790.2014.915679>
- Feng, Y.-L., Guo, S.-L., Tao, W.-Q., & Sagaut, P. (2018). Regularized thermal Lattice Boltzmann method for natural convection with large temperature differences. *International Journal of Heat and Mass Transfer*, 125, 1379–1391. <https://doi.org/10.1016/j.ijheatmasstransfer.2018.05.051>
- Frapolli, N., Chikatamarla, S., & Karlin, I. (2020). Theory, analysis, and applications of the entropic Lattice Boltzmann model for compressible flows. *Entropy*, 22(3), 370. <https://doi.org/10.3390/e22030370>
- Frapolli, N., Chikatamarla, S. S., & Karlin, I. V. (2015). Entropic Lattice Boltzmann model for compressible flows. *Physical Review E*, 92(6), Article 061301. <https://doi.org/10.1103/PhysRevE.92.061301>
- Frapolli, N., Chikatamarla, S. S., & Karlin, I. V. (2016a). Entropic Lattice Boltzmann model for gas dynamics: Theory, boundary conditions, and implementation. *Physical Review E*, 93(6), Article 063302. <https://doi.org/10.1103/PhysRevE.93.063302>
- Frapolli, N., Chikatamarla, S. S., & Karlin, I. V. (2016b). Lattice kinetic theory in a comoving Galilean reference frame. *Physical Review Letters*, 117(1), Article 010604. <https://doi.org/10.1103/PhysRevLett.117.010604>
- Ghadyani, M., & Esfahanian, V. (2014). A more robust compressible Lattice Boltzmann model by using the numerical filters. *Journal of Mechanics*, 30(5), 515–525. <https://doi.org/10.1017/jmech.2014.45>
- Ghadyani, M., Esfahanian, V., & Taeibi-Rahni, M. (2015). The use of shock-detecting sensor to improve the stability of Lattice Boltzmann model for high Mach number flows. *International Journal of Modern Physics C*, 26(1), 1550006. <https://doi.org/10.1142/S0129183115500060>
- Ghaffarian, A., & Hejranfar, K. (2019). An implicit dual-time stepping spectral difference Lattice Boltzmann method for simulation of viscous compressible flows on structured meshes. *Meccanica*, 54(10), 1561–1581. <https://doi.org/10.1007/s11012-019-01036-w>
- Goodarzi, M., D’Orazio, A., Keshavarzi, A., Mousavi, S., & Karimipour, A. (2018). Develop the nano scale method of Lattice Boltzmann to predict the fluid flow and heat transfer of air in the inclined lid driven cavity with a large heat source inside, two case studies: Pure natural convection & mixed convection. *Physica A: Statistical Mechanics and its Applications*, 509, 210–233. <https://doi.org/10.1016/j.physa.2018.06.013>
- Guangwu, Y., Yaosong, C., & Shouxin, H. (1999). Simple Lattice Boltzmann model for simulating flows with shock wave. *Physical Review E*, 59(1), 454. <https://doi.org/10.1103/PhysRevE.59.454>
- Gunstensen, A. K., Rothman, D. H., Zaleski, S., & Zanetti, G. (1991). Lattice Boltzmann model of immiscible fluids. *Physical Review A*, 43(8), 4320. <https://doi.org/10.1103/PhysRevA.43.4320>
- Guo, S., Feng, Y., Jacob, J., Renard, F., & Sagaut, P. (2020). An efficient Lattice Boltzmann method for compressible aerodynamics on D3Q19 lattice. *Journal of Computational Physics*, 418, Article 109570. <https://doi.org/10.1016/j.jcp.2020.109570>
- Guo, S., Feng, Y., & Sagaut, P. (2020). Improved standard thermal Lattice Boltzmann model with hybrid recursive regularization for compressible laminar and turbulent flows. *Physics of Fluids*, 32(12), 126108. <https://doi.org/10.1063/5.0033364>
- Guo, S., Feng, Y., & Sagaut, P. (2021). On the use of conservative formulation of energy equation in hybrid compressible Lattice Boltzmann method. *Computers & Fluids*, 219, Article 104866. <https://doi.org/10.1016/j.compfluid.2021.104866>
- Guo, Z., Wang, L.-P., & Qi, Y. (2023). Discrete unified gas kinetic scheme for continuum compressible flows. *Physical Review E*, 107(2), Article 025304. <https://doi.org/10.1103/PhysRevE.107.025304>
- Guo, Z., Wang, R., & Xu, K. (2015). Discrete unified gas kinetic scheme for all Knudsen number flows. II. Thermal compressible case. *Physical Review E*, 91(3), Article 033313. <https://doi.org/10.1103/PhysRevE.91.033313>
- Gurevitch, J., Koricheva, J., Nakagawa, S., & Stewart, G. (2018). Meta-analysis and the science of research synthesis. *Nature*, 555(7695), 175–182. <https://doi.org/10.1038/nature25753>
- Hanada, T., & Kataoka, T. (2015). Numerical simulation of two-dimensional supersonic flows using the free-molecular-type kinetic scheme. *Theoretical and Applied Mechanics Japan*, 63, 3–8.
- Hanada, T., & Kataoka, T. (2021). Lattice Boltzmann method for compressible Euler equations based on exact kinetic system. *International Journal for Numerical Methods in Fluids*, 93(8), 2554–2569. <https://doi.org/10.1002/fld.4987>
- He, B., Chen, Y., Feng, W., Li, Q., Song, A., Wang, Y., Zhang, M., & Zhang, W. (2012). Compressible Lattice Boltzmann method and applications. *International Journal of Numerical Analysis & Modeling*, 9(2), 410–418.
- He, X., Chen, S., & Doolen, G. D. (1998). A novel thermal model for the Lattice Boltzmann method in incompressible limit. *Journal of Computational Physics*, 146(1), 282–300. <https://doi.org/10.1006/jcph.1998.6057>

- He, X., Chen, S., & Zhang, R. (1999). A Lattice Boltzmann scheme for incompressible multiphase flow and its application in simulation of Rayleigh–Taylor instability. *Journal of Computational Physics*, 152(2), 642–663. <https://doi.org/10.1006/jcph.1999.6257>
- He, Y.-L., Liu, Q., & Li, Q. (2013). Three-dimensional finite-difference Lattice Boltzmann model and its application to inviscid compressible flows with shock waves. *Physica A: Statistical Mechanics and its Applications*, 392(20), 4884–4896. <https://doi.org/10.1016/j.physa.2013.06.021>
- Hejranfar, K., & Ghaffarian, A. (2016). A spectral difference Lattice Boltzmann method for solution of inviscid compressible flows on structured grids. *Computers & Mathematics with Applications*, 72(5), 1341–1368. <https://doi.org/10.1016/j.camwa.2016.06.043>
- Hejranfar, K., & Ghaffarian, A. (2020). A high-order accurate unstructured spectral difference Lattice Boltzmann method for computing inviscid and viscous compressible flows. *Aerospace Science and Technology*, 98, Article 105661. <https://doi.org/10.1016/j.ast.2019.105661>
- Hejranfar, K., Hashemi Nasab, H., & Azampour, M. H. (2020). Arbitrary Lagrangian-Eulerian unstructured finite-volume lattice-Boltzmann method for computing two-dimensional compressible inviscid flows over moving bodies. *Physical Review E*, 101(2), Article 023308. <https://doi.org/10.1103/PhysRevE.101.023308>
- Hosseini, S. A., Coreixas, C., Darabiha, N., & Thévenin, D. (2019). Extensive analysis of the Lattice Boltzmann method on shifted stencils. *Physical Review E*, 100(6), Article 063301. <https://doi.org/10.1103/PhysRevE.100.063301>
- Hosseini, S. A., Darabiha, N., & Thévenin, D. (2020). Compressibility in Lattice Boltzmann on standard stencils: Effects of deviation from reference temperature. *Philosophical Transactions of the Royal Society A*, 378(2175), 20190399. <https://doi.org/10.1098/rsta.2019.0399>
- Huang, H.-Y., Jin, K., Li, K., & Zheng, X.-J. (2023). Application of shifted lattice model to 3D compressible Lattice Boltzmann method. *Chinese Physics B*, 32(9), 094701. <https://doi.org/10.1088/1674-1056/acc78d>
- Huang, X., Chen, J., Zhang, J., Wang, L., & Wang, Y. (2023). An adaptive mesh refinement–rotated Lattice Boltzmann flux solver for numerical simulation of two and three-dimensional compressible flows with complex shock structures. *Symmetry*, 15(10), 1909. <https://doi.org/10.3390/sym15101909>
- Jalali Khouzani, H., & Kamali Moghadam, R. (2020). A novel approach of unsteady adjoint Lattice Boltzmann method based on circular function scheme. *Journal of Scientific Computing*, 85(2), 38. <https://doi.org/10.1007/s10915-020-01318-6>
- Jalali Khouzani, H., & Kamali-Moghadam, R. (2023). Airfoil inverse design based on laminar compressible adjoint Lattice Boltzmann method. *International Journal for Numerical Methods in Fluids*, 95(8), 1197–1219. <https://doi.org/10.1002/fld.5192>
- Jiakuan, X., Jiang, X., Yutian, W., Ningjuan, D., Lei, Q., & Junqiang, B. (2025). Progress in local-variable-based transition-turbulence models for subsonic and transonic boundary layers. *Chinese Journal of Aeronautics*, 39(4), 103688.
- Kallikounis, N., & Karlin, I. (2024). Particles on demand method: Theoretical analysis, simplification techniques, and model extensions. *Physical Review E*, 109(1), Article 015304. <https://doi.org/10.1103/PhysRevE.109.015304>
- Kallikounis, N. G., Dorschner, B., & Karlin, I. (2021). Multi-scale semi-Lagrangian Lattice Boltzmann method. *Physical Review E*, 103(6), Article 063305. <https://doi.org/10.1103/PhysRevE.103.063305>
- Kallikounis, N. G., Dorschner, B., & Karlin, I. V. (2022). Particles on demand for flows with strong discontinuities. *Physical Review E*, 106(1), Article 015301. <https://doi.org/10.1103/PhysRevE.106.015301>
- Kamali Moghadam, R., Sahranavard Fard, N., & Jalali, H. (2021). New approach of axisymmetric compressible finite-volume Lattice Boltzmann method for numerical simulation of supersonic inviscid flow. *Fluid Dynamics*, 56(1), 121–133. <https://doi.org/10.1134/S0015462821010080>
- Karimipour, A., Nezhad, A. H., D’Orazio, A., Esfe, M. H., Safaei, M. R., & Shirani, E. (2015). Simulation of copper–water nanofluid in a microchannel in slip flow regime using the Lattice Boltzmann method. *European Journal of Mechanics-B/Fluids*, 49, 89–99. <https://doi.org/10.1016/j.euromechflu.2014.08.004>
- Karimipour, A., Nezhad, A. H., D’Orazio, A., & Shirani, E. (2012). Investigation of the gravity effects on the mixed convection heat transfer in a microchannel using Lattice Boltzmann method. *International Journal of Thermal Sciences*, 54, 142–152. <https://doi.org/10.1016/j.ijthermalsci.2011.11.015>
- Karlin, I., Sichau, D., & Chikatamarla, S. (2013). Consistent two-population Lattice Boltzmann model for thermal flows. *Physical Review E*, 88(6), Article 063310. <https://doi.org/10.1103/PhysRevE.88.063310>
- Kataoka, T., & Hanada, T. (2019). New Lattice Boltzmann model for the compressible Navier–Stokes equations. *International Journal for Numerical Methods in Fluids*, 91(4), 183–197. <https://doi.org/10.1002/fld.4748>
- Kataoka, T., & Tsutahara, M. (2004). Lattice Boltzmann model for the compressible Navier–Stokes equations with flexible specific-heat ratio. *Physical Review E*, 69(3), Article 035701. <https://doi.org/10.1103/PhysRevE.69.035701>
- Krivovichev, G. V., & Bezrukova, E. S. (2023). Analysis of discrete velocity models for Lattice Boltzmann simulations of compressible flows at arbitrary specific heat ratio. *Computation*, 11(7), 138. <https://doi.org/10.3390/computation11070138>
- Krivovichev, G. V., & Marnopolskaya, E. S. (2020). The approach to optimization of finite-difference schemes for the advective stage of finite-difference-based Lattice Boltzmann method. *International Journal of Modeling, Simulation, and Scientific Computing*, 11(1), 2050002. <https://doi.org/10.1142/S1793962320500026>
- Kruger, T. (2016). *Lattice Boltzmann method-principles and practice*. Springer International.
- Kumar, G. S., & Maheshwari, N. (2020). Viscous multi-species Lattice Boltzmann solver for simulating shock-wave structure. *Computers & Fluids*, 203, Article 104539. <https://doi.org/10.1016/j.compfluid.2020.104539>
- Ladd, A. J. (1994). Numerical simulations of particulate suspensions via a discretized Boltzmann equation. Part 2. Numerical results. *Journal of Fluid Mechanics*, 271, 311–339. <https://doi.org/10.1017/S0022112094001783>

- Lallemand, P., & Luo, L.-S. (2003). Theory of the Lattice Boltzmann method: Acoustic and thermal properties in two and three dimensions. *Physical Review E*, 68(3), Article 036706. <https://doi.org/10.1103/PhysRevE.68.036706>
- Latt, J., Coreixas, C., Beny, J., & Parmigiani, A. (2020). Efficient supersonic flow simulations using Lattice Boltzmann methods based on numerical equilibria. *Philosophical Transactions of the Royal Society A*, 378(2175), 20190559. <https://doi.org/10.1098/rsta.2019.0559>
- Li, K., & Zhong, C. (2015b). A Lattice Boltzmann model for simulation of compressible flows. *International Journal for Numerical Methods in Fluids*, 77(6), 334–357. <https://doi.org/10.1002/fld.3984>
- Li, K., & Zhong, C. (2016). Aeroacoustic simulations using compressible Lattice Boltzmann method. *Advances in Applied Mathematics and Mechanics*, 8(5), 795–809. <https://doi.org/10.4208/aamm.2015.m1083>
- Li, K., & Zhong, C.-W. (2015a). A multiple-relaxation-time Lattice Boltzmann method for high-speed compressible flows. *Chinese Physics B*, 24(5), 050501. <https://doi.org/10.1088/1674-1056/24/5/050501>
- Li, W., Wang, J., Gopalakrishnan, P., Li, Y., Zhang, R., & Chen, H. (2023). Hybrid Lattice Boltzmann approach for simulation of high-speed flows. *AIAA Journal*, 61(12), 5548–5557. <https://doi.org/10.2514/1.J063199>
- Li, Y., Niu, X.-D., Yuan, H.-Z., Khan, A., & Li, X. (2018). A numerical study for WENO scheme-based on different Lattice Boltzmann flux solver for compressible flows. *Acta Mechanica Sinica*, 34(6), 995–1014. <https://doi.org/10.1007/s10409-018-0785-9>
- Li, Y., Yuan, H.-Z., Niu, X.-D., Yang, Y.-Y., & Shu, S. (2018). WENO scheme-based Lattice Boltzmann flux solver for simulation of compressible flows. *Computer Physics Communications*, 23(4), 1012–1036. <https://doi.org/10.4208/cicp.OA-2016-0210>
- Li, Z., & Shan, X. (2023). Body-force modelling in thermal compressible flows with the Lattice Boltzmann method. *Journal of Fluid Mechanics*, 964, A14. <https://doi.org/10.1017/jfm.2023.328>
- Liberati, A., Altman, D. G., Tetzlaff, J., Mulrow, C., Gøtzsche, P. C., Ioannidis, J. P., Clarke, M., Devereaux, P. J., Kleijnen, J., & Moher, D. (2009). The PRISMA statement for reporting systematic reviews and meta-analyses of studies that evaluate health care interventions: Explanation and elaboration. *Annals of Internal Medicine*, 151(4), W-65–W-94. <https://doi.org/10.7326/0003-4819-151-4-200908180-00136>
- Lin, C., Luo, K. H., Fei, L., & Succi, S. (2017). A multi-component discrete Boltzmann model for nonequilibrium reactive flows. *Scientific Reports*, 7(1), 14580. <https://doi.org/10.1038/s41598-017-14824-9>
- Lin, C., Xu, A., Zhang, G., Li, Y., & Succi, S. (2014). Polar-coordinate Lattice Boltzmann modeling of compressible flows. *Physical Review E*, 89(1), Article 013307. <https://doi.org/10.1103/PhysRevE.89.013307>
- Liu, Q., Feng, X.-B., & Lu, C.-W. (2019). Simulating high-Mach-number compressible flows with shock waves via Hermite-expansion-based Lattice Boltzmann method. *Physica A: Statistical Mechanics and its Applications*, 533, Article 122062. <https://doi.org/10.1016/j.physa.2019.122062>
- Liu, X., Cai, X., Huang, Z., Hou, Y., Qin, J., & Chen, Z. (2025). Comparative study on the oblique water-entry of high-speed projectile based on rigid-body and elastic-plastic body model. *Defence Technology*, 46, 133–155. <https://doi.org/10.1016/j.dt.2024.11.007>
- Liu, Z., Chen, R., & Xu, L. (2022). Parallel unstructured finite volume Lattice Boltzmann method for high-speed viscous compressible flows. *International Journal of Modern Physics C*, 33(5), 2250066. <https://doi.org/10.1142/S0129183122500668>
- Liyue, W., Cong, W., Shuyue, W., Sheng, Q., Gang, S., Bo, Y., & Zhong, Y. (2022). A boundary surrogate model for micro/nano grooved surface structure applied in turbulence flow control over airfoil. *Chinese Journal of Aeronautics*, 35(2), 62–73. <https://doi.org/10.1016/j.cja.2021.04.026>
- Mahendra, A. K., & Singh, R. K. (2013). Onsager reciprocity principle for kinetic models and kinetic schemes. arXiv preprint arXiv:1308.4119.
- Nasab, H. H., Hejranfar, K., & Azampour, M. H. (2023). Application of arbitrary Lagrangian–Eulerian unstructured finite volume Lattice Boltzmann method to simulate compressible viscous flows over moving bodies. *Meccanica*, 58(12), 2329–2346. <https://doi.org/10.1007/s11012-023-01729-3>
- Nejat, A., & Abdollahi, V. (2013). A critical study of the compressible Lattice Boltzmann methods for Riemann problem. *Journal of Scientific Computing*, 54(1), 1–20. <https://doi.org/10.1007/s10915-012-9596-5>
- Noah, K., Lien, F. S., & Yee, E. (2021). Large-eddy simulation of subsonic turbulent jets using the compressible Lattice Boltzmann method. *International Journal for Numerical Methods in Fluids*, 93(4), 927–952. <https://doi.org/10.1002/fld.4914>
- Pae, K.-N., Kim, Y.-J., Kim, W.-J., & Kim, S.-J. (2023). Simulation of acoustical field of ballistic shock therapy device by the Lattice Boltzmann method. *Physics of Wave Phenomena*, 31(1), 36–43. <https://doi.org/10.3103/S1541308X23010053>
- Page, M. J., McKenzie, J. E., Bossuyt, P. M., Boutron, I., Hoffmann, T. C., Mulrow, C. D., Shamseer, L., Tetzlaff, J. M., Akl, E. A., & Brennan, S. E. (2021). The PRISMA 2020 statement: An updated guideline for reporting systematic reviews. *bmj*, 372, n71. <https://doi.org/10.1136/bmj.n71>
- Perumal, D. A., & Dass, A. K. (2015). A review on the development of Lattice Boltzmann computation of macro fluid flows and heat transfer. *Alexandria Engineering Journal*, 54(4), 955–971. <https://doi.org/10.1016/j.aej.2015.07.015>
- Qiao, Z., Yang, X., & Zhang, Y. (2024). A free-energy based multiple-distribution-function Lattice Boltzmann method for multi-component and multi-phase flows. *Applied Thermal Engineering*, 257, Article 124241. <https://doi.org/10.1016/j.applthermaleng.2024.124241>
- Qin, J., Wu, J., Gu, X., & Ma, C. (2022). A combined flux reconstruction and Lattice Boltzmann flux solver for inviscid compressible flow simulation. *International Journal for Numerical Methods in Fluids*, 94(12), 2087–2103. <https://doi.org/10.1002/fld.5137>
- Qin, J., Wu, J., & Ma, C. (2022). A high-order scheme based on Lattice Boltzmann flux solver for viscous compressible flow simulations. *Applied Mathematics and Mechanics*, 43(10), 1601–1614. <https://doi.org/10.1007/s10483-022-2913-7>
- Qiu, R., Chen, R., Zhu, C., & You, Y. (2018). A hermite-based Lattice Boltzmann model with artificial viscosity for compressible viscous flows. *International Journal of Modern Physics B*, 32(13), 1850157. <https://doi.org/10.1142/S0217979218501576>

- Qiu, R.-F., Che, H.-H., Zhou, T., Zhu, J.-F., & You, Y.-C. (2020). Lattice Boltzmann simulation for unsteady shock wave/boundary layer interaction in a shock tube. *Computers & Mathematics with Applications*, 80(10), 2241–2257. <https://doi.org/10.1016/j.camwa.2020.07.012>
- Qiu, R. F., Wang, H. N., Zhu, J. F., Chen, R. Q., Zhu, C. X., & You, Y. C. (2019). Compressible Lattice Boltzmann simulations on high-performance and low-cost GeForce GPU. *Concurrency and Computation: Practice and Experience*, 31(20), e5341. <https://doi.org/10.1002/cpe.5341>
- Qiu, R.-F., You, Y.-C., Zhu, C.-X., Chen, R.-Q., & Zhu, J.-F. (2017). Three-dimensional coupled double-distribution-function Lattice Boltzmann models for compressible Navier–Stokes equations. *Pramana*, 89(6), 1–11. <https://doi.org/10.1007/s12043-017-1480-2>
- Rathnayake, S. (2024). Compressible Lattice Boltzmann solver for supersonic flow.
- Renard, F., Feng, Y., Boussuge, J.-F., & Sagaut, P. (2021). Improved compressible hybrid Lattice Boltzmann method on standard lattice for subsonic and supersonic flows. *Computers & Fluids*, 219, Article 104867. <https://doi.org/10.1016/j.compfluid.2021.104867>
- Renard, F., Wissocq, G., Boussuge, J.-F., & Sagaut, P. (2021). A linear stability analysis of compressible hybrid Lattice Boltzmann methods. *Journal of Computational Physics*, 446, Article 110649. <https://doi.org/10.1016/j.jcp.2021.110649>
- Saadat, M. H., Bösch, F., & Karlin, I. V. (2019). Lattice Boltzmann model for compressible flows on standard lattices: Variable Prandtl number and adiabatic exponent. *Physical Review E*, 99(1), Article 013306. <https://doi.org/10.1103/PhysRevE.99.013306>
- Saadat, M. H., Bösch, F., & Karlin, I. V. (2020). Semi-Lagrangian Lattice Boltzmann model for compressible flows on unstructured meshes. *Physical Review E*, 101(2), Article 023311. <https://doi.org/10.1103/PhysRevE.101.023311>
- Saadat, M. H., Hosseini, S. A., Dorschner, B., & Karlin, I. (2021). Extended Lattice Boltzmann model for gas dynamics. *Physics of Fluids*, 33(4), 046104. <https://doi.org/10.1063/5.0048029>
- Saadat, M. H., & Karlin, I. V. (2020). Arbitrary Lagrangian–Eulerian formulation of Lattice Boltzmann model for compressible flows on unstructured moving meshes. *Physics of Fluids*, 32(4), 046105. <https://doi.org/10.1063/5.0004024>
- Safdari Shadloo, M. (2019). Numerical simulation of compressible flows by Lattice Boltzmann method. *Numerical Heat Transfer, Part A: Applications*, 75(3), 167–182. <https://doi.org/10.1080/10407782.2019.1580053>
- Sbragaglia, M., Benzi, R., Biferale, L., Succi, S., Sugiyama, K., & Toschi, F. (2007). Generalized Lattice Boltzmann method with multirange pseudopotential. *Physical Review E*, 75(2), 026702. <https://doi.org/10.1103/PhysRevE.75.026702>
- Şeker, A., Zafer, B., Gonzalez-Martino, I., & Yüce-tepe, M. (2023). Aeroacoustic investigation of transonic flow behavior in M219 deep cavity with passive flow control configurations. *Physics of Fluids*, 35(10), 105117. <https://doi.org/10.1063/5.0163526>
- Shan, X. (2019). Central-moment-based Galilean-invariant multiple-relaxation-time collision model. *Physical Review E*, 100(4), Article 043308. <https://doi.org/10.1103/PhysRevE.100.043308>
- Shan, X., & Chen, H. (1993). Lattice Boltzmann model for simulating flows with multiple phases and components. *Physical Review E*, 47(3), 1815. <https://doi.org/10.1103/PhysRevE.47.1815>
- Shim, J. W. (2013). Multidimensional on-lattice higher-order models in the thermal Lattice Boltzmann theory. *Physical Review E – Statistical, Nonlinear, and Soft Matter Physics*, 88(5), Article 053310. <https://doi.org/10.1103/PhysRevE.88.053310>
- Shirsat, A. U., Nayak, S. G., & Patil, D. V. (2022). Simulation of high-Mach-number inviscid flows using a third-order Runge-Kutta and fifth-order WENO-based finite-difference Lattice Boltzmann method. *Physical Review E*, 106(2), 025314. <https://doi.org/10.1103/PhysRevE.106.025314>
- Siebert, D., Hegele, L., Jr., & Philippi, P. C. (2008). Lattice Boltzmann equation linear stability analysis: Thermal and athermal models. *Physical Review E – Statistical, Nonlinear, and Soft Matter Physics*, 77(2), Article 026707. <https://doi.org/10.1103/PhysRevE.77.026707>
- Sofonea, V., Lamura, A., Gonnella, G., & Cristea, A. (2004). Finite-difference Lattice Boltzmann model with flux limiters for liquid-vapor systems. *Physical Review E*, 70(4), Article 046702. <https://doi.org/10.1103/PhysRevE.70.046702>
- Succi, S. (2001). *The Lattice Boltzmann equation: For fluid dynamics and beyond*. Oxford University Press.
- Succi, S., Foti, E., & Higuera, F. (1989). Three-dimensional flows in complex geometries with the Lattice Boltzmann method. *Europhysics Letters*, 10(5), 433. <https://doi.org/10.1209/0295-5075/10/5/008>
- Sun, C. (1998). Lattice-Boltzmann models for high speed flows. *Physical Review E*, 58(6), 7283. <https://doi.org/10.1103/PhysRevE.58.7283>
- Sun, C., & Hsu, A. T. (2003). Three-dimensional Lattice Boltzmann model for compressible flows. *Physical Review E*, 68(1), Article 016303. <https://doi.org/10.1103/PhysRevE.68.016303>
- Sun, Y., Cai, J., & Qin, W. (2018). High-order discontinuous Galerkin solution of compressible flows with a hybrid Lattice Boltzmann flux. *Transactions of Nanjing University of Aeronautics and Astronautics*, 35, 413–422.
- Swift, M. R., Osborn, W., & Yeomans, J. (1995). Lattice Boltzmann simulation of nonideal fluids. *Physical Review Letters*, 75(5), 830. <https://doi.org/10.1103/PhysRevLett.75.830>
- Thyagarajan, K., Coreixas, C., & Latt, J. (2023). Exponential distribution functions for positivity-preserving Lattice Boltzmann schemes: Application to 2D compressible flow simulations. *Physics of Fluids*, 35(12), 126115. <https://doi.org/10.1063/5.0175908>
- Timm, K., Kusumaatmaja, H., Kuzmin, A., Shardt, O., Silva, G., & Viggien, E. (2016). *The Lattice Boltzmann method: Principles and practice*. Springer International Publishing AG.
- Tong, S., Liu, Y., Gao, X., Cheng, N., Du, X., & Li, P. (2025). Analysis on the compression process of multi-stage wobble plate compressor under full pressure condition. *Applied Thermal Engineering*, 268, Article 125798. <https://doi.org/10.1016/j.applthermaleng.2025.125798>
- Tran, S. B. Q., Leong, F. Y., Le, Q. T., & Le, D. V. (2022). Lattice Boltzmann method for high Reynolds number compressible flow. *Computers & Fluids*, 249, Article 105701. <https://doi.org/10.1016/j.compfluid.2022.105701>
- Tsutahara, M. (2012). The finite-difference Lattice Boltzmann method and its application in computational aero-acoustics. *Fluid Dynamics Research*, 44(4), Article 045507. <https://doi.org/10.1088/0169-5983/44/4/045507>

- Vahala, G., Keating, B., Soe, M., Yopez, J., Vahala, L., Carter, J., & Ziegeler, S. (2008). MHD turbulence studies using Lattice Boltzmann algorithms. *Computer Physics Communications*, 4, 624–646.
- Wang, D., & Duan, P. (2025). Numerical study of Leidenfrost effect suppression on structured thermal pillars using a Lattice Boltzmann method. *Engineering Applications of Computational Fluid Mechanics*, 19(1), Article 2499251. <https://doi.org/10.1080/19942060.2025.2499251>
- Wang, P., & Zhang, S. (2013). A simple method to construct local equilibrium function for Lattice Boltzmann method. *International Journal of Modern Physics C*, 24(6), 1350038. <https://doi.org/10.1142/S0129183113500381>
- Wang, Y., Yang, L., & Shu, C. (2015). From Lattice Boltzmann method to Lattice Boltzmann flux solver. *Entropy*, 17(11), 7713–7735. <https://doi.org/10.3390/e17117713>
- Watari, M. (2016). Is the Lattice Boltzmann method applicable to rarefied gas flows? Comprehensive evaluation of the higher-order models. *Journal of Fluids Engineering*, 138(1), 011202. <https://doi.org/10.1115/1.4031000>
- Watari, M., & Tsutahara, M. (2003). Two-dimensional thermal model of the finite-difference Lattice Boltzmann method with high spatial isotropy. *Physical Review E*, 67(3), Article 036306. <https://doi.org/10.1103/PhysRevE.67.036306>
- Wilde, D., Krämer, A., Küllmer, K., Foyi, H., & Reith, D. (2019). Multistep Lattice Boltzmann methods: Theory and applications. *International Journal for Numerical Methods in Fluids*, 90(3), 156–169. <https://doi.org/10.1002/fld.4716>
- Wilde, D., Krämer, A., Reith, D., & Foyi, H. (2020). Semi-Lagrangian Lattice Boltzmann method for compressible flows. *Physical Review E*, 101(5), Article 053306. <https://doi.org/10.1103/PhysRevE.101.053306>
- Wilde, D., Krämer, A., Reith, D., & Foyi, H. (2021). High-order semi-Lagrangian kinetic scheme for compressible turbulence. *Physical Review E*, 104(2), Article 025301. <https://doi.org/10.1103/PhysRevE.104.025301>
- Wissocq, G., Taileb, S., Zhao, S., & Boivin, P. (2023). A hybrid Lattice Boltzmann method for gaseous detonations. *Journal of Computational Physics*, 494, Article 112525. <https://doi.org/10.1016/j.jcp.2023.112525>
- Xu, A., Gonnella, G., & Lamura, A. (2003). Phase-separating binary fluids under oscillatory shear. *Physical Review E*, 67(5), Article 056105. <https://doi.org/10.1103/PhysRevE.67.056105>
- Xu, A., Gonnella, G., & Lamura, A. (2006). Morphologies and flow patterns in quenching of lamellar systems with shear. *Physical Review E*, 74(1), Article 011505. <https://doi.org/10.1103/PhysRevE.74.011505>
- Xu, A., Gonnella, G., Lamura, A., Amati, G., & Massaioli, F. (2005). Scaling and hydrodynamic effects in lamellar ordering. *Europhysics Letters*, 71(4), 651. <https://doi.org/10.1209/epl/i2005-10130-3>
- Xu, A., Lin, C., Zhang, G., & Li, Y. (2015). Multiple-relaxation-time Lattice Boltzmann kinetic model for combustion. *Physical Review E*, 91(4), Article 043306. <https://doi.org/10.1103/PhysRevE.91.043306>
- Xu, A.-G., Zhang, G.-C., Gan, Y.-B., Chen, F., & Yu, X.-J. (2012). Lattice Boltzmann modeling and simulation of compressible flows. *Frontiers of Physics*, 7(5), 582–600. <https://doi.org/10.1007/s11467-012-0269-5>
- Xu, K., & Huang, J.-C. (2010). A unified gas-kinetic scheme for continuum and rarefied flows. *Journal of Computational Physics*, 229(20), 7747–7764. <https://doi.org/10.1016/j.jcp.2010.06.032>
- Xu, T., Luo, Y., Hou, Z., Huang, Q., & Pan, G. (2025). Hydrodynamics of jet-flapping combinatorial propulsion in a squid-like swimmer: A study on different coordination modes. *Physics of Fluids*, 37(6), 061909. <https://doi.org/10.1063/5.0271736>
- Xu, X., Hu, Y., He, Y., Han, J., & Zhu, J. (2023). High-order analysis of Lattice Boltzmann models for the conservative Allen-Cahn equation. *Computers & Mathematics with Applications*, 146, 106–125. <https://doi.org/10.1016/j.camwa.2023.06.034>
- Yang, L., Shu, C., & Wu, J. (2012). Development and comparative studies of three non-free parameter Lattice Boltzmann models for simulation of compressible flows. *Advances in Applied Mathematics and Mechanics*, 4(4), 454–472. <https://doi.org/10.4208/aamm.10-m11146>
- Yang, L., Shu, C., & Wu, J. (2013). A moment conservation-based non-free parameter compressible Lattice Boltzmann model and its application for flux evaluation at cell interface. *Computers & Fluids*, 79, 190–199. <https://doi.org/10.1016/j.compfluid.2013.03.020>
- Yang, L., Shu, C., & Wu, J. (2016a). A hybrid Lattice Boltzmann flux solver for simulation of viscous compressible flows. *Advances in Applied Mathematics and Mechanics*, 8(6), 887–910. <https://doi.org/10.4208/aamm.2015.m1172>
- Yang, L., Shu, C., & Wu, J. (2016b). Extension of Lattice Boltzmann flux solver for simulation of 3D viscous compressible flows. *Computers & Mathematics with Applications*, 71(10), 2069–2081. <https://doi.org/10.1016/j.camwa.2016.03.027>
- Yang, T., Wang, J., Yang, L., & Shu, C. (2019). Development of multicomponent Lattice Boltzmann flux solver for simulation of compressible viscous reacting flows. *Physical Review E*, 100(3), 033315. <https://doi.org/10.1103/PhysRevE.100.033315>
- Yazdi, H., Rahimian, M. H., & Safari, H. (2020). A Lattice Boltzmann model for computing compressible two-phase flows with high density ratio. *SN Applied Sciences*, 2(1), 1–14. <https://doi.org/10.1007/s42452-019-1872-7>
- Yoo, H., Wissocq, G., Jacob, J., Favier, J., & Sagaut, P. (2023). Compressible Lattice Boltzmann method with rotating over-set grids. *Physical Review E*, 107(4), Article 045306. <https://doi.org/10.1103/PhysRevE.107.045306>
- You-Sheng, X., Yang, L., Xiao-Feng, Y., & Feng-Min, W. (2008). Lattice Boltzmann simulation of convection in a porous medium with temperature jump and velocity slip boundary conditions. *Communications in Theoretical Physics*, 49(5), 1319. <https://doi.org/10.1088/0253-6102/49/5/51>
- Zhan, N., Chen, R., Liu, J., Qiu, R., & You, Y. (2021). Mesh-free Lattice Boltzmann flux solver for compressible inviscid flows. *International Journal for Numerical Methods in Fluids*, 93(5), 1378–1395. <https://doi.org/10.1002/fld.4933>
- Zhao, S., Farag, G., Boivin, P., & Sagaut, P. (2020). Toward fully conservative hybrid Lattice Boltzmann methods for compressible flows. *Physics of Fluids*, 32(12), 126118. <https://doi.org/10.1063/5.0033245>
- Zhao, X., Yang, L., & Shu, C. (2022). An implicit Lattice Boltzmann flux solver for simulation of compressible flows.

- Computers & Mathematics with Applications*, 107, 82–94. <https://doi.org/10.1016/j.camwa.2021.12.014>
- Zipunova, E., Perepelkina, A., Zakirov, A., & Khilkov, S. (2021). Regularization and the particles-on-demand method for the solution of the discrete Boltzmann equation. *Journal of Computational Science*, 53, Article 101376. <https://doi.org/10.1016/j.jocs.2021.101376>
- Zou, H., Zhang, Q., Lu, H., Song, M., Hu, Z., Jiang, D., Li, T., & Liu, J. (2025). A comprehensive GPU-accelerated Lattice Boltzmann framework for high-Reynolds-number flow in complex rod bundles. *Engineering Applications of Computational Fluid Mechanics*, 19(1), Article 2555335. <https://doi.org/10.1080/19942060.2025.2555335>